

# CENTRAL QUEENSLAND UNIVERSITY

## FIRE MAIN UPGRADE

### ROCKHAMPTON NORTH CAMPUS

# 12622621



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| TOTAL: 42                          |                                           |

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Project No.  
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Drawing Title  
HYDRAULIC SERVICES  
COVER SHEET & DRAWING INDEX

Drawing No.  
12622621-GHD-00-00-DRG-HY-00001

Size  
A1

Rev  
0



GENERAL NOTES

SCOPE OF WORK

1. THE SCOPE OF WORK INCLUDES THE SUPPLY, DELIVERY, INSTALLATION, TESTING AND COMMISSIONING OF THE SYSTEMS AS SHOWN ON THE SERVICES DRAWINGS AND DETAILED WITHIN THE PROJECT SPECIFICATION. THE CONTRACTOR IS REQUIRED TO PROVIDE AND INSTALL ALL THE REQUIRED MATERIALS TO ENSURE THE FULL, CORRECT, COMPLIANT AND SAFE OPERATION OF THE SYSTEMS.

RELEVANT LEGISLATION, ACTS AND REGULATIONS

1. ALL WORKS TO BE CONSTRUCTED IN ACCORDANCE WITH QUEENSLAND LEGISLATION, ACTS AND REGULATIONS.
- QUEENSLAND BUILDING ACT 1975
  - BUILDING CODE OF AUSTRALIA 2019
  - QUEENSLAND DEVELOPMENT CODE
  - PLUMBING AND DRAINAGE ACT 2018
  - PLUMBING AND DRAINAGE REGULATION 2022
  - QUEENSLAND PLUMBING AND WASTEWATER CODE 2019
  - QUEENSLAND BUILDING AND CONSTRUCTION COMMISSION ACT 1991
  - QUEENSLAND BUILDING AND CONSTRUCTION COMMISSION REGULATION 2018.
2. THE FOLLOWING LIST CONTAINS A NON-EXHAUSTIVE LIST OF THE LATEST GAZETTED STANDARDS REQUIRED FOR THE COMPLIANT DESIGN AND INSTALLATION OF HYDRAULIC SERVICES.
- | STANDARD      | YEAR | TITLE                                                                                                                    |
|---------------|------|--------------------------------------------------------------------------------------------------------------------------|
| AS 1851       | 2012 | ROUTINE SERVICE OF FIRE PROTECTION SYSTEMS AND EQUIPMENT                                                                 |
| AS 2419.1     | 2021 | FIRE HYDRANT INSTALLATION - SYSTEM DESIGN                                                                                |
| AS 2419.2     | 2021 | FIRE HYDRANT VALVES                                                                                                      |
| AS 2419.3     | 2021 | FIRE BRIGADE BOOSTER CONNECTIONS                                                                                         |
| AS 2441       | 2005 | FIRE HOSE REEL INSTALLATIONS                                                                                             |
| AS 2444       | 2001 | FIRE EXTINGUISHERS INSTALLATIONS                                                                                         |
| AS 2941       | 2013 | FIRE PUMPSET SYSTEMS                                                                                                     |
| AS 4072.1     | 2005 | COMPONENTS FOR THE PROTECTION OF OPENINGS IN FIRE-RESISTANT SEPARATING ELEMENTS SERVICE PENETRATIONS AND CONTROL JOINTS. |
| AS/NZS 2648.1 | 1995 | UNDERGROUND MARKING TAPE.                                                                                                |
| AS/NZS 3000   | 2018 | ELECTRICAL INSTALLATIONS                                                                                                 |
| AS/NZS 3500.1 | 2018 | WATER SERVICES                                                                                                           |

DRAWINGS

1. THESE DRAWINGS ARE GUIDED ONLY FOR THE LOCATION AND LAYOUT OF THE SERVICE PIPES.
2. DO NOT SCALE OFF THESE DRAWINGS.

CO-ORDINATION

1. THESE DRAWINGS ARE TO BE READ IN CONJUNCTION WITH THE OTHER CONTRACT DOCUMENTS, SPECIFICATIONS, AND DRAWINGS.
2. THROUGHOUT THE PROCESS OF EXECUTION OF THE WORKS, THE HYDRAULIC CONTRACTOR WILL BE RESPONSIBLE FOR THE COORDINATION OF THE INSTALLATION WITH OTHER TRADES.

FIRE HYDRANTS AND HOSE REELS

STANDARD

- FH01. FIRE HYDRANT SERVICE TO COMPLY WITH AS 2419.1,2 & 3
- FH02. FIRE HOSE REEL SERVICE TO COMPLY WITH AS/NZS 1221 & AS 2441.
- FH03. CONTRACTOR SHALL ALLOW FOR FLOW AND PRESSURE TESTING OF THE HYDRANT & HOSE REEL SYSTEMS BY ACCREDITED TESTERS ON COMPLETION OF THE WORKS. PROVIDE A CERTIFICATE THAT THE SYSTEMS IS INSTALLED TO THE REQUIREMENTS OF AS2419.1

VALVES

- FH04. ALL FIRE SERVICES ISOLATION VALVES SHALL BE LOCKED IN THE OPEN POSITION ON COMPLETION.
- FH05. PROVIDE SLUICE VALVES AS REQUIRED TO AS 2638.2-1999. SLUICE VALVES SHALL BE KEY HEAD TYPE, ANTI-CLOCKWISE CLOSING, BURIED SERVICE, RESILIENT SEATED, FLANGED AND "RILSAN" COATED. SILICONE CAPS OVER VALVE BODY BOLTS SHALL BE NEUTRAL CURE AND FREE FROM DEFECTS.
- FH06. PROVIDE VALVES SUITABLE FOR A SERVICE RATINGS OF 1 600 kPa AT TEMPERATURES FROM 10°C TO 40°C, AND A MINIMUM "OPEN VALVE" PIPELINE TEST PRESSURE OF 3 000 kPa.

SAFETY

- FH07. LIAISE WITH THE LOCAL AUTHORITY AND OBTAIN ALL PERMITS AS REQUIRED FOR THE GENERAL CONSTRUCTION OF THE PLUMBING WORKS. THESE INCLUDE BUT ARE NOT LIMITED TO ROAD OPENING PERMITS, PLUMBING PERMITS AND PAYMENT OF AUTHORITY FEES.
- FH08. ENSURE THAT APPROPRIATE SAFETY MEASURES ARE TAKEN TO PROTECT TRAFFIC, PROPERTY AND PEDESTRIANS FOR THE FULL DURATION OF THE WORKS.

MARKING SERVICES

- FH09. UNDERGROUND MARKING TAPE: TO AS/NZS 2648.1. LAY PLASTIC WARNING TAPE, 350 mm ± 50 mm ABOVE ALL BURIED PIPING, FOR THE FULL LENGTH OF THE PIPING. COVER THE TAPE WITH APPROXIMATELY 100 mm DEEP SAND.
- FH10. PROVIDE TRENCH MARKER TAPE THAT CLEARLY IDENTIFIES THE SERVICE BELOW.

SAFETY IN DESIGN

GENERAL

1. THE SAFETY RISK MITIGATION ITEMS BELOW ARE BASED ON GHD'S DESIGN OFFICE EXPERIENCE AND DO NOT NECESSARILY ACCOUNT FOR ALL CONSTRUCTION, OPERATION, MAINTENANCE, AND DEMOLITION SAFETY RISKS. BASED ON INFORMATION AVAILABLE WHEN THIS DRAWING WAS MADE, IN ITS CAPACITY AS DESIGNER ONLY, GHD HAS TRIED TO IDENTIFY SAFETY RISKS PERTAINING TO CONSTRUCTION, OPERATION, MAINTENANCE AND DEMOLITION PHASES OF THE ASSET. INCLUSION (OR NOT) OF ANY ITEM DOES NOT REDUCE OR LIMIT OBLIGATIONS OF CONSTRUCTOR, USER, MAINTAINER AND DEMOLISHER TO UNDERTAKE APPROPRIATE RISK MANAGEMENT ACTIVITIES TO REDUCE RISK AND IS NOT AN ADMISSION BY GHD THAT INCLUSION OF ANY ITEM IS THE DESIGNER'S RESPONSIBILITY.
2. CARRY OUT WORK IN A SAFE MANNER IN ACCORDANCE WITH APPLICABLE LEGISLATION, STATUTORY REGULATIONS, BY-LAWS OR RULES. CONTRACTOR IS RESPONSIBLE FOR OCCUPATIONAL HEALTH AND SAFETY OF SITE PERSONNEL AND GENERAL PUBLIC, IN ACCORDANCE WITH WORK HEALTH AND SAFETY ACT 2010, LEGISLATIVE REQUIREMENTS, ASSOCIATED REGULATIONS AND CODES OF PRACTICE, INDUSTRIAL AGREEMENTS AND ACCEPTED INDUSTRY PRACTICE, INDUSTRIAL AGREEMENTS AND ACCEPTED INDUSTRY PRACTICE.
3. CONSTRUCT BUILDING ELEMENTS THAT CONTRIBUTE TO SAFETY, SUCH AS HANDRAILS AND TO BOARDS, FALL ARREST SYSTEMS, ACCESS STAIRS ETC. AS EARLY AS POSSIBLE.
4. PROVIDE SAFETY BARRIERS AT EDGES OF OPENINGS AND ELEVATED AREAS.
5. REVIEW ADEQUACY OF WORKING SPACE AVAILABLE FOR CONSTRUCTION ACTIVITIES. ENSURE SEPARATION OF PLANT AND PERSONNEL ON SITE, INCLUDING MOVEMENTS OF BOTH.
6. LOCATE LIFTING SLEW AND LAY DOWN AREAS AWAY FROM REGULAR CONSTRUCTION TRAFFIC.
7. PROVIDE PROTECTION TO PERSONNEL FROM PLANT AND EQUIPMENT.
8. ENSURE ISOLATION SAFE SYSTEMS OF WORK OR PROTECTIVE MEASURES ARE INSTALLED BEFORE WORKING NEAR LIVE ELECTRICAL INFRASTRUCTURE. PROVIDE PROTECTION OF ELECTRICAL OVERHEAD WIRING SYSTEMS DURING CONSTRUCTION.
9. WRITTEN RISK ASSESSMENTS ARE ADVISED FOR ACCESS TO OPEN EXCAVATIONS.
10. PROVIDE ACCESS AND EGRESS TO EXCAVATIONS APPROPRIATE IN CASE OF INUNDATION, COLLAPSE, OR ENGULFMENT.
11. LOCATE STOCKPILES AND HEAVY EQUIPMENT INCLUDING CRANES AWAY FROM BURIED SERVICES AND BUILDING BOUNDARIES WHERE ADJACENT BASEMENTS ARE PRESENT.
12. SEEK ADVICE FROM SUITABLY QUALIFIED GEOTECHNICAL OR STRUCTURAL ENGINEER PRIOR TO OPERATION OF HEAVY SURFACE PLANT AND EQUIPMENT OR STOCKPILING MATERIAL NEAR OPEN EXCAVATIONS OR EXISTING RETAINING STRUCTURES.
13. DO NOT STOCKPILE MATERIAL BEHIND OR EXCAVATE IN FRONT OF EXISTING RETAINING WALLS UNTIL WALL STABILITY HAS BEEN REVIEWED BY SUITABLY QUALIFIED STRUCTURAL ENGINEER.
14. SEEK ADVICE FROM SUITABLY QUALIFIED STRUCTURAL ENGINEER BEFORE EXCAVATING BELOW EXISTING FOOTING LEVELS.
15. SEEK ADVICE FROM SUITABLY QUALIFIED STRUCTURAL ENGINEER IF PLANNING CRANE LIFTS OR HOIST INSTALLATION ON PARTIALLY ERECTED OR SUSPENDED STRUCTURES.
16. SEEK ADVICE FROM SUITABLY QUALIFIED STRUCTURAL ENGINEER BEFORE CORING, CHASING, CUTTING OR REMOVAL OF EXISTING CONCRETE AND ENFORCEMENT.
17. HAVE SUITABLY QUALIFIED STRUCTURAL ENGINEER UNDERTAKE STRUCTURAL CHECK OF EXISTING CONCRETE, MASONRY, AND STUD WALLS WHERE FIXINGS OR EQUIPMENT IS TO BE ATTACHED.
18. MINIMISED SITE-BASED TREATMENTS (E.G. WELDING, CUTTING, SPRAY PAINTING, GRIT BLASTING ETC.), PROVIDE ADEQUATE PROTECTION SCREENING AND VENTILATION TO MINIMISE HAZARDS TO PERSONNEL IF SITE-BASED TREATMENT IS UNAVOIDABLE.
19. TRY TO AVOID WORKING IN CONFINED SPACES. IF CONFINED SPACES WORK CAN'T BE AVOIDED, PROVIDE SAFE WORK METHOD STATEMENT (SWMS) ADDRESSING MITIGATION OF RISKS. PROVIDE ADEQUATE SIGNAGE TO TEMPORARY AND PERMANENT CONFINED SPACES TO AS 2865.
20. AVOID HOT WORKS ON SITE PARTICULARLY IN TIMBER FRAMED STRUCTURES. HOT WORKS TO COMPLY WITH CLIENT PROCEDURES FOR "APPLICABLE HOT WORKS" PERMITS.

ABBREVIATIONS

|      |                            |      |                               |
|------|----------------------------|------|-------------------------------|
| AAV  | AIR ADMITTANCE VALVE       | IL   | INVERT LEVEL                  |
| AFFL | ABOVE FINISHED FLOOR LEVEL | IO   | INSPECTION OPENING            |
| AHD  | AUSTRALIAN HEIGHT DATUM    | IOS  | INSPECTION OPENING TO SURFACE |
| AP   | ACCESS PANEL               | MH   | MAINTENANCE HOLE              |
| AS   | AUSTRALIAN STANDARD        | NP   | NON-POTABLE                   |
| B    | BASIN                      | NRV  | NON-RETURN VALVE              |
| BG   | BOX GUTTER                 | NTS  | NOT TO SCALE                  |
| BTH  | BATH                       | OD   | OUTSIDE DIAMETER              |
| CV   | CHECK VALVE                | OF   | OVERFLOW                      |
| COS  | CLEAN OUT TO SERVICES      | ORG  | OVERFLOW RELIEF GULLY         |
| CP   | CHROME PLATED              | PC   | PROPERTY CONNECTION           |
| CU   | COPPER                     | PE   | POLYTHYLENE                   |
| CW   | COLD WATER                 | PEX  | CROSS-LINKED POLYEHTYLENE     |
| DD   | DRAINAGE DROPPER           | PRV  | PRESSURE REDUCING VALVE       |
| DG   | DISCONNECTOR GULLY         | PW   | POTABLE WATER                 |
| DIA  | DIAMETER                   | RL   | RELATIVE LEVEL                |
| DN   | NOMINAL DIAMETER           | RPZD | REDUCED PRESSURE ZONE DEVICE  |
| DP   | DOWNPIPE                   | RV   | RELIEF VALVE                  |
| DT   | DISCONNECTED TRAP          | RW   | RAINWATER                     |
| DW   | DISHWASHER                 | SHR  | SHOWER                        |
| DWV  | DRAINAGE WASTE VENT        | SK   | SINK                          |
| E    | EXISTING                   | SK-C | CLEANERS SINK                 |
| EG   | EAVES GUTTER               | SMH  | SEWER MANHOLE                 |
| EJ   | EXPANSION JOINT            | SS   | STAINLESS STEEL               |
| FE   | FIRE EXTINGUISHER          | STV  | STACK VENT                    |
| FFL  | FINISHED FLOOR LEVEL       | SW   | STORMWATER                    |
| FH   | FIRE HYDRANT               | SWP  | STORMWATER PIT                |
| FHR  | FIRE HOSE REEL             | TD   | TUNDISH                       |
| FSL  | FINISHED SURFACE LEVEL     | TMV  | THERMOSTATIC MIXING VALVE     |
| FU   | FIXTURE UNIT               | TY   | TYPICAL                       |
| FW   | FLOOR WASTE                | UNO  | UNLESS NOTED OTHERWISE        |
| FWG  | FLOOR WASTE GULLY          | UN   | URINAL                        |
| GP   | GULLY PIT                  | VC   | VENT COWL                     |
| HT   | HOSE TAP VACUUM BREAKER    | VP   | VENT PIPE                     |
| HW   | HOT WATER                  | WC   | WATER CLOSET                  |
| HW-R | HOT WATER RETURN           | WM   | WASHING MACHINE               |
| HWU  | HOT WATER UNIT             | WM   | WATER METER.                  |
| IC   | INSPECTION CHAMBER         |      |                               |
| ID   | INSIDE DIAMETER            |      |                               |

| PIPEWORK MATERIALS SCHEDULE |                                   |          |             |        |                    |                                                                               |
|-----------------------------|-----------------------------------|----------|-------------|--------|--------------------|-------------------------------------------------------------------------------|
| SERVICES                    | APPLICATION                       | MATERIAL | STANDARD    | CLASS  | JOINTING           | COMMENTS                                                                      |
| FIRE SERVICES               |                                   |          |             |        |                    |                                                                               |
| FIRE                        | IN-GROUND                         | PE100    | AS/NZS 4130 | PN16   | FUSION WELDED      |                                                                               |
|                             | ABOVE GROUND                      | STEEL    | AS 1074     | MEDIUM | ROLLED GROOVE      | RIDGED JOINTS. HOT DIPPED GALVANIZED.                                         |
|                             | PILLAR HYDRANT & BOOSTER ASSEMBLY | STEEL    | AS 1074     | MEDIUM | WELDED AND FLANGED | WELDING TO AS 4041 CLASS 3. PIPEWORK HOT DIPPED GALVANISED AFTER FABRICATION. |
|                             | HOSE REELS                        | STEEL    | AS 1074     | MEDIUM | SCREWED            | IN-LINE GALVANISING                                                           |
|                             | HOSE REELS                        | COPPER   | AS 1432     | TYPE B | BRAZED / PRESS-FIT | FHR CONNECTED TO DOMESTIC WATER SERVICE                                       |

| TESTABLE BACKFLOW PREVENTION DEVICES |       |                         |           |            |
|--------------------------------------|-------|-------------------------|-----------|------------|
| DEVICE No.                           | MAKE  | TYPE                    | SIZE (mm) | SERIAL No. |
| RPZD #1                              | AMES  | RPZD - IN-LINE          | 150       | 1031601220 |
| RPZD #2                              | AMES  | RPZD - BYPASS           | 150       | 1031581229 |
| SCVT                                 | WATTS | DEDICATED FIRE MAIN SCV | 150       | AN0033     |

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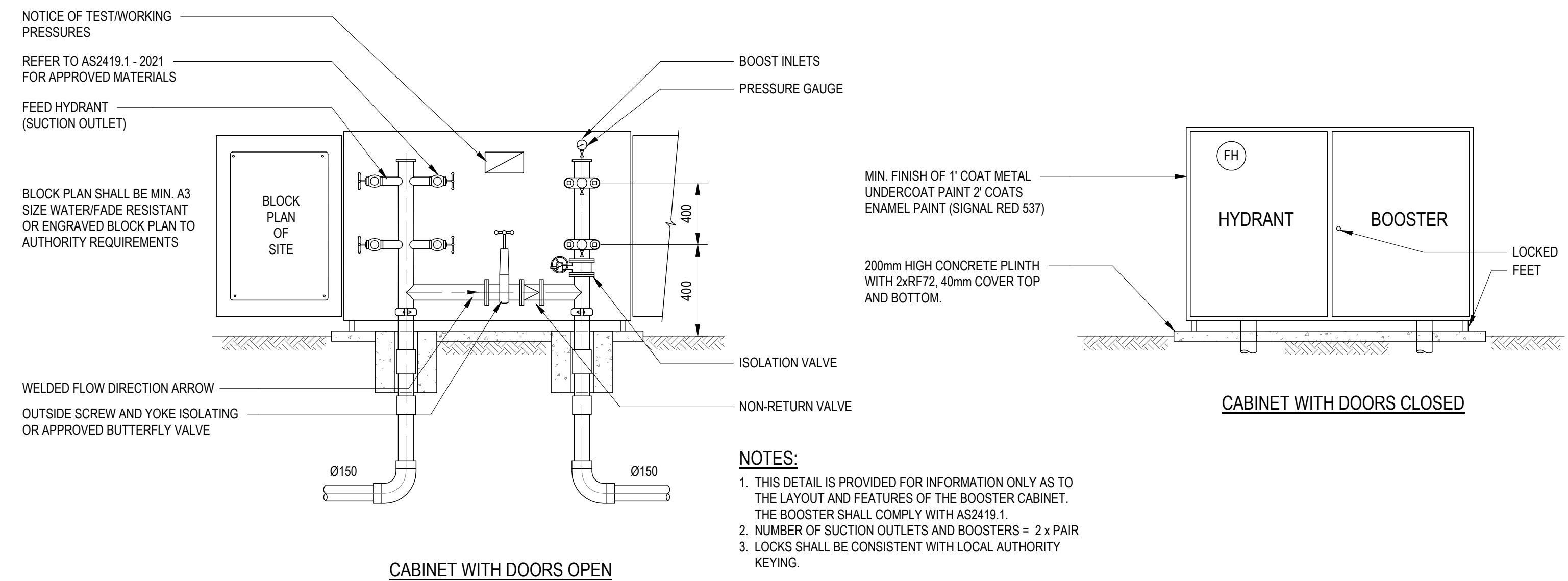
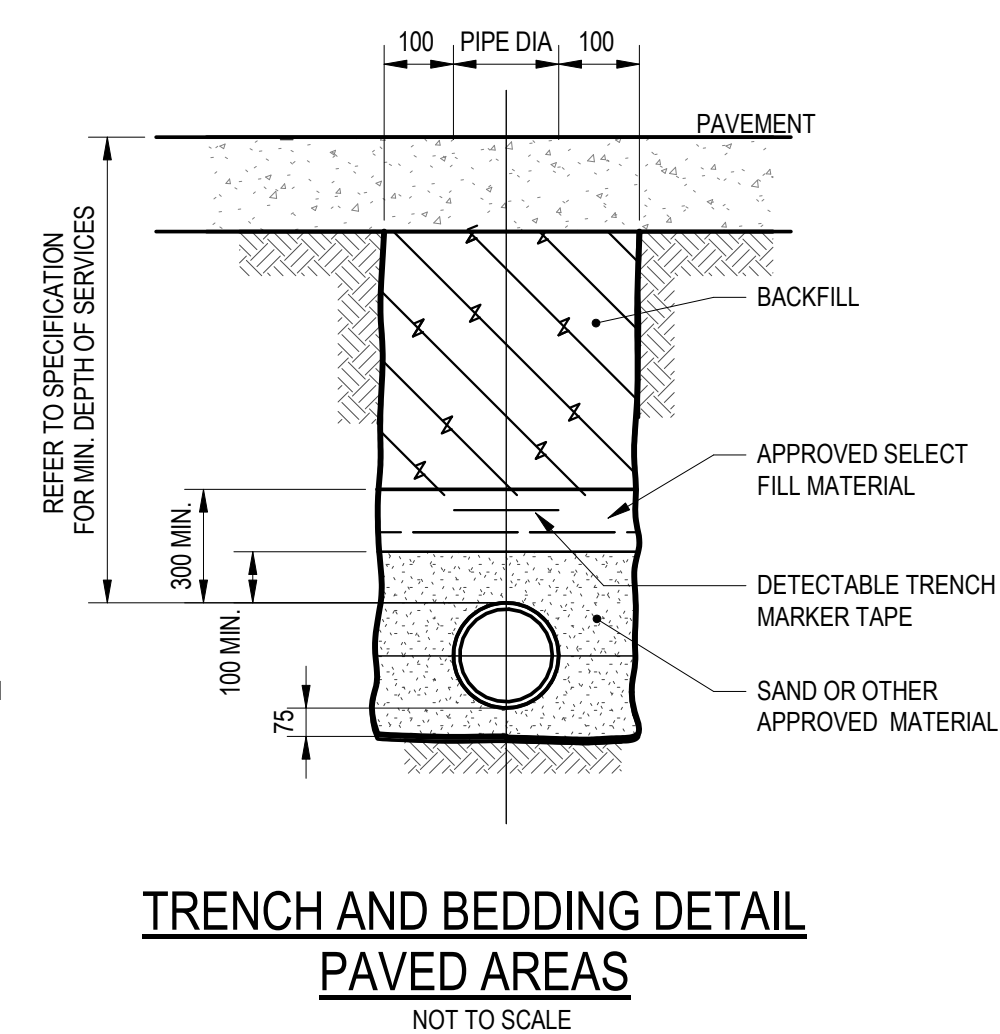
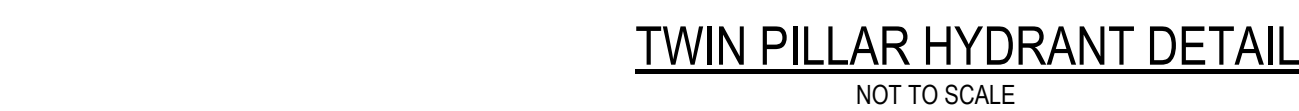
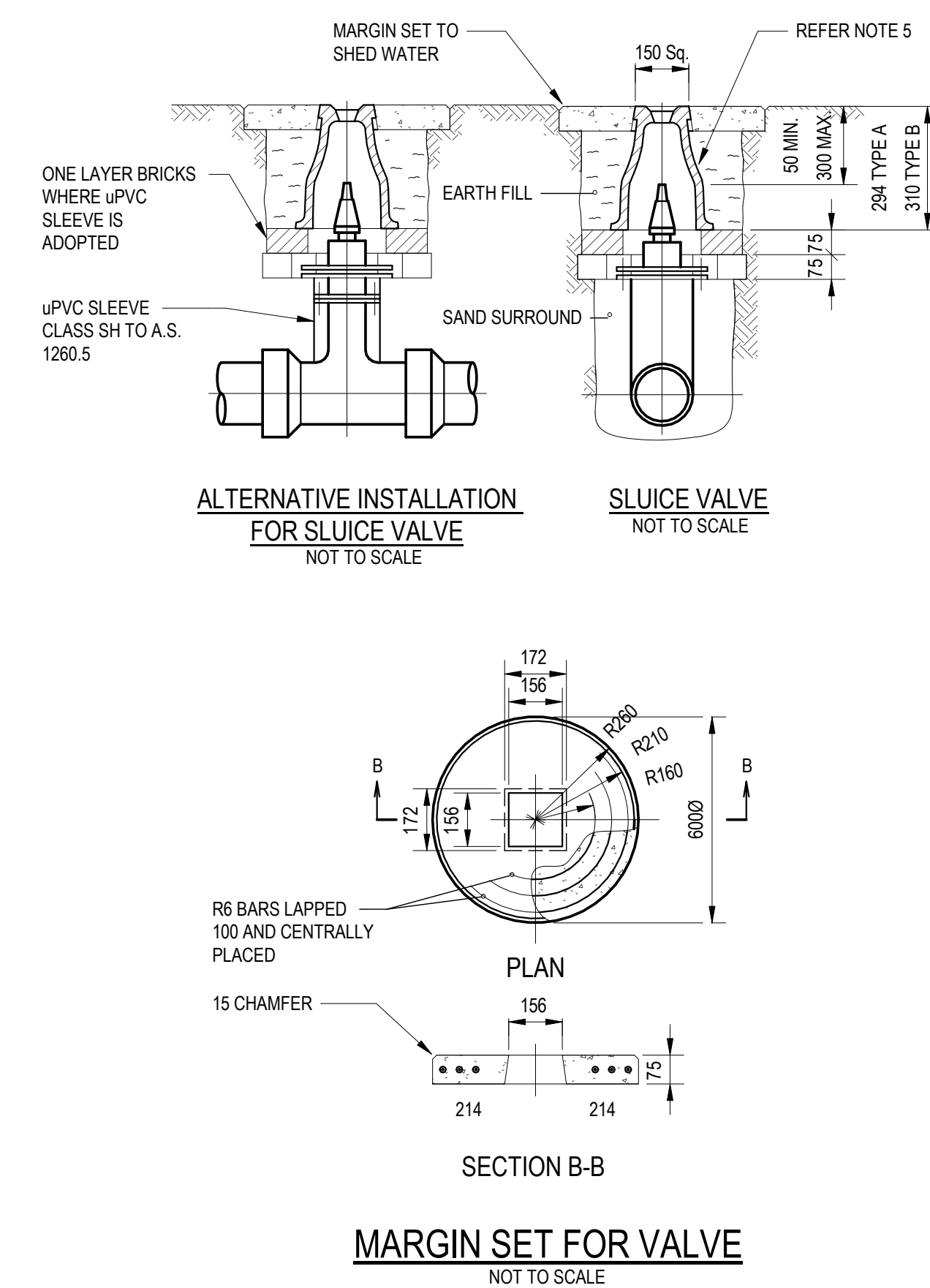
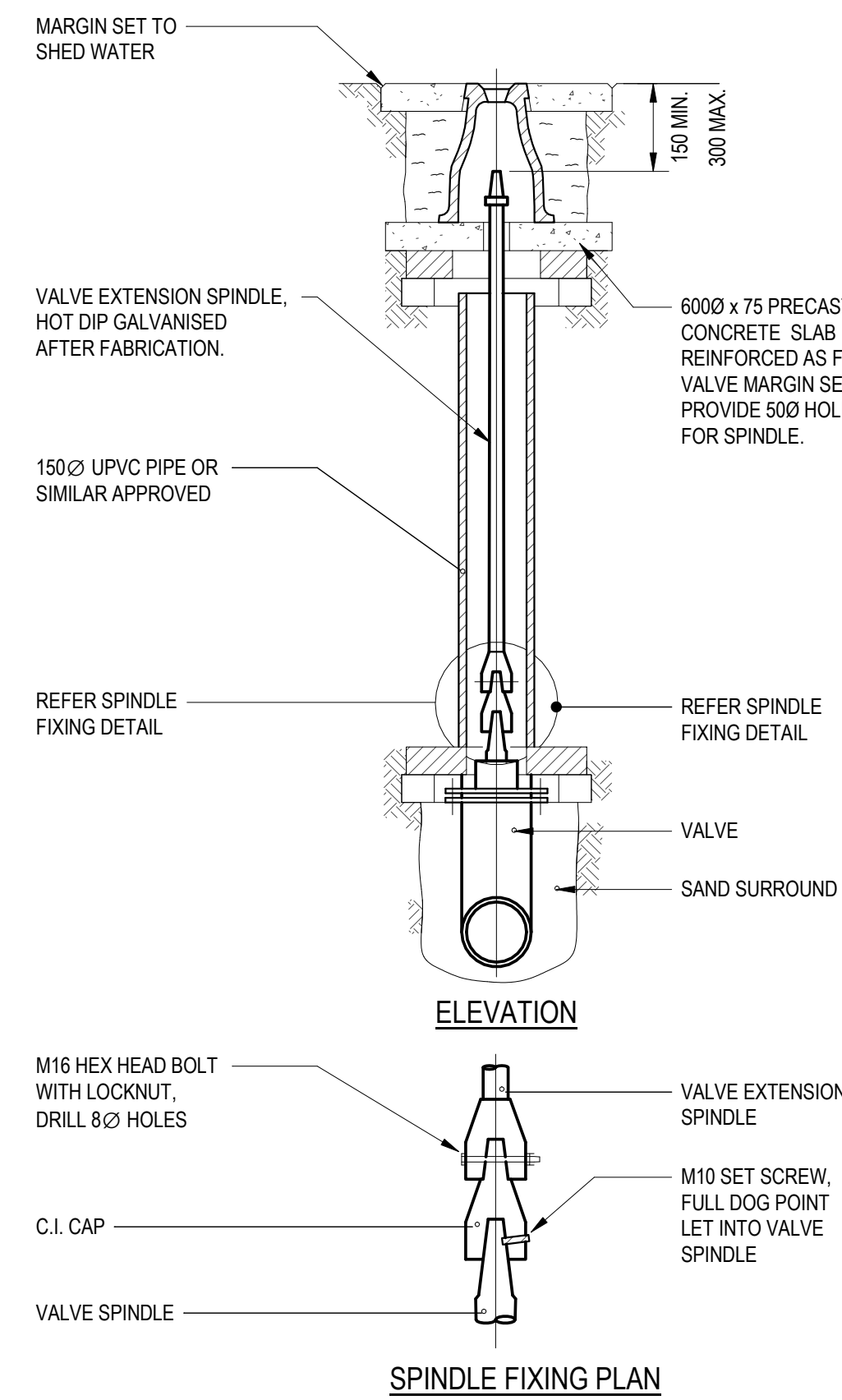
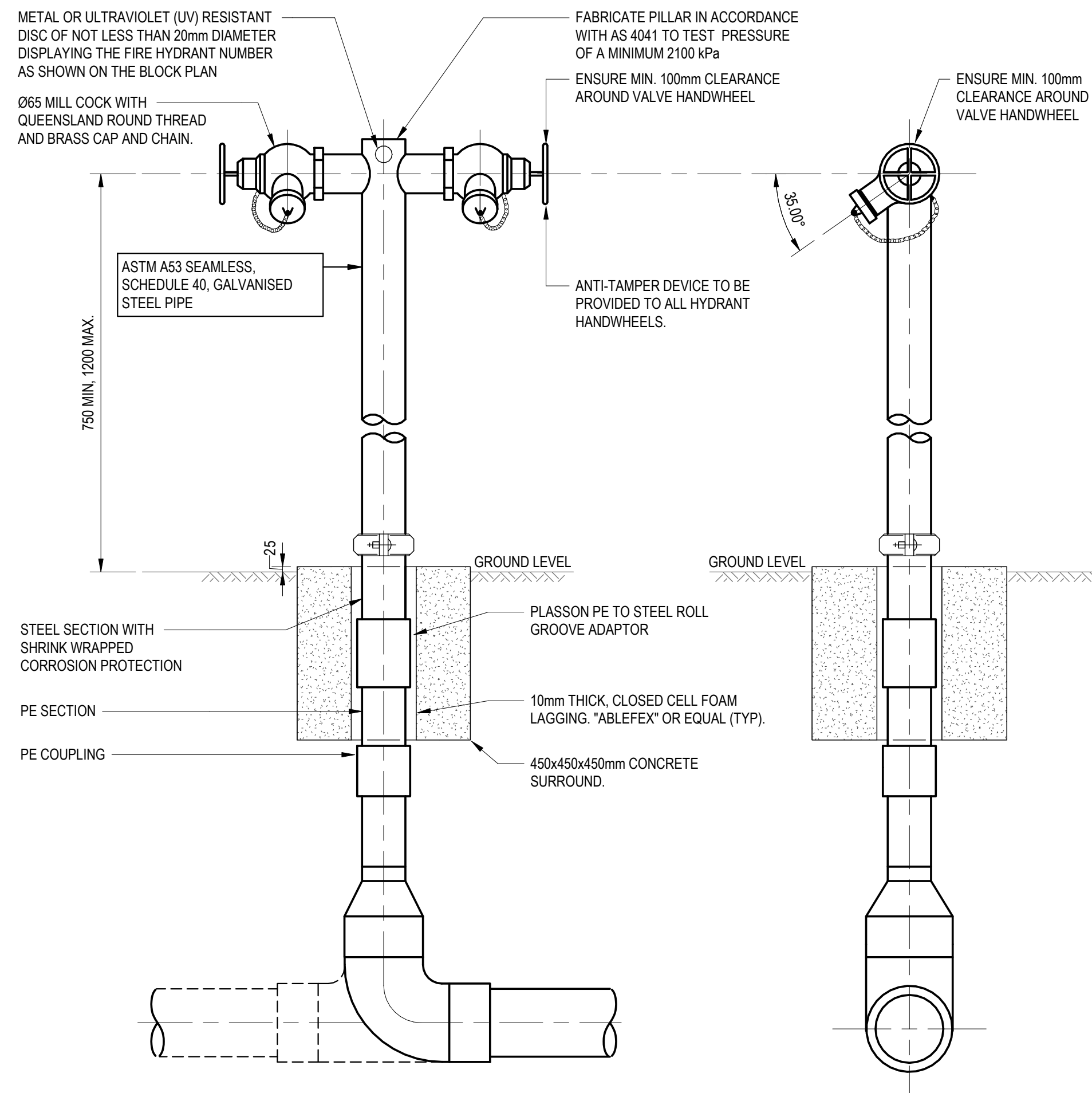
Drawing Title  
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HYDRAULIC NOTES AND SCHEDULES

Drawing No.  
12622621-GHD-00-00-DRG-HY-00002

Size  
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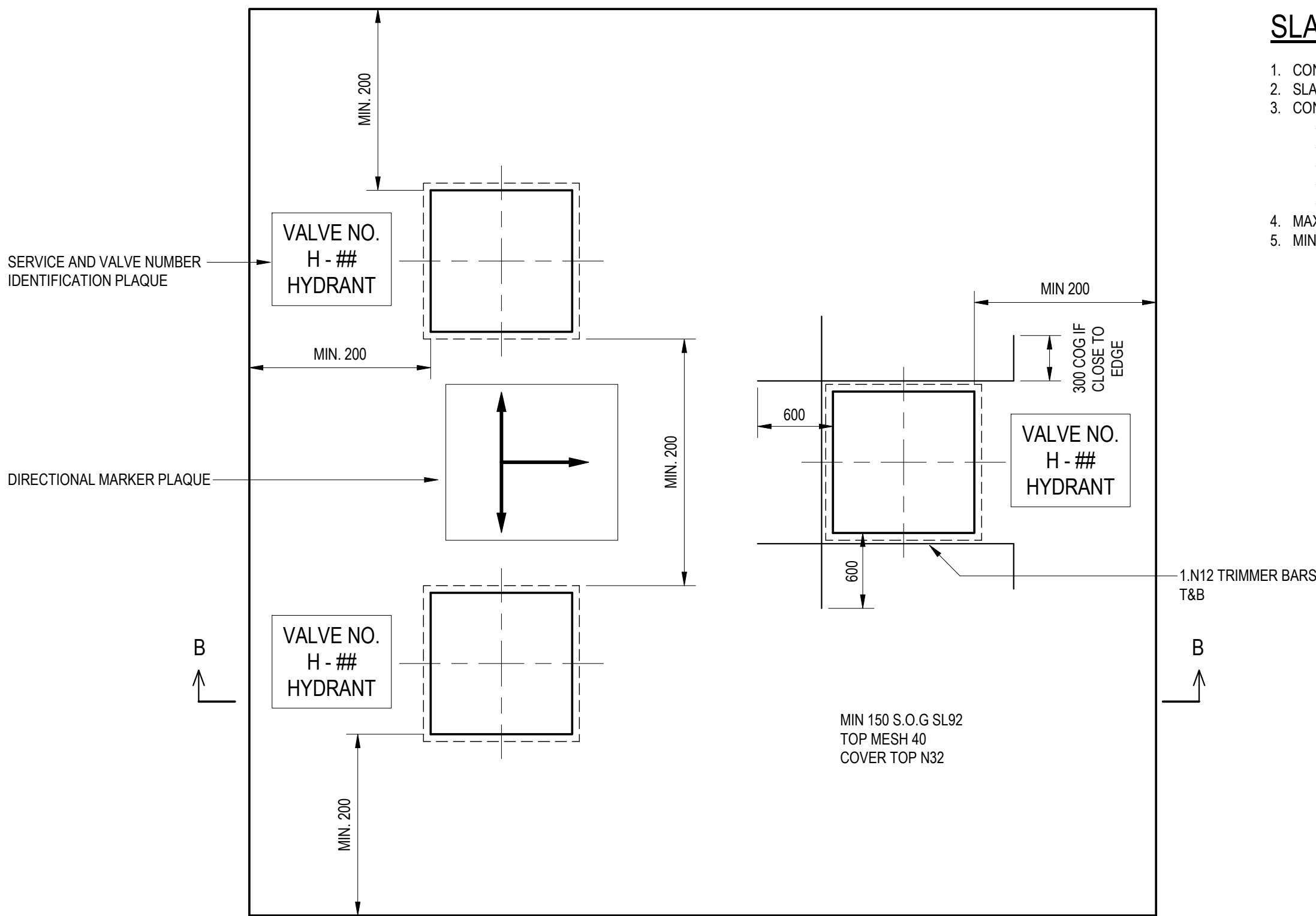
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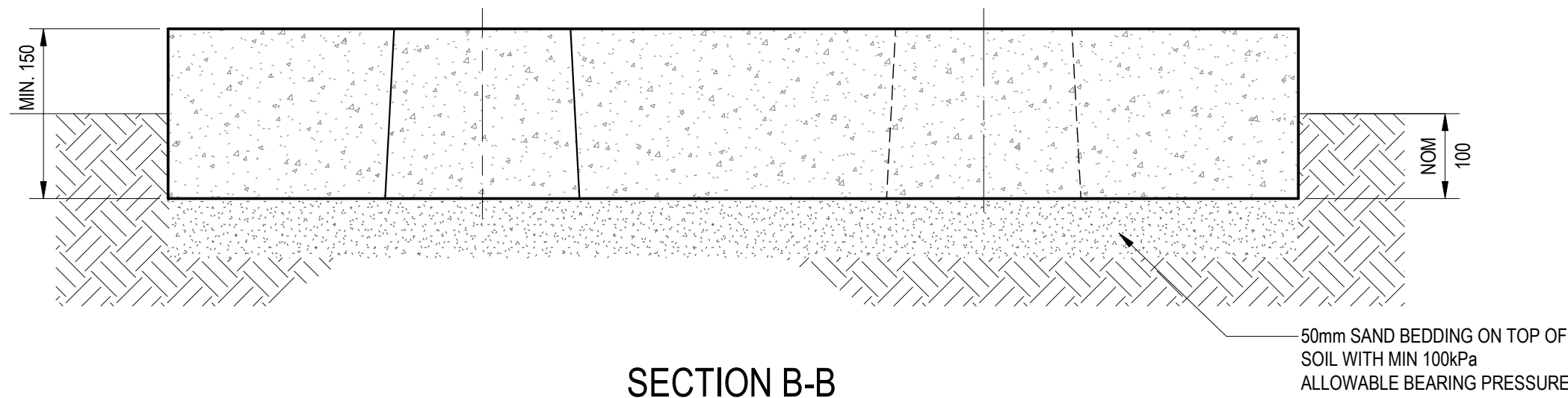


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PLAN

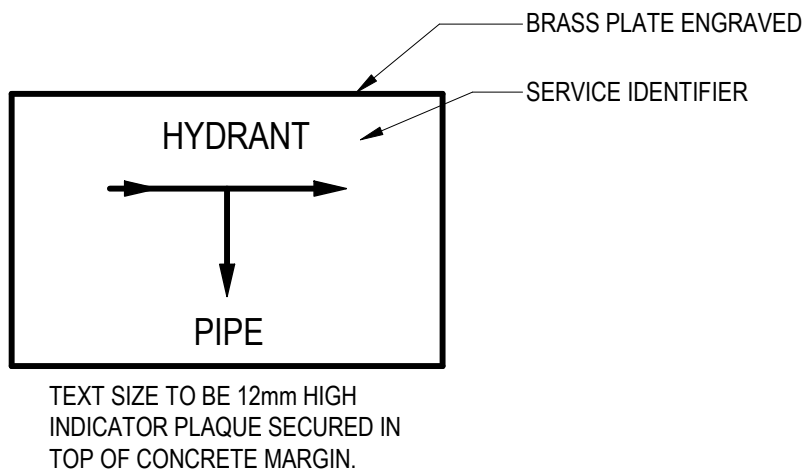


SECTION B-B

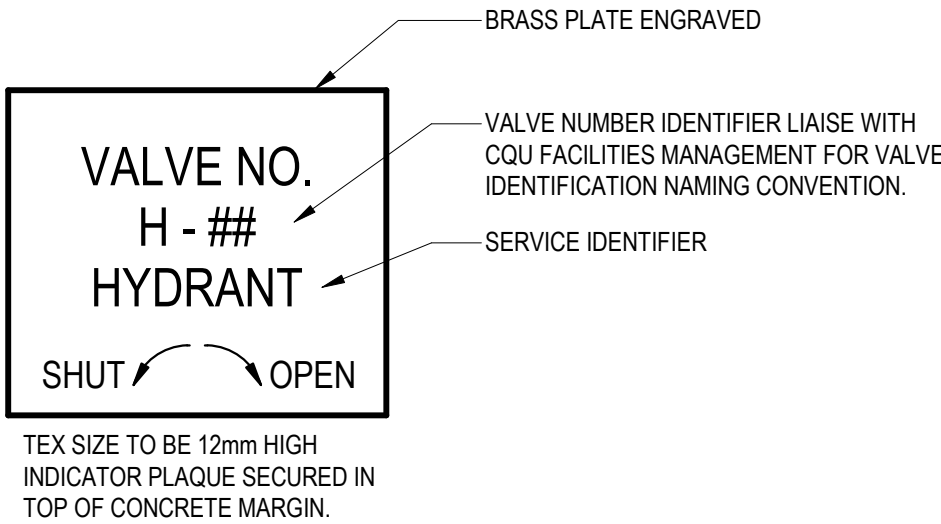
MARGIN SETT FOR THREE WAY VALVE  
NOT TO SCALE

SLAB ON GROUND NOTES

1. CONCRETE SLAB SHALL BE CONSTRUCTED IN ACCORDANCE WITH AS3600
2. SLAB DESIGN LOADS: 3kPa
3. CONCRETE SLAB DETAILS:
  - CONCRETE STRENGTH: 32MPa
  - EXPOSURE CLASS: B1
  - CONCRETE COVER TOP: 40mm, SIDE: 45mm
  - SLAB REINFORCEMENT: SL92 TOP MESH
  - TRIMMER BARS 1.N12 T&B AROUND PENO
4. MAX PENO SIZE 800x800
5. MIN DISTANCE BETWEEN PENO & PENO TO SLAB EDGE: 200mm



DIRECTIONAL MARKER DETAIL



VALVE MARKER DETAIL

|          |                |         |          |          |
|----------|----------------|---------|----------|----------|
|          |                |         |          |          |
|          |                |         |          |          |
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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

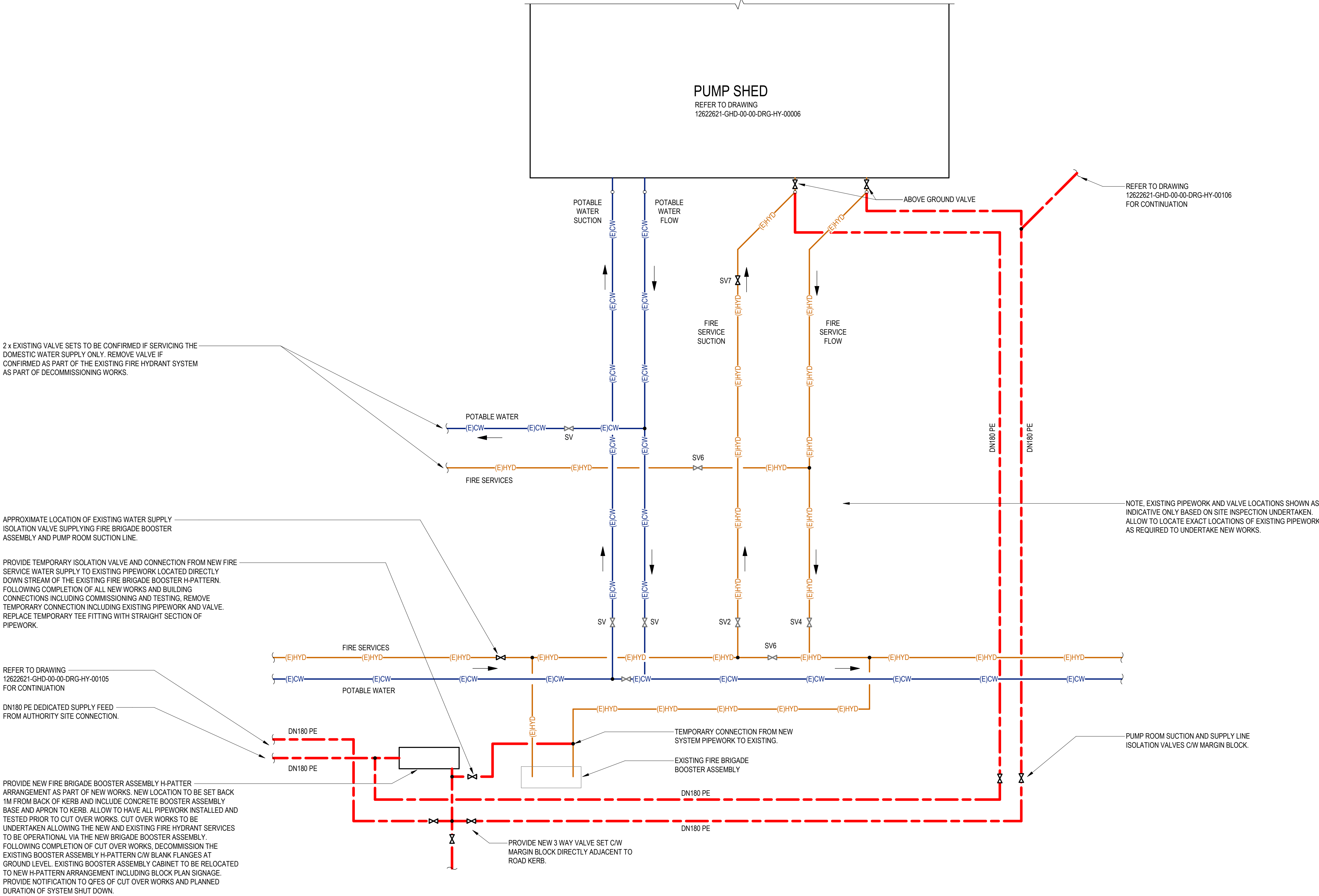
Drawing Title  
HYDRAULIC SERVICES  
DETAIL SHEET 2

Drawing No.  
12622621-GHD-00-00-DRG-HY-00004

Size  
A1

Rev  
0





FIRE PUMP SHED - PIPE / VALVE SCHEMATIC  
EXTERNAL  
NOT TO SCALE

|          |                |                       |
|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |



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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SCHEMATIC

Drawing No.  
12622621-GHD-00-00-DRG-HY-00005

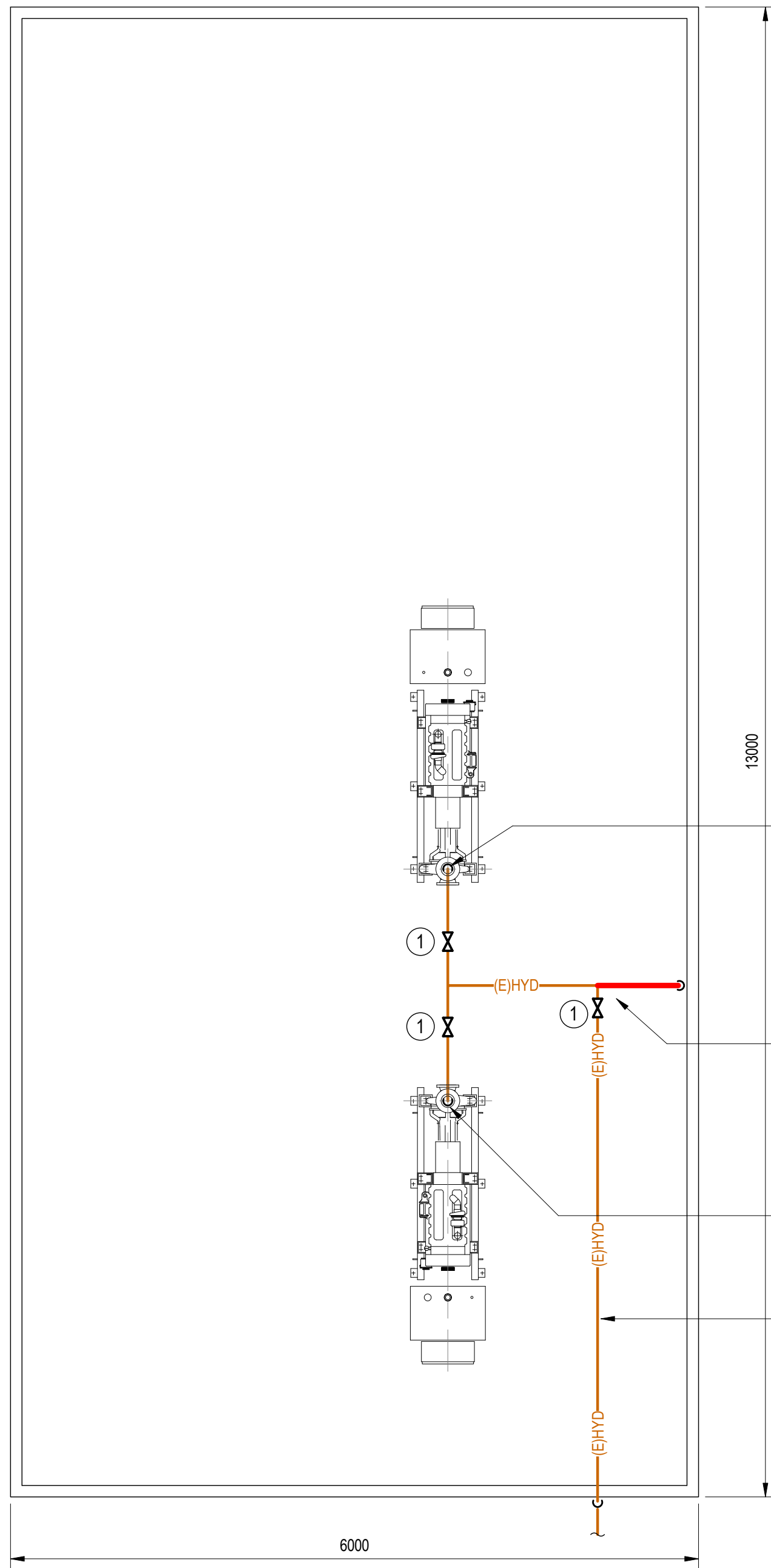
Size  
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Rev  
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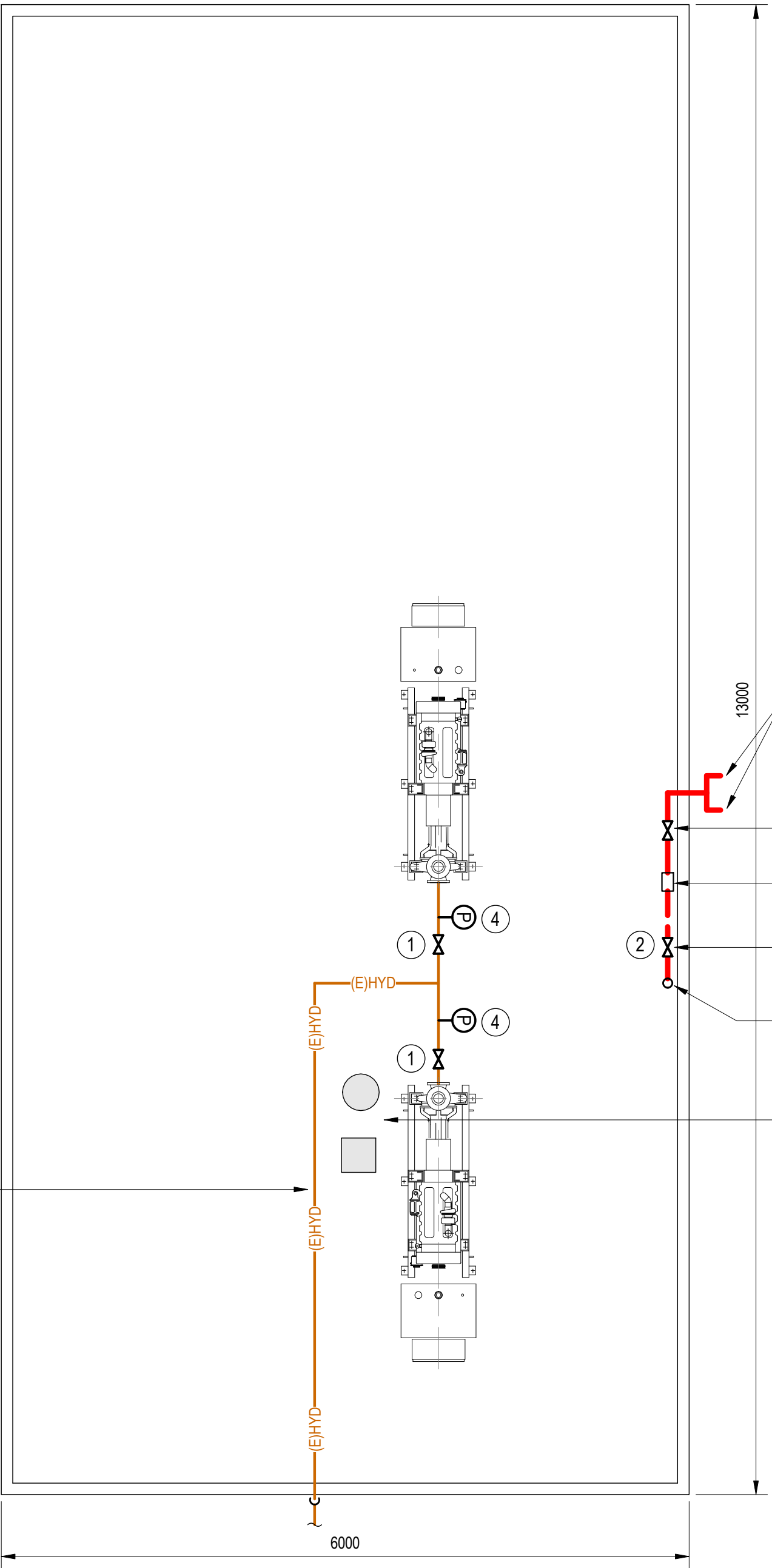


LEGENDS:

- ① LOCKED OPEN ISOLATING VALVE  
② LOCKED CLOSED ISOLATING VALVE  
③ NON-RETURN VALVE  
④ PRESSURE GAUGE



PUMP SHED HIGH LEVEL DETAIL  
NOT TO SCALE



PUMP SHED LOW LEVEL DETAIL  
NOT TO SCALE

| Rev                                            | Description    | Checked | Approved | Date     |
|------------------------------------------------|----------------|---------|----------|----------|
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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

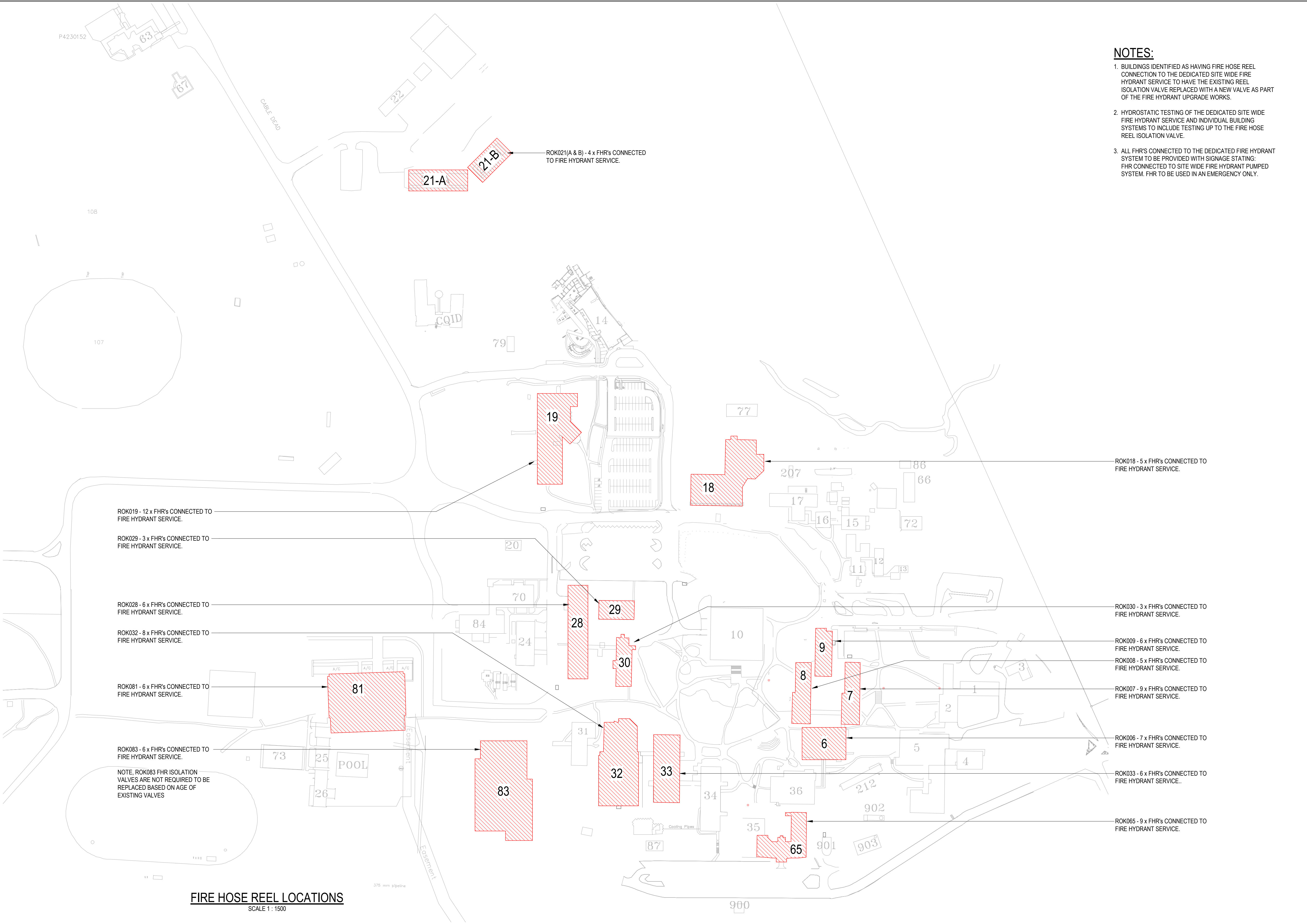
Drawing Title  
HYDRAULIC SERVICES  
PUMP SHED DETAILS

Drawing No.  
12622621-GHD-00-00-DRG-HY-00006

Size  
A1

Rev  
0





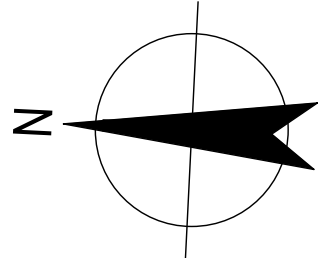
- NOTES:**
1. BUILDINGS IDENTIFIED AS HAVING FIRE HOSE REEL CONNECTION TO THE DEDICATED SITE WIDE FIRE HYDRANT SERVICE TO HAVE THE EXISTING REEL ISOLATION VALVE REPLACED WITH A NEW VALVE AS PART OF THE FIRE HYDRANT UPGRADE WORKS.
  2. HYDROSTATIC TESTING OF THE DEDICATED SITE WIDE FIRE HYDRANT SERVICE AND INDIVIDUAL BUILDING SYSTEMS TO INCLUDE TESTING UP TO THE FIRE HOSE REEL ISOLATION VALVE.
  3. ALL FHR'S CONNECTED TO THE DEDICATED FIRE HYDRANT SYSTEM TO BE PROVIDED WITH SIGNAGE STATING: FHR CONNECTED TO SITE WIDE FIRE HYDRANT PUMPED SYSTEM. FHR TO BE USED IN AN EMERGENCY ONLY.

**FIRE HOSE REEL LOCATIONS**  
SCALE 1 : 1500

|          |                |                |          |      |          |
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| Rev      | Description    | Checked        | Approved | Date |          |
| Author   |                | Drafting Check |          |      |          |
| Designer |                | Design Check   |          |      |          |

0 5 10 15 20 25 m  
SCALE 1:500 AT ORIGINAL SIZE

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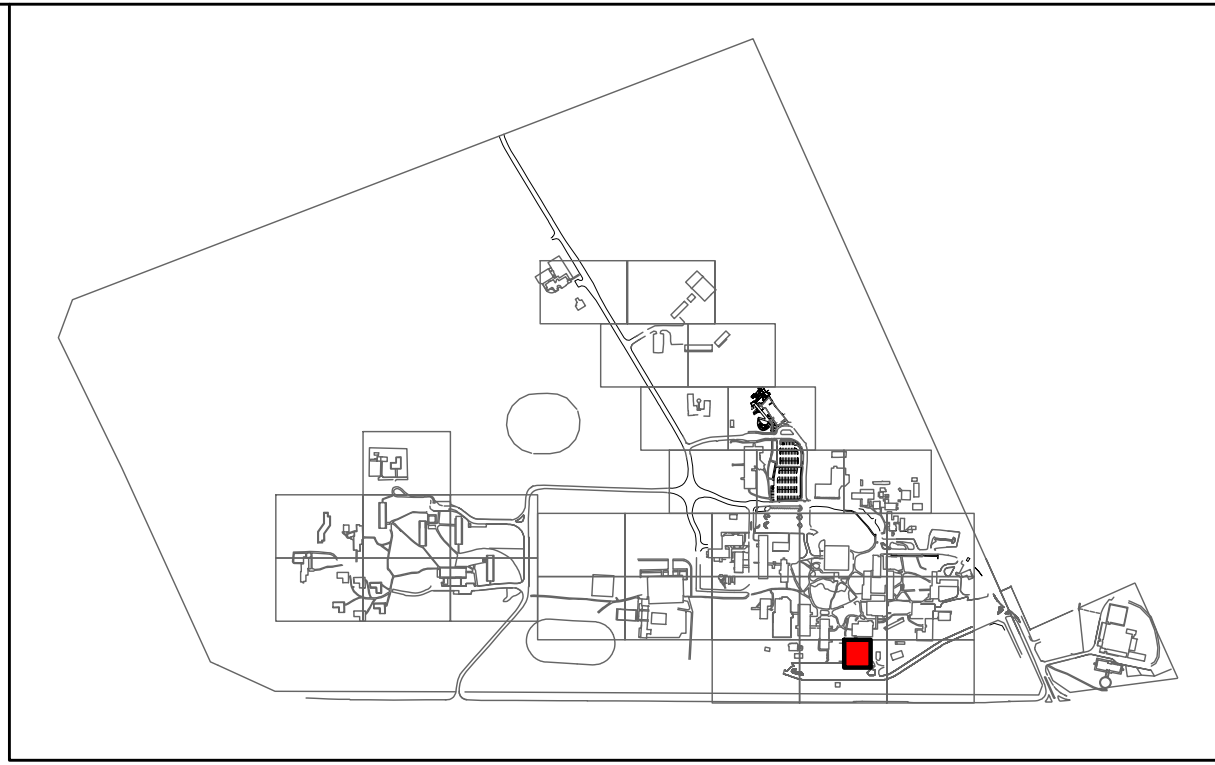
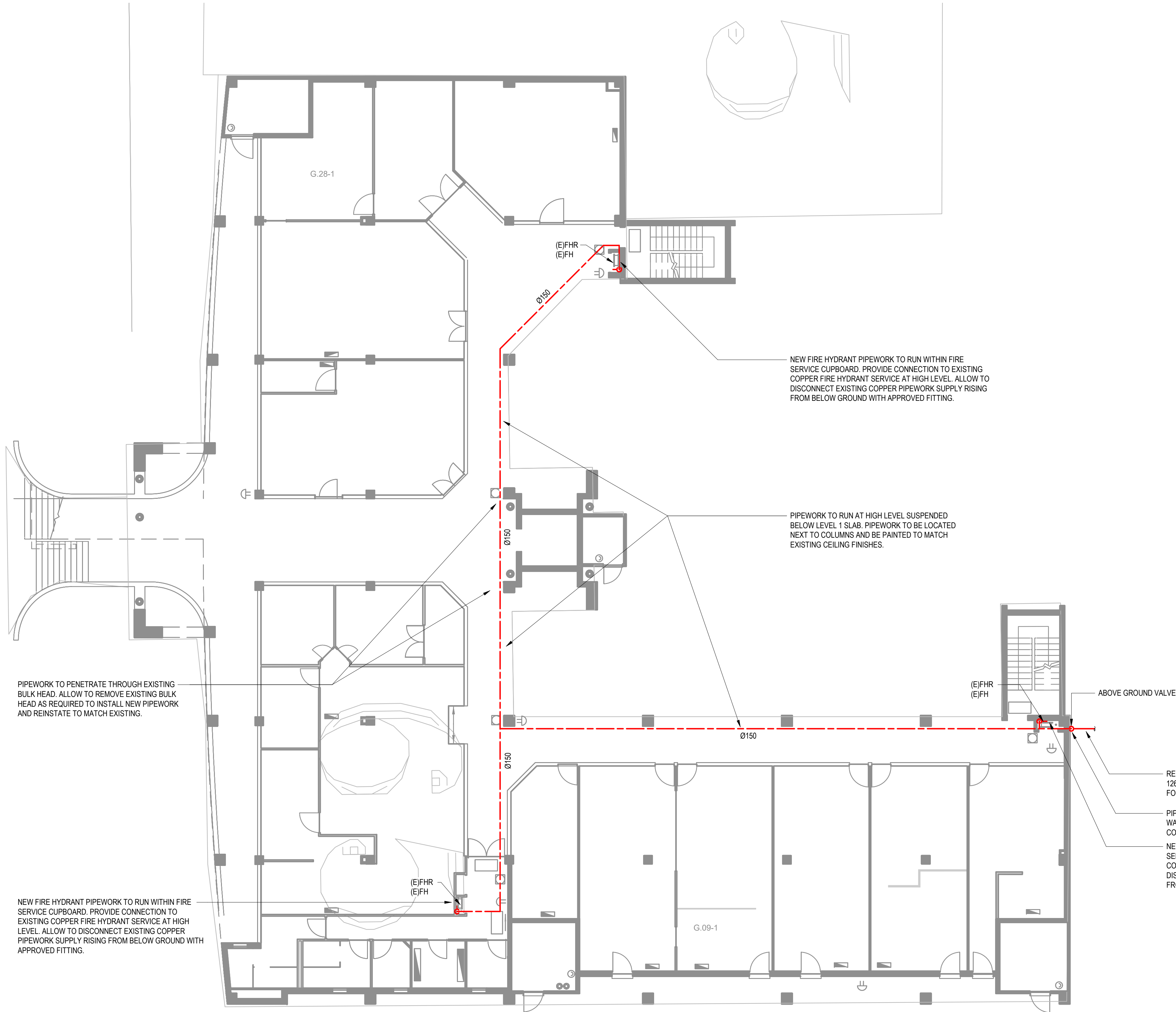
Drawing Title  
**HYDRAULIC SERVICES  
FIRE HOSE REEL LOCATIONS**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00007

Size  
A1

Rev  
0





KEY PLAN

LEGEND

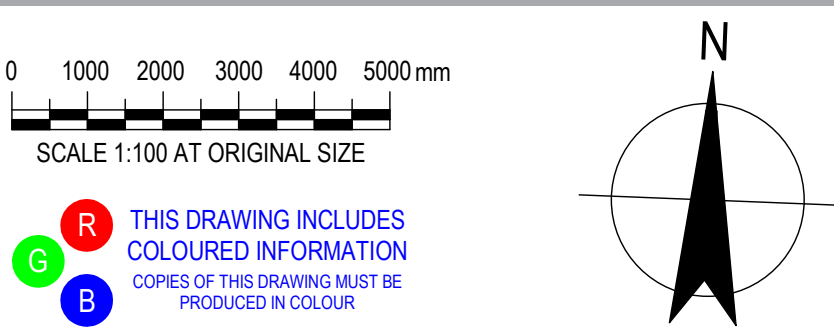
- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

NOTES:

- EXISTING SERVICES INCLUDING THE EXISTING FIRE HYDRANT SERVICE IS SHOWN FROM EXISTING RECORDS ONLY AND MAY NOT BE ACCURATE.
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LEVEL 1 FIRE HYDRANT LAYOUT  
SCALE 1 : 100

|          |                |                       |
|----------|----------------|-----------------------|
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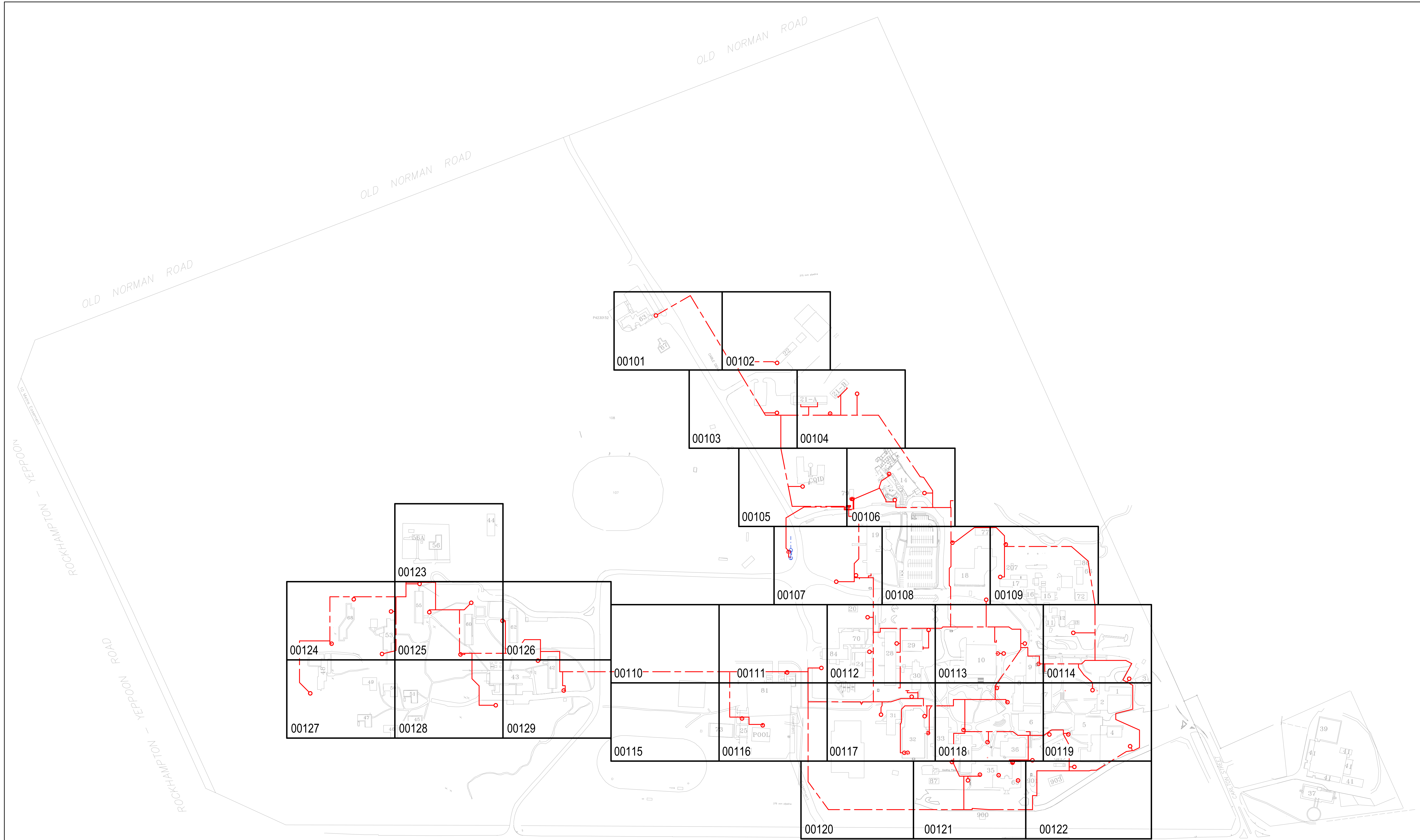
Project No.  
12622621

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|---------------|-----------------------------------------------------------------|
| Drawing Title | HYDRAULIC SERVICES<br>BUILDING 65 - LEVEL 1 FIRE HYDRANT LAYOUT |
| Drawing No.   | 12622621-GHD-00-00-DRG-HY-00008                                 |

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| Size | A1 |
| Rev  | 0  |



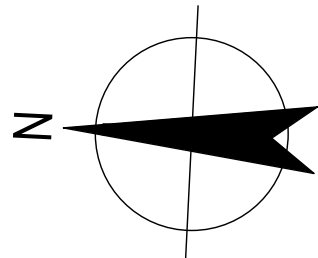


KEY PLAN - OVERALL  
SCALE 1 : 2500

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| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Designing      | Designing             |
| Designer | Designing      | Designing             |

0 25 50 75 100 125m  
SCALE 1:2500 AT ORIGINAL SIZE

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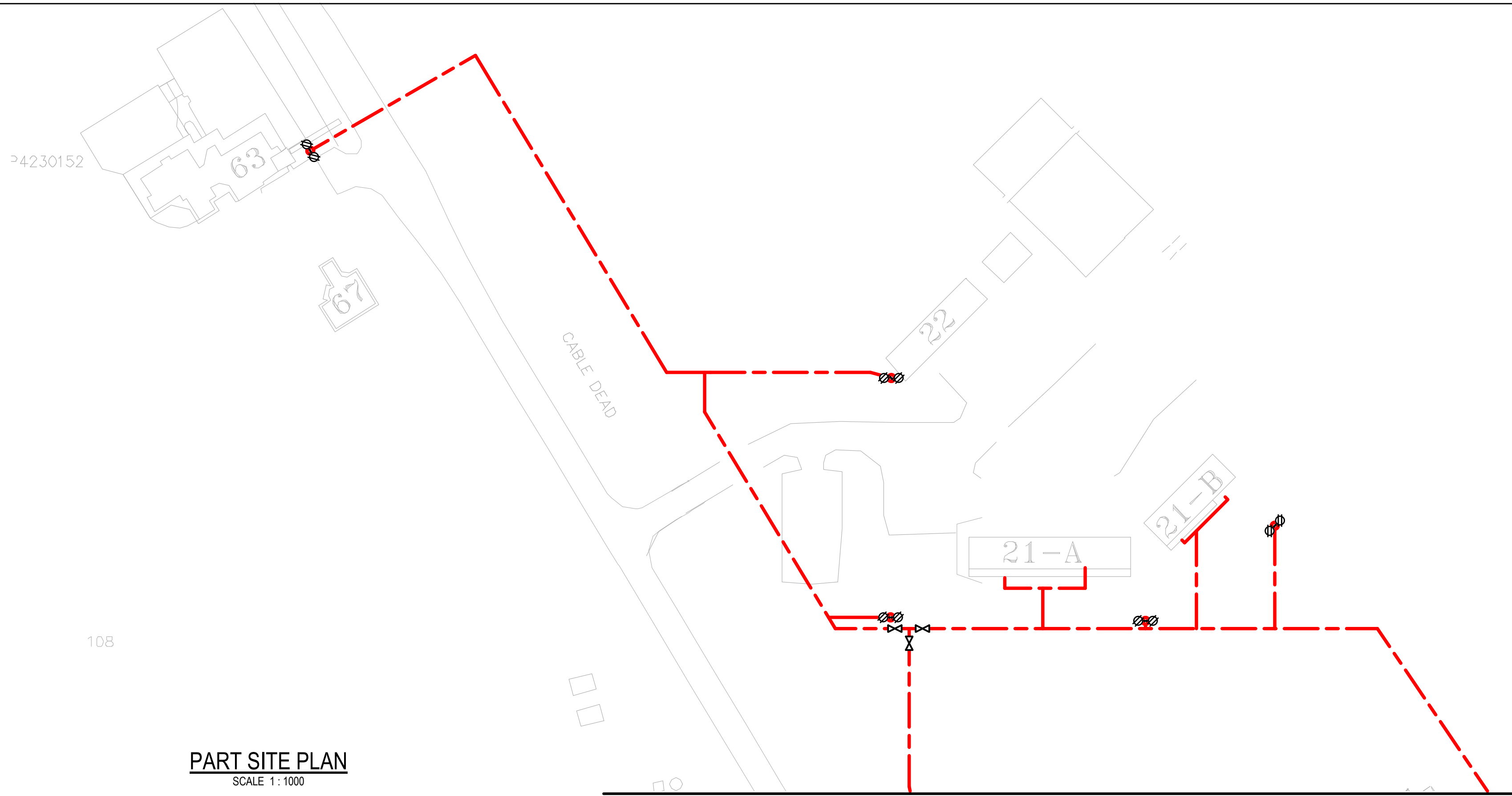
Client **CENTRAL QUEENSLAND UNIVERSITY**  
Project **FIRE MAIN UPGRADE**  
Status **AS-CONSTRUCTED**

Drawing Title  
**HYDRAULIC SERVICES  
KEYPLAN - OVERALL**

Drawing No.  
**12622621-GHD-00-00-DRG-HY-00010**

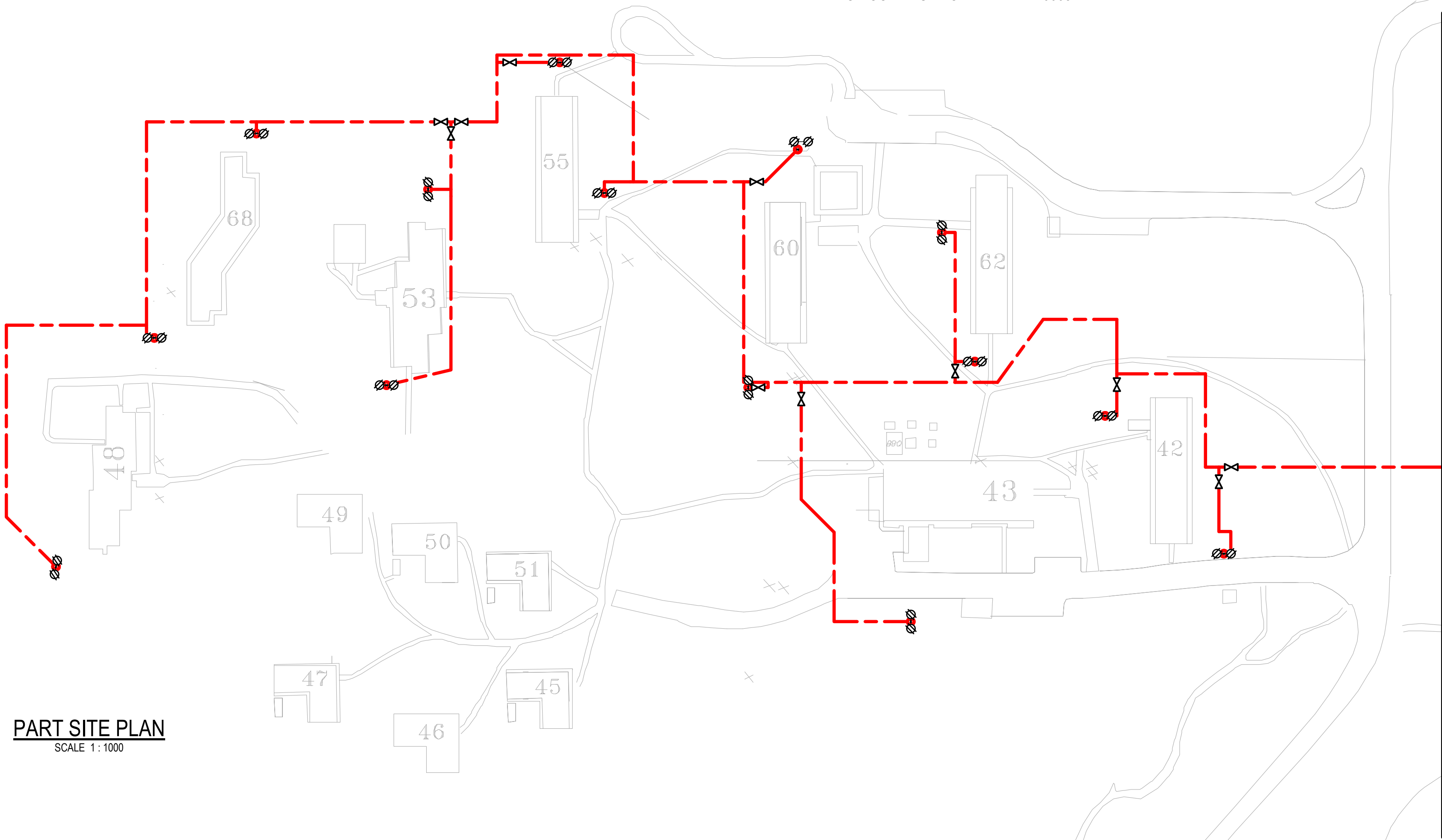
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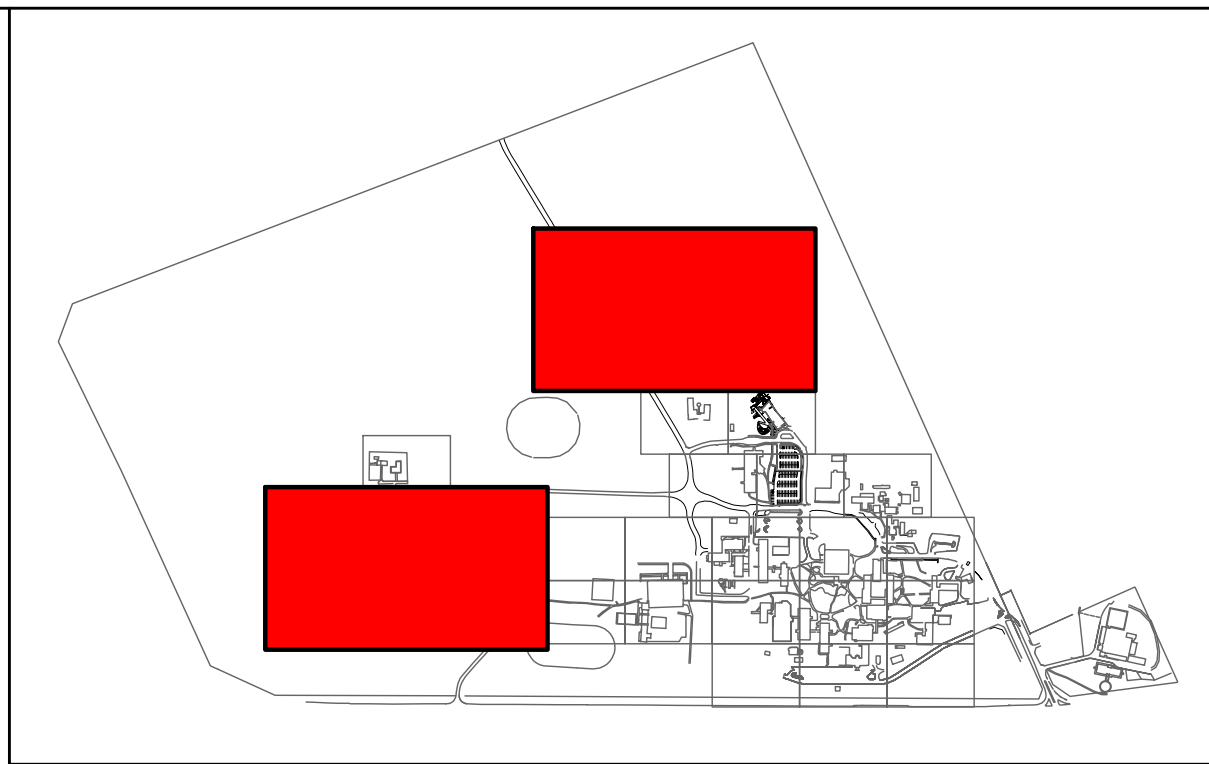
PART SITE PLAN  
SCALE 1:1000

FOR CONTINUATION REFER HY-00052



PART SITE PLAN  
SCALE 1:1000

FOR CONTINUATION REFER HY-00052



KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

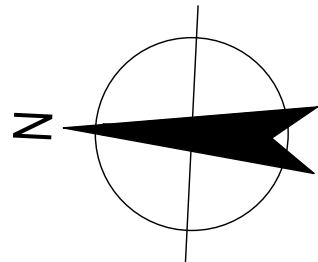
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0 10 20 30 40 50 m  
SCALE 1:1000 AT ORIGINAL SIZE

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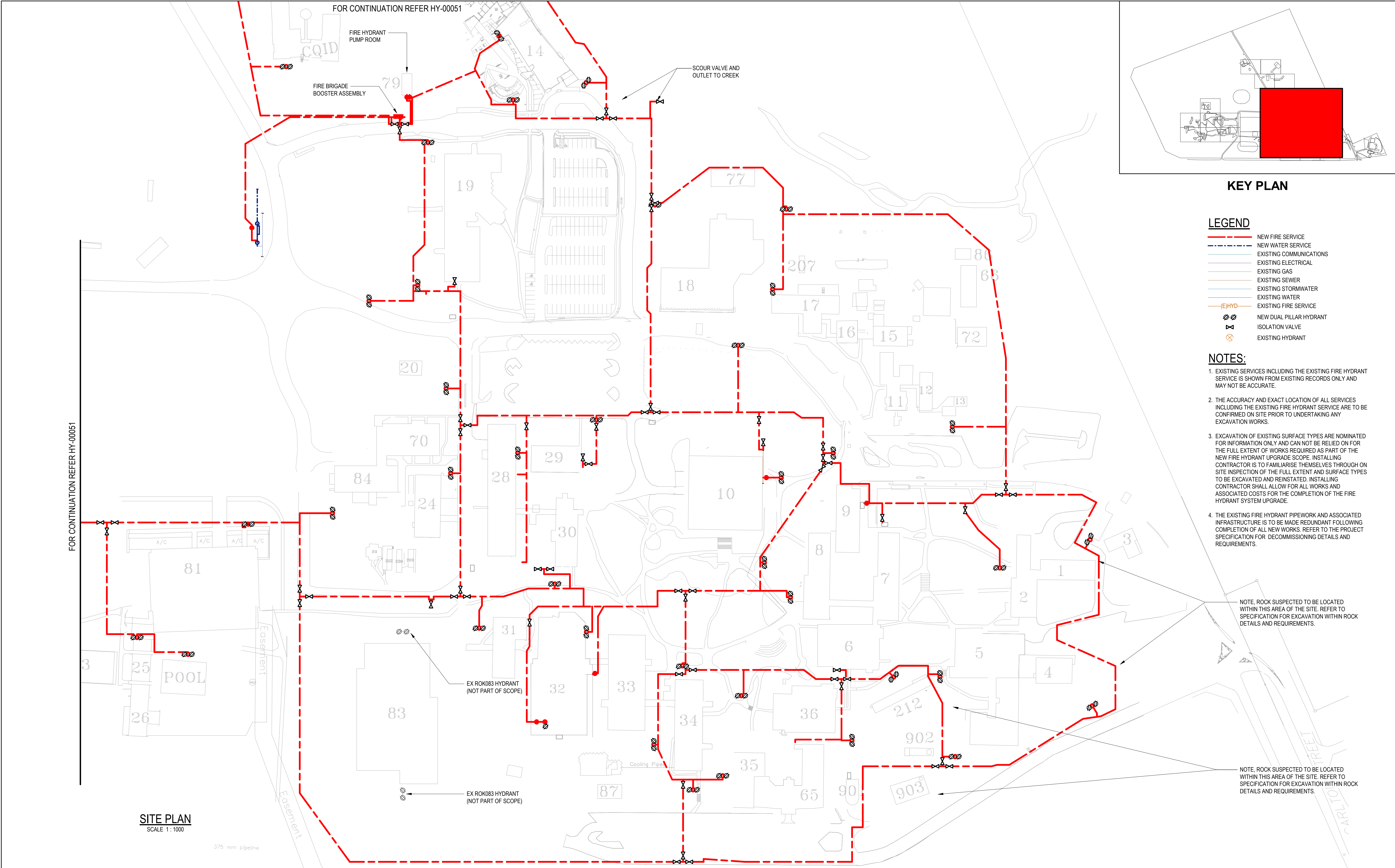
Drawing Title  
HYDRAULIC SERVICES  
SITE PLAN SHEET 1

Drawing No.  
12622621-GHD-00-00-DRG-HY-00051

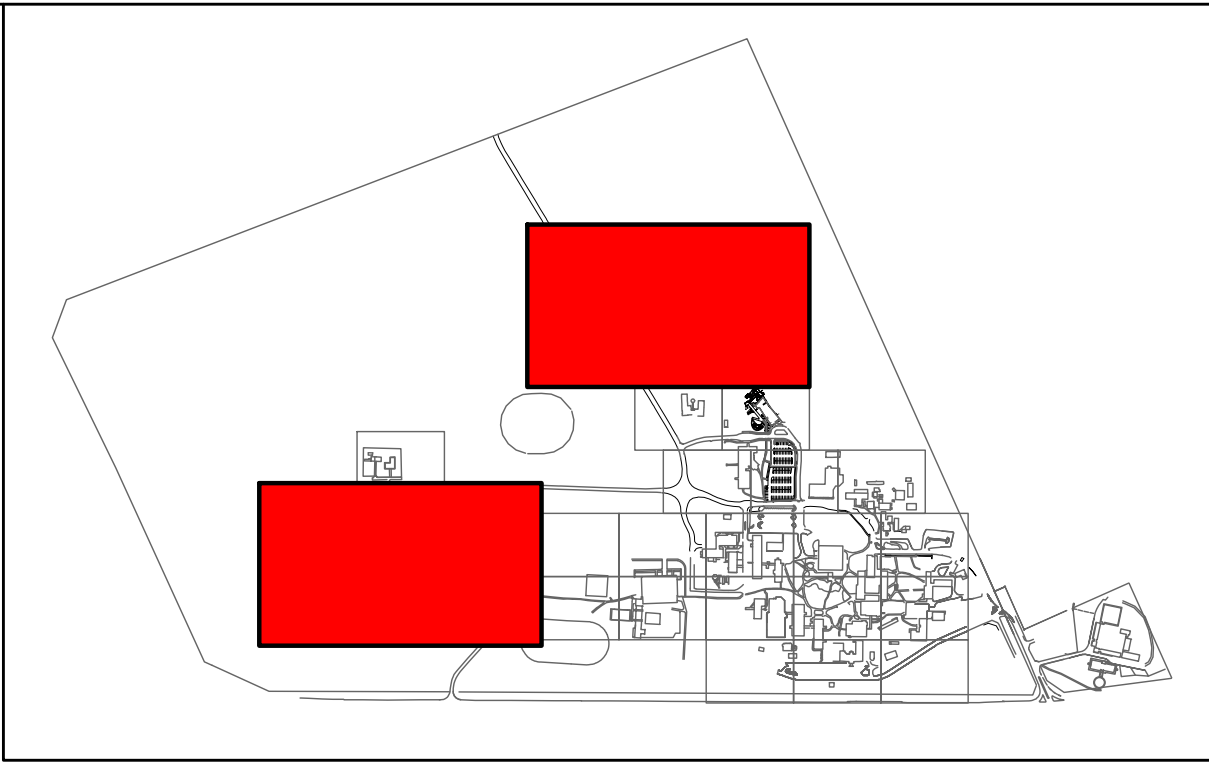
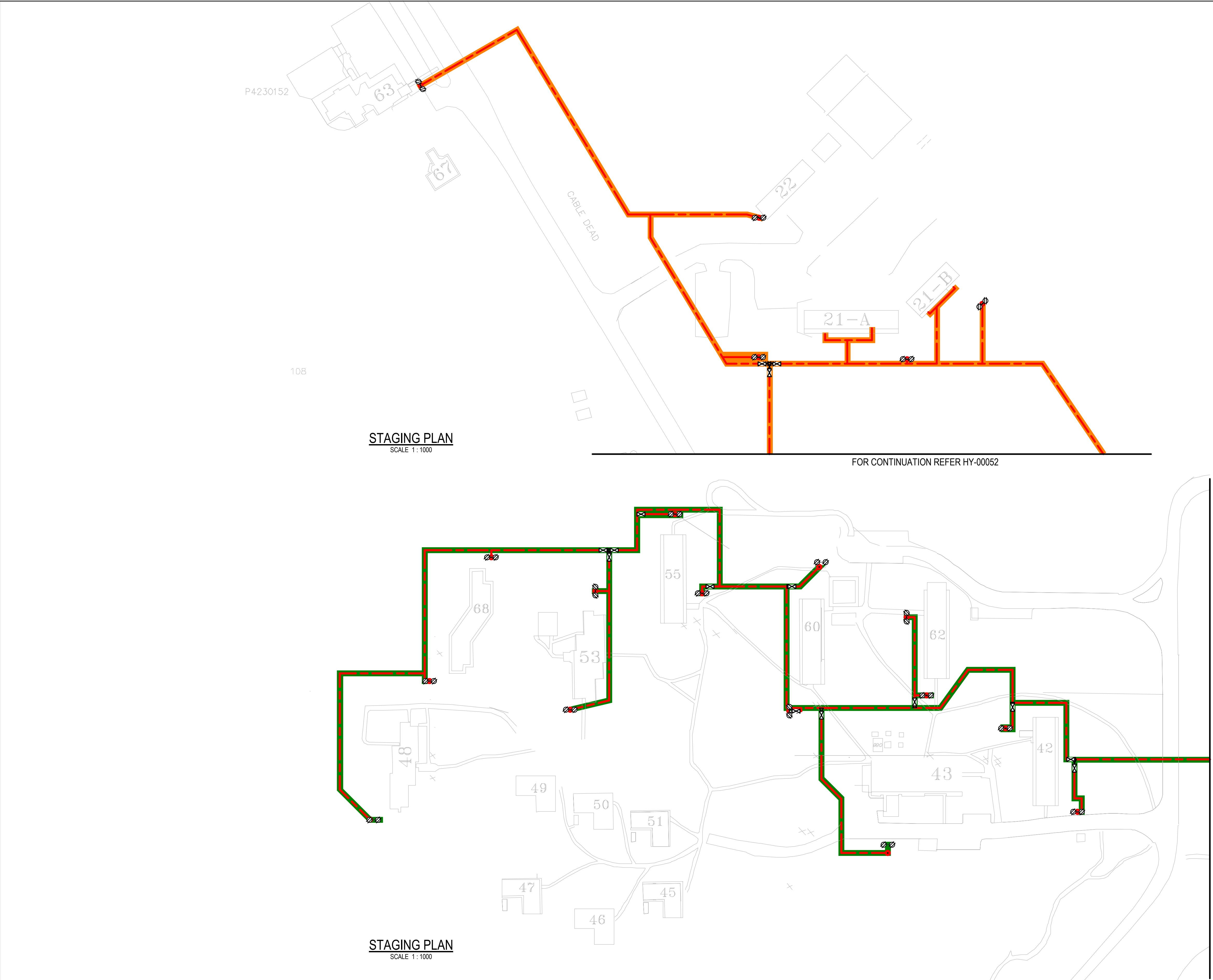
Size  
A1

Rev  
0









KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

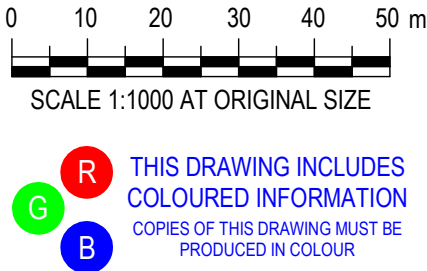
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STAGING LEGEND

- STAGE 3
- STAGE 4

|          |                |                       |
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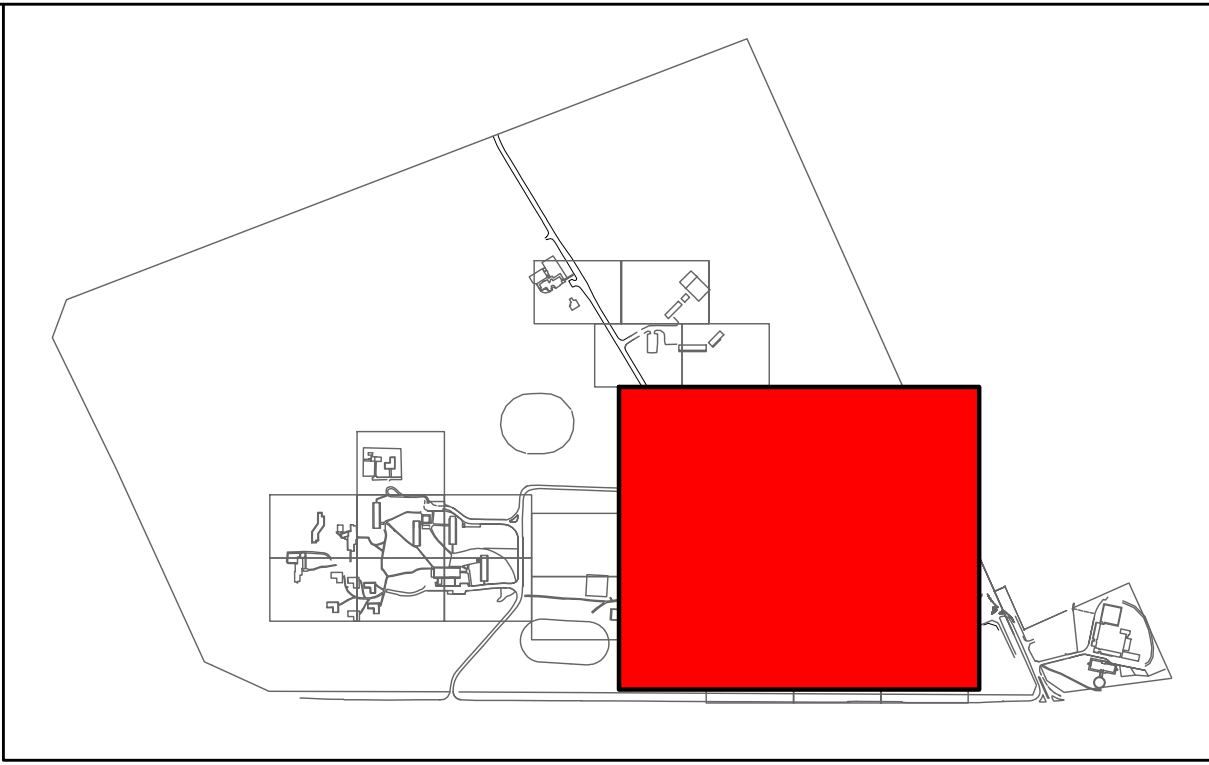
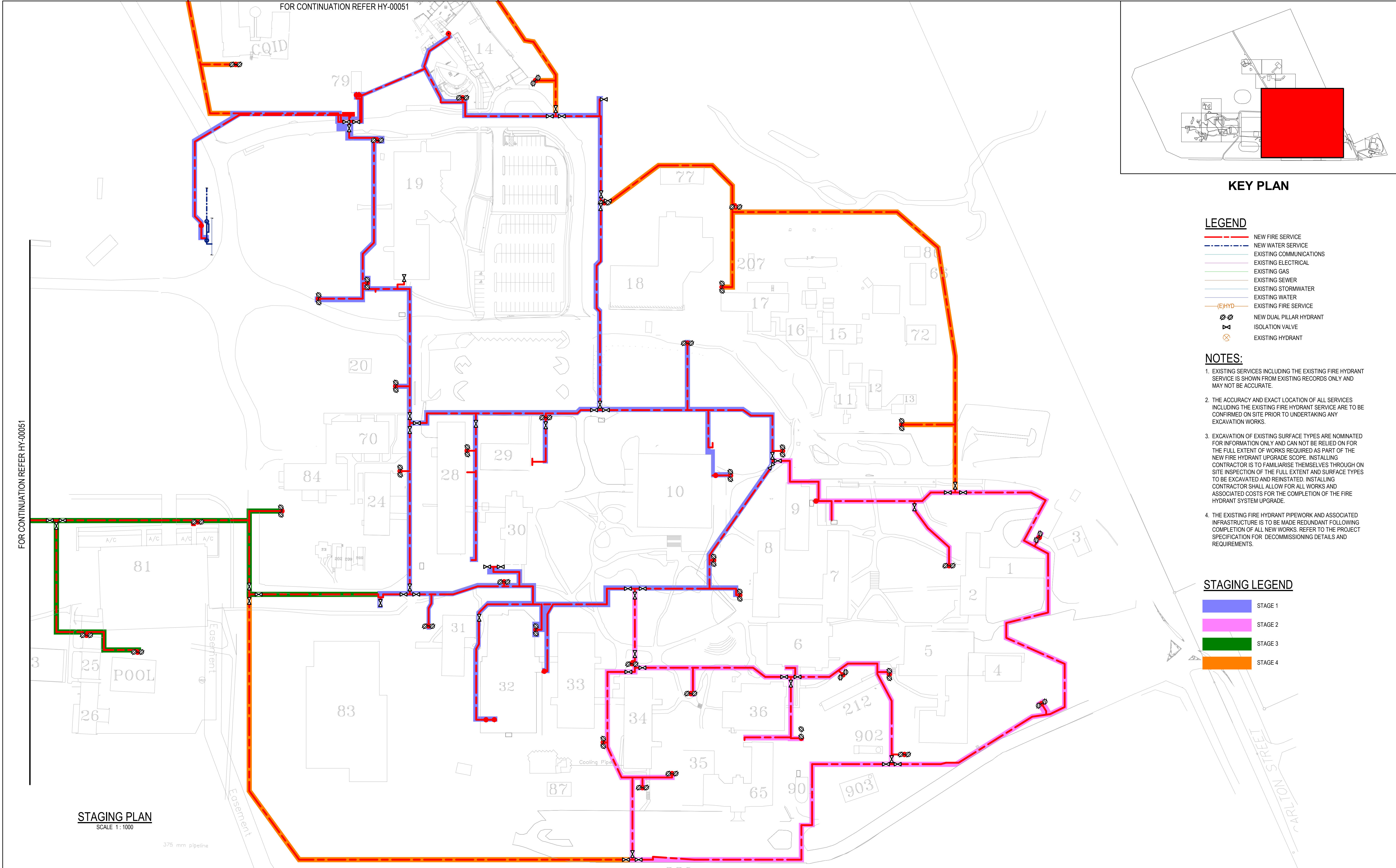
Drawing Title  
HYDRAULIC SERVICES  
STAGING PLAN SHEET 1

Drawing No.  
12622621-GHD-00-00-DRG-HY-00061

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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STAGING LEGEND

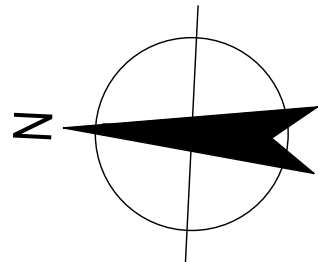
- STAGE 1
- STAGE 2
- STAGE 3
- STAGE 4

STAGING PLAN  
SCALE 1:1000

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| Designer         | Design Check   |         |          |          |

0 10 20 30 40 50 m  
SCALE 1:1000 AT ORIGINAL SIZE

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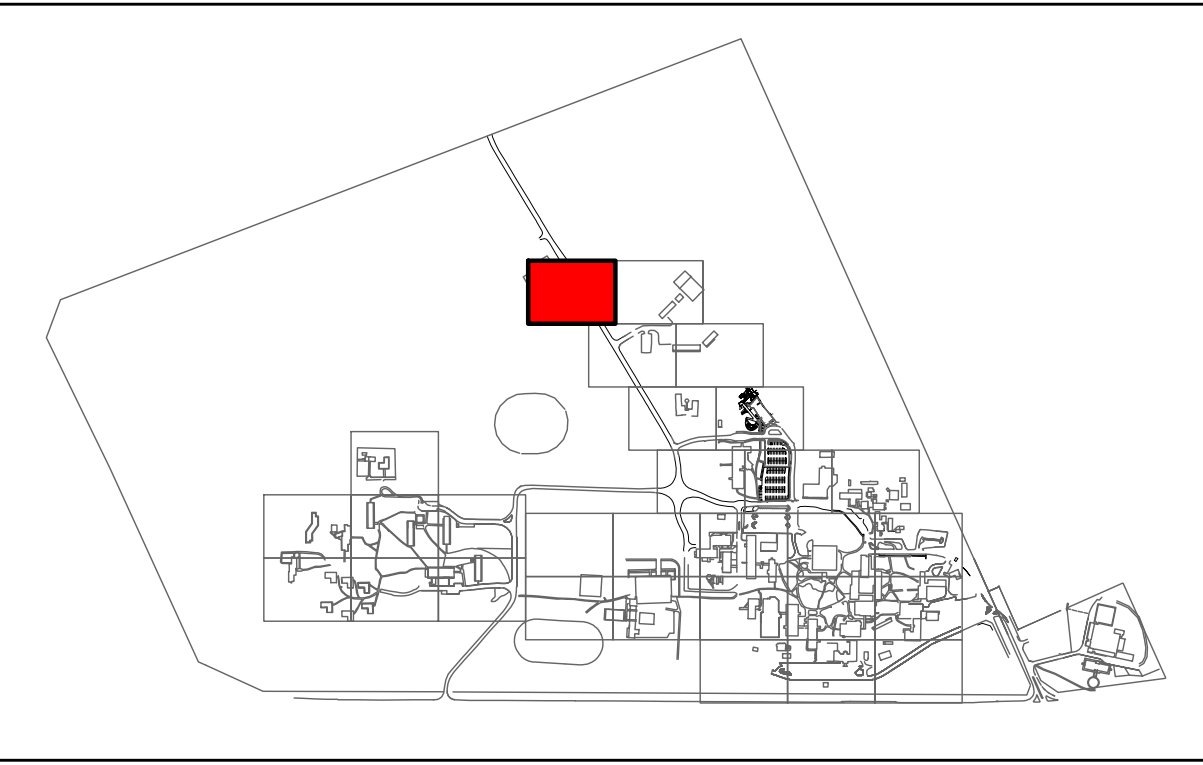
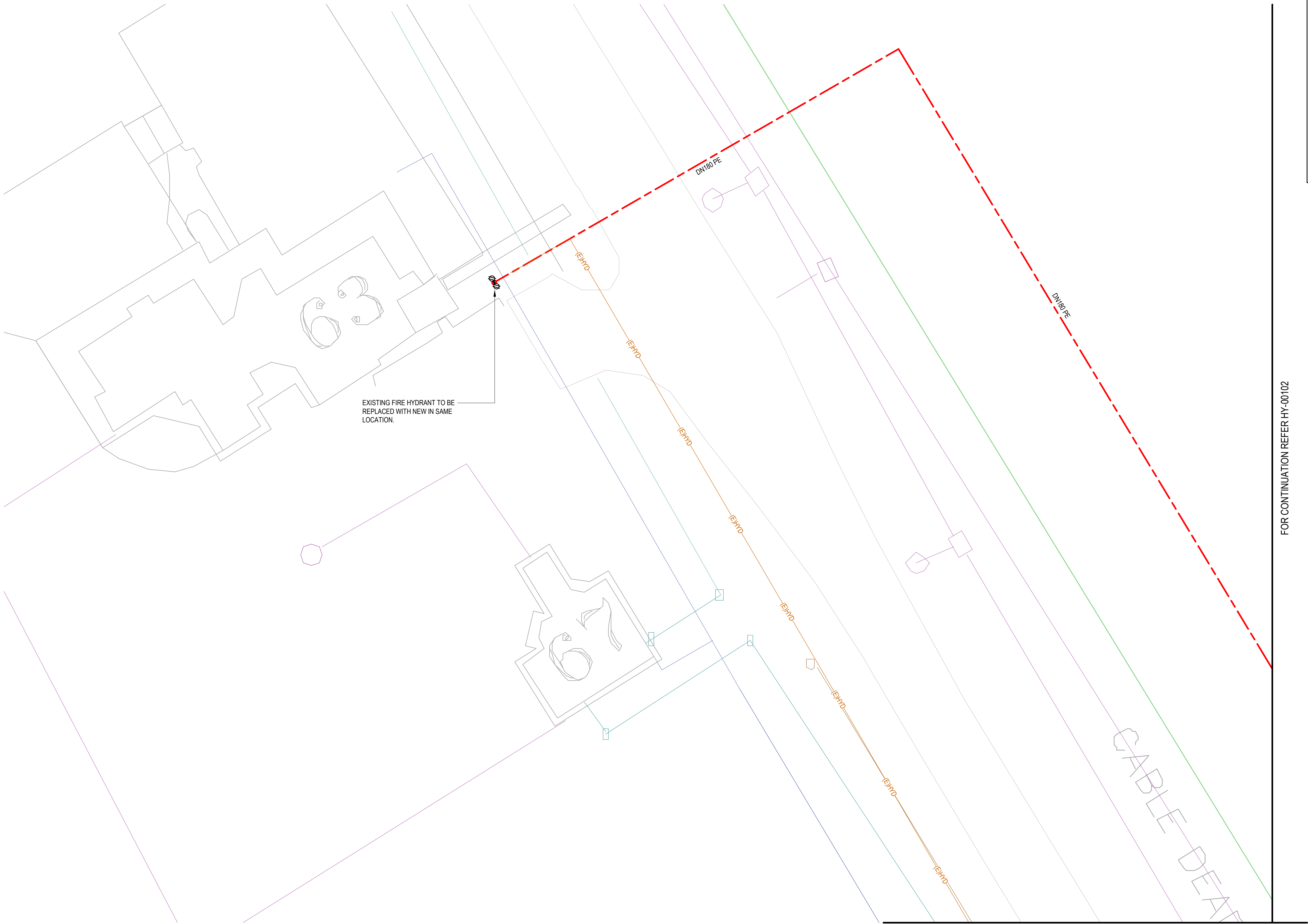
Drawing Title  
HYDRAULIC SERVICES  
STAGING PLAN SHEET 2

Drawing No.  
12622621-GHD-00-00-DRG-HY-00062

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

NOTES:

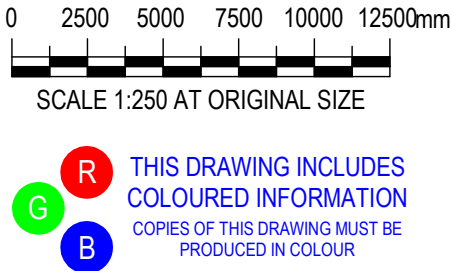
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FOR CONTINUATION REFER HY-00102

FOR CONTINUATION REFER HY-00103

SITE SERVICES LAYOUT - SHEET 01  
SCALE 1 : 250

|          |                |                       |
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Project No.  
12622621

|         |                               |
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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

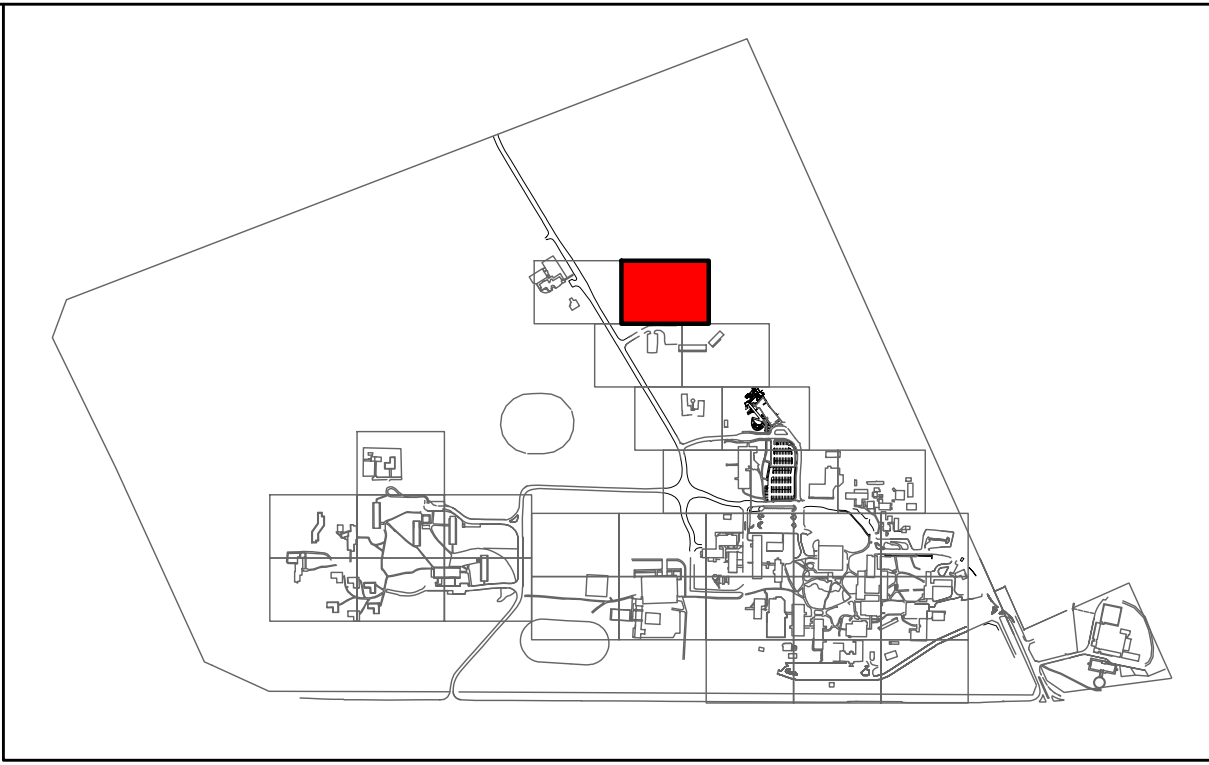
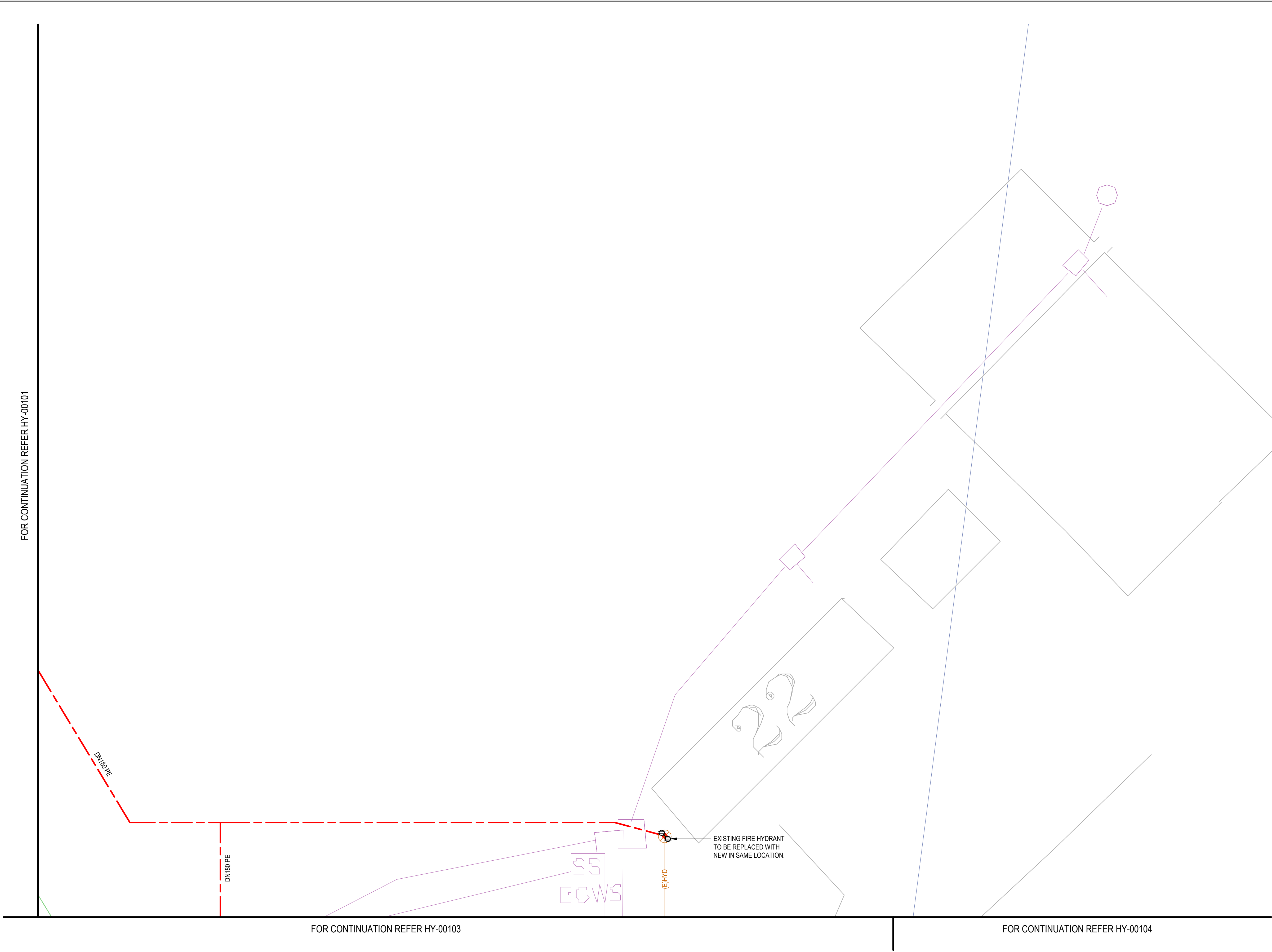
Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 01

Drawing No.  
12622621-GHD-00-00-DRG-HY-00101

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

NOTES:

- EXISTING SERVICES INCLUDING THE EXISTING FIRE HYDRANT SERVICE IS SHOWN FROM EXISTING RECORDS ONLY AND MAY NOT BE ACCURATE.
- THE ACCURACY AND EXACT LOCATION OF ALL SERVICES INCLUDING THE EXISTING FIRE HYDRANT SERVICE ARE TO BE CONFIRMED ON SITE PRIOR TO UNDERTAKING ANY EXCAVATION WORKS.
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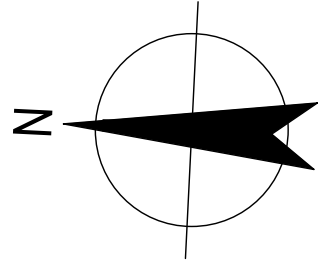
SITE SERVICES LAYOUT - SHEET 02

SCALE 1 : 250

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|          |                |         |          |          |
| 0        | AS-CONSTRUCTED |         |          | 11.06.25 |
| Rev      | Description    | Checked | Approved | Date     |
| Author   | Drafting Check |         |          |          |
| Designer | Design Check   |         |          |          |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 02

Drawing No.  
12622621-GHD-00-00-DRG-HY-00102

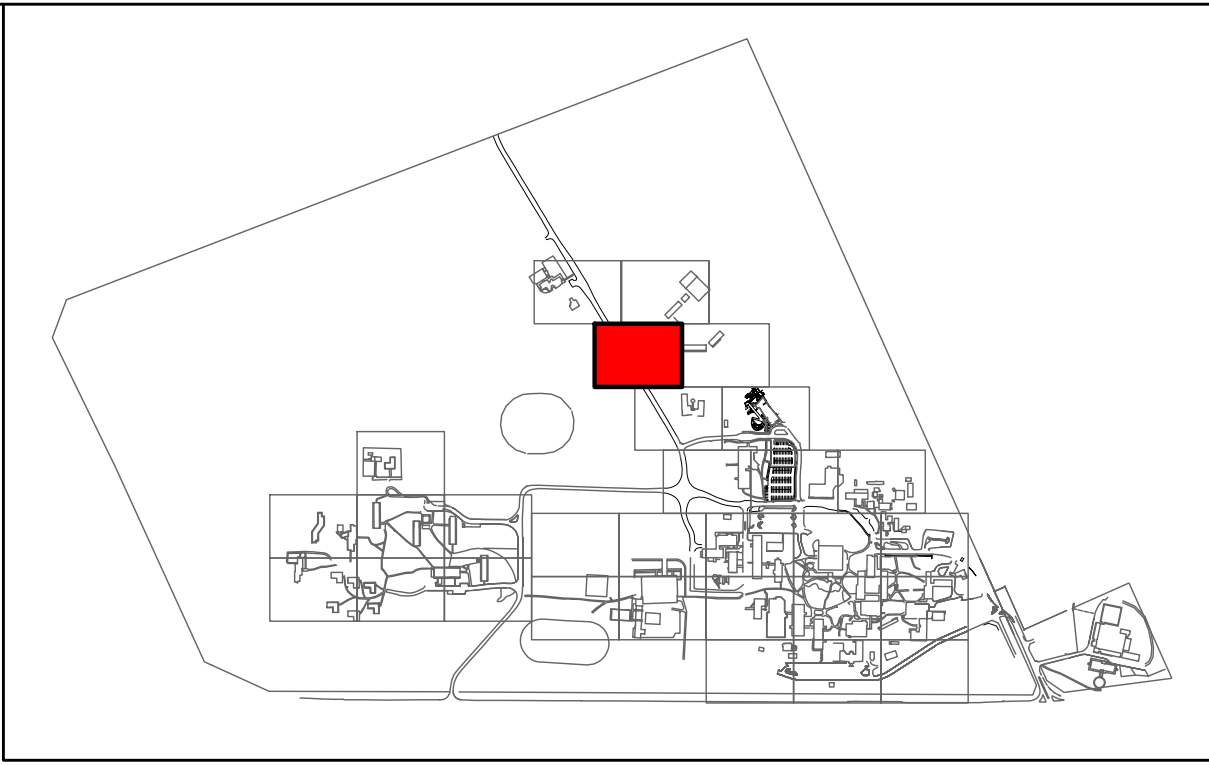
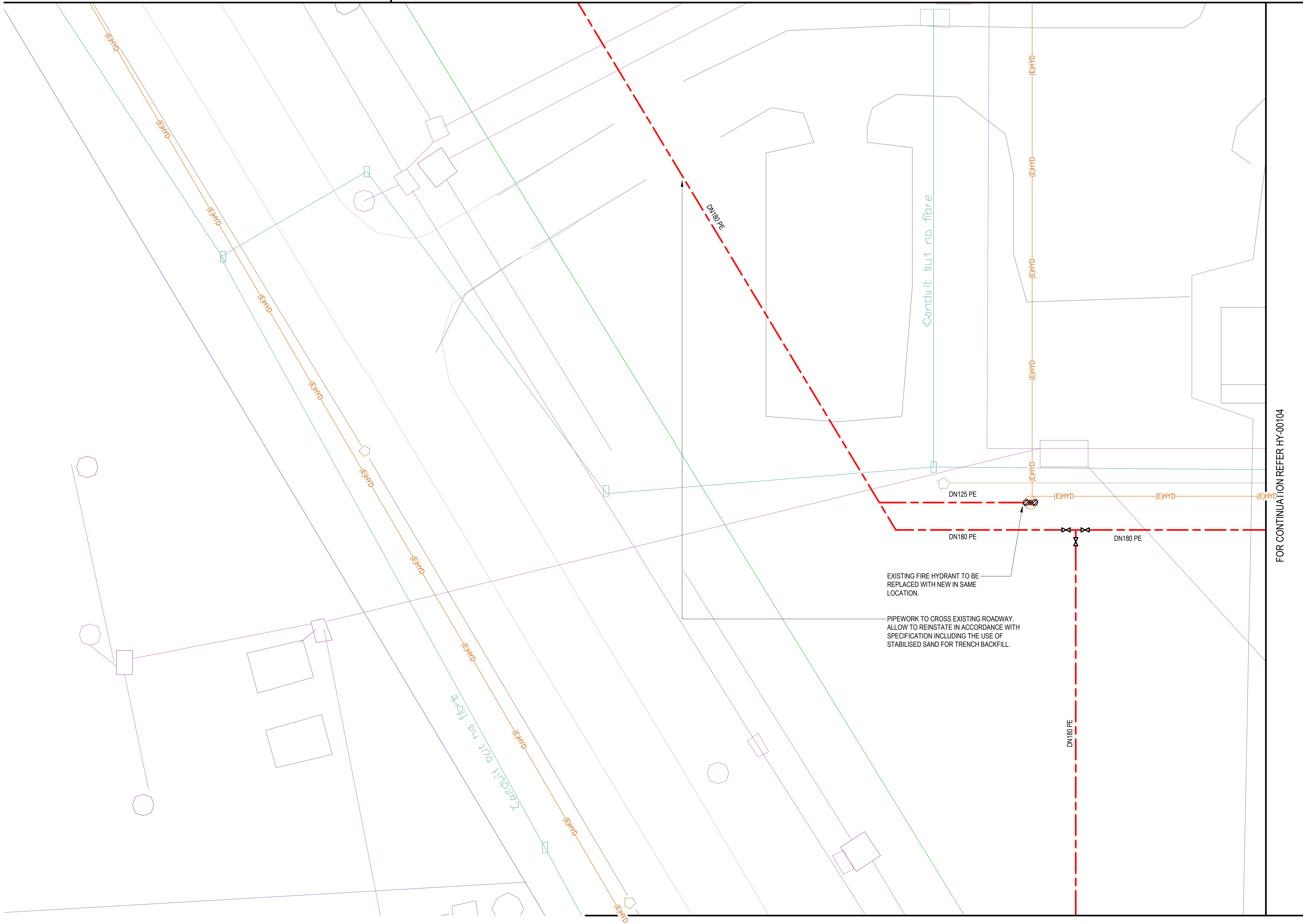
Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00101

FOR CONTINUATION REFER HY-00102



KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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FOR CONTINUATION REFER HY-00104

FOR CONTINUATION REFER HY-00105

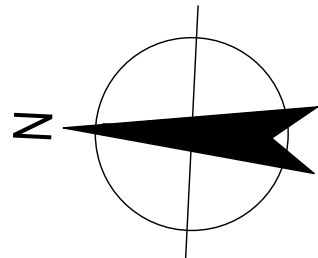
SITE SERVICES LAYOUT - SHEET 03

SCALE 1 : 250

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| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
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Project No.  
12622621

Client **CENTRAL QUEENSLAND UNIVERSITY**  
Project **FIRE MAIN UPGRADE**  
Status **AS-CONSTRUCTED**

Drawing Title **HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 03**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00103

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00102

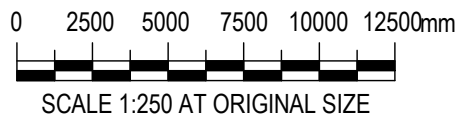
FOR CONTINUATION REFER HY-00103

FOR CONTINUATION REFER HY-00105

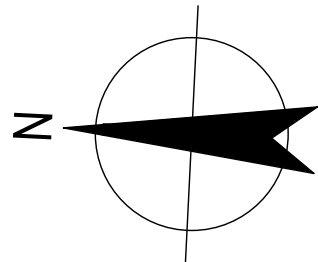
FOR CONTINUATION REFER HY-00106

SITE SERVICES LAYOUT - SHEET 04

SCALE 1 : 250



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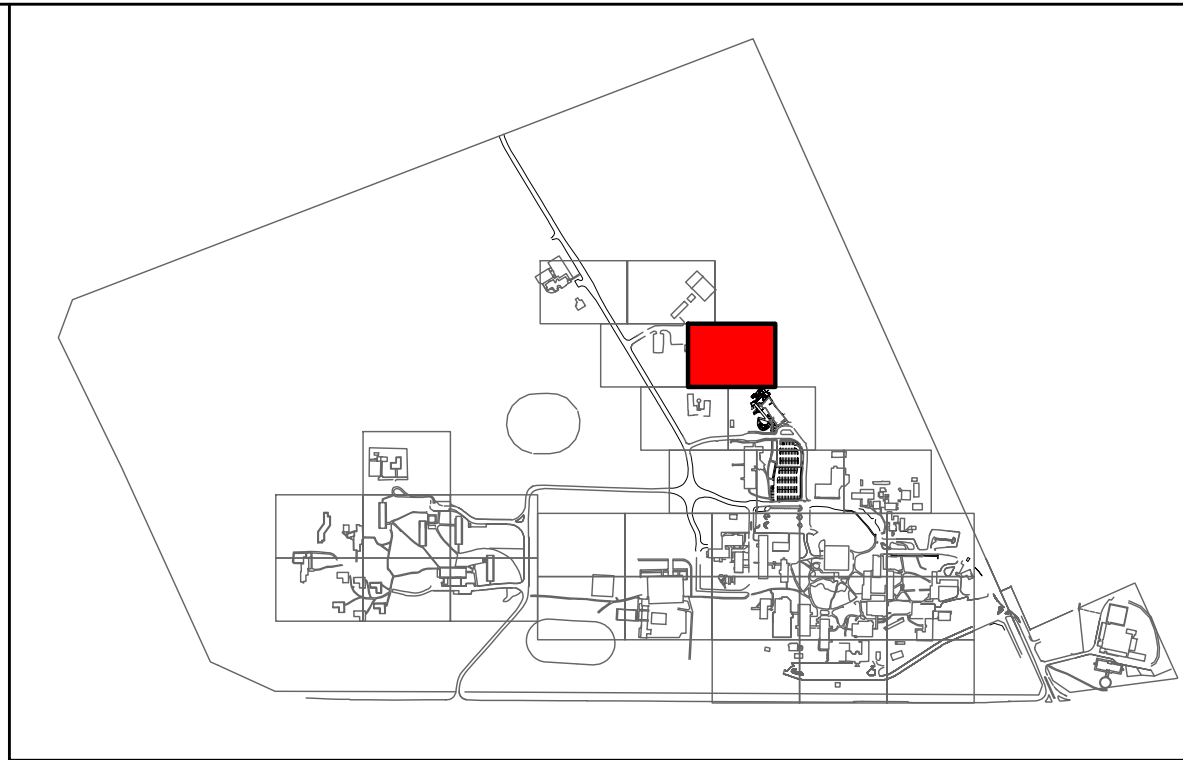
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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 04

Drawing No.  
12622621-GHD-00-00-DRG-HY-00104

Size  
A1

Rev  
0



KEY PLAN

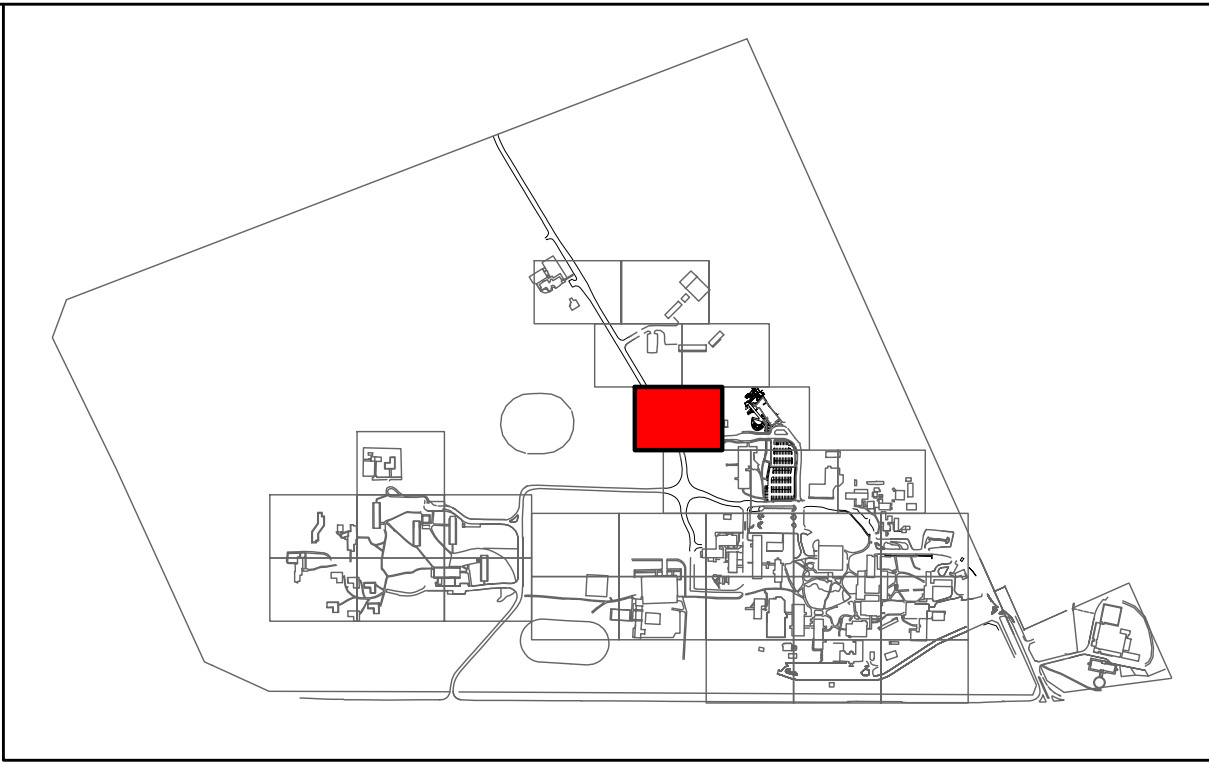
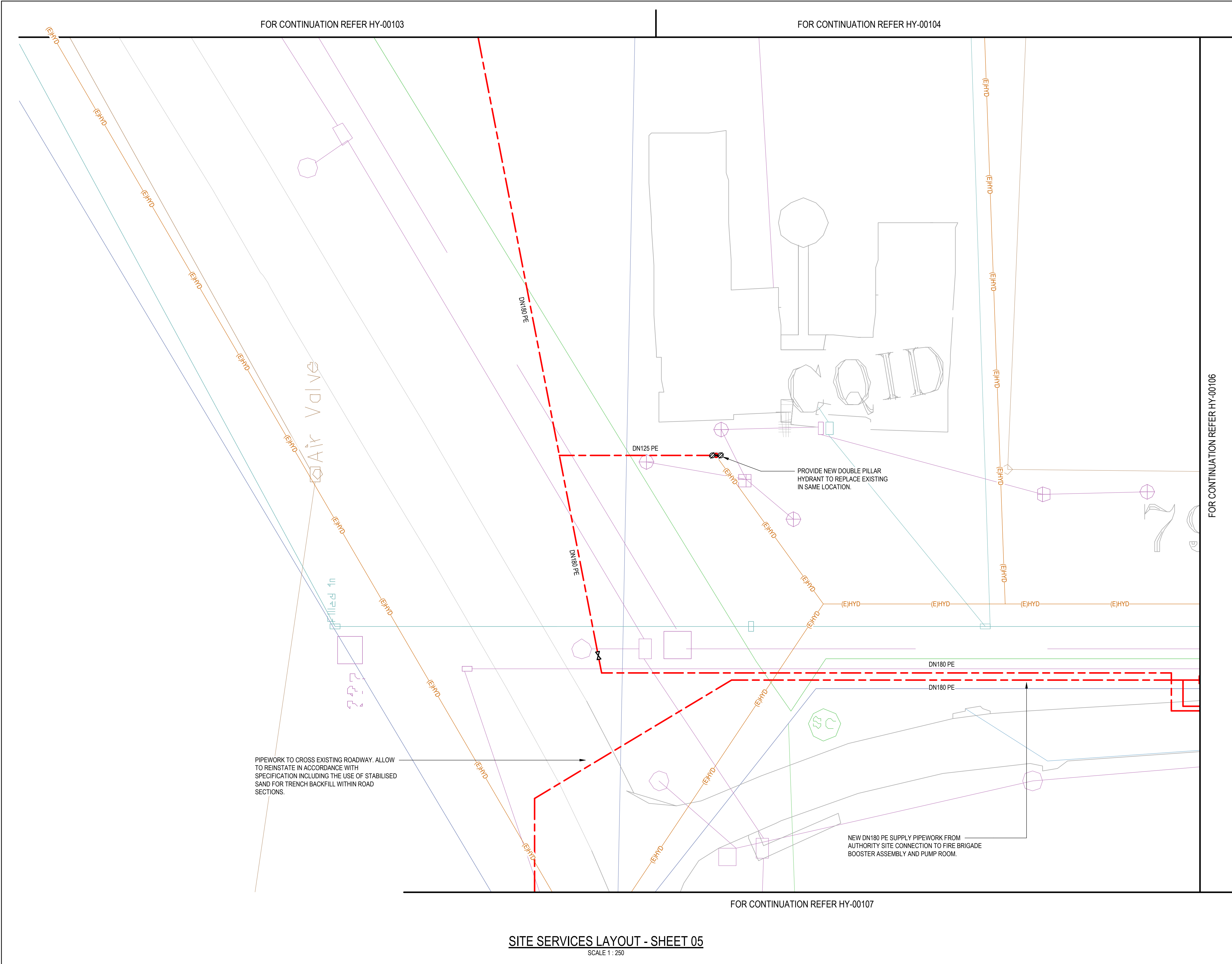
LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

NOTES:

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- THE EXISTING FIRE HYDRANT PIPEWORK AND ASSOCIATED INFRASTRUCTURE IS TO BE MADE REDUNDANT FOLLOWING COMPLETION OF ALL NEW WORKS. REFER TO THE PROJECT SPECIFICATION FOR DECOMMISSIONING DETAILS AND REQUIREMENTS.





**KEY PLAN**

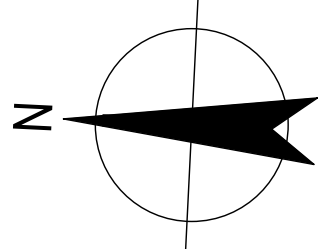
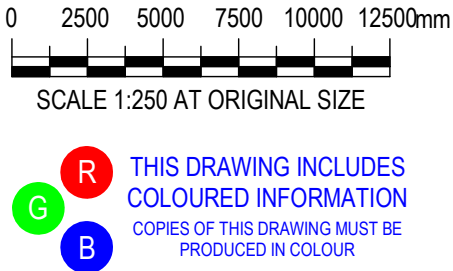
**LEGEND**

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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|          |                |                       |
|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |



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Project No.  
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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
**HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 05**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00105

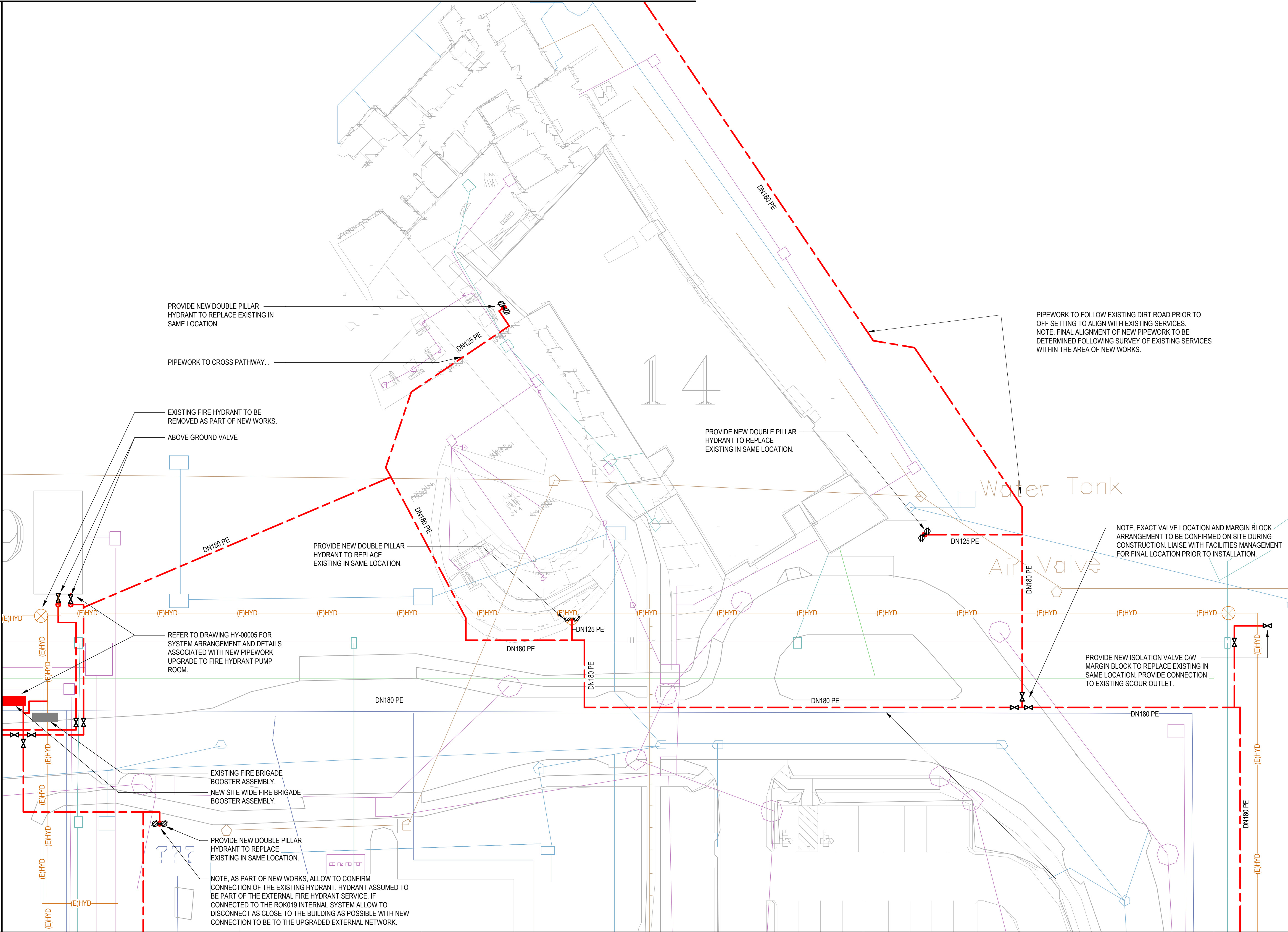
Size  
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Rev  
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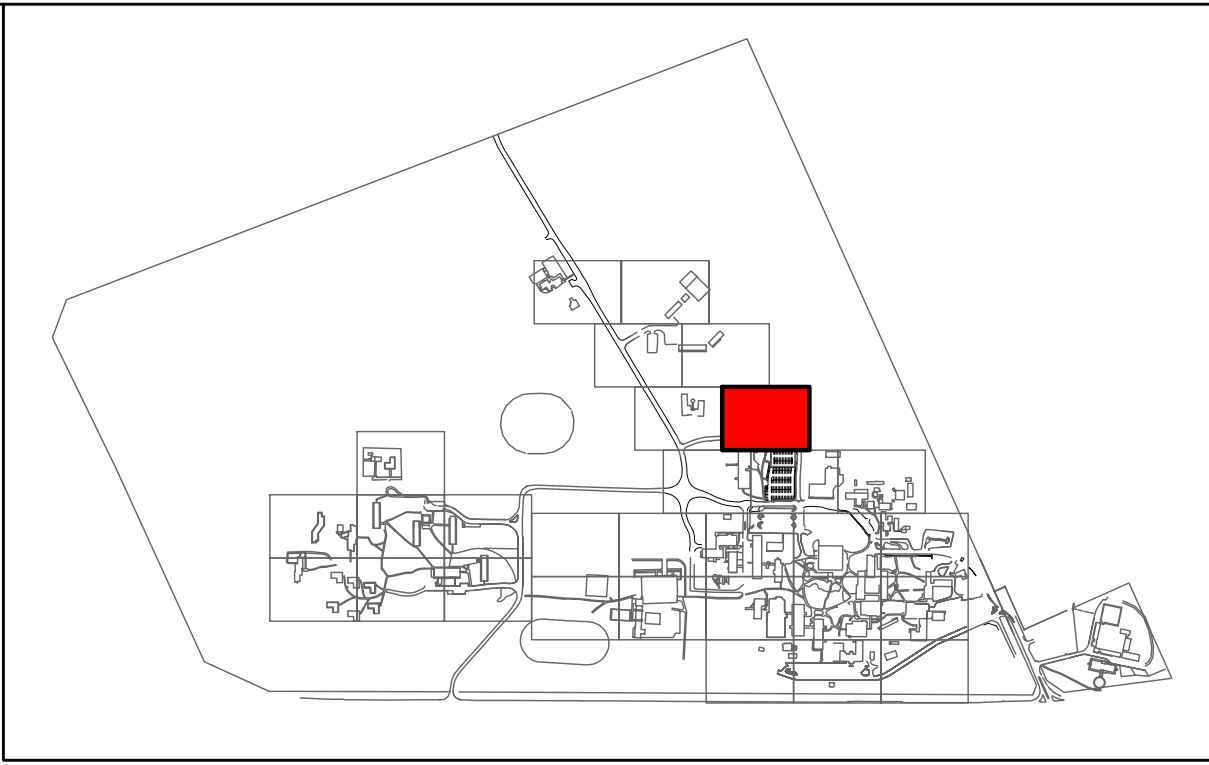
FOR CONTINUATION REFER HY-00104

FOR CONTINUATION REFER HY-00105



FOR CONTINUATION REFER HY-00107

FOR CONTINUATION REFER HY-00108



KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

NOTES:

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- THE EXISTING FIRE HYDRANT PIPEWORK AND ASSOCIATED INFRASTRUCTURE IS TO BE MADE REDUNDANT FOLLOWING COMPLETION OF ALL NEW WORKS. REFER TO THE PROJECT SPECIFICATION FOR DECOMMISSIONING DETAILS AND REQUIREMENTS.

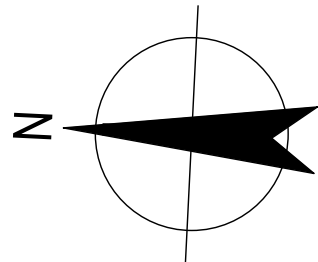
SITE SERVICES LAYOUT - SHEET 06

SCALE 1 : 250

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|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
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| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 06

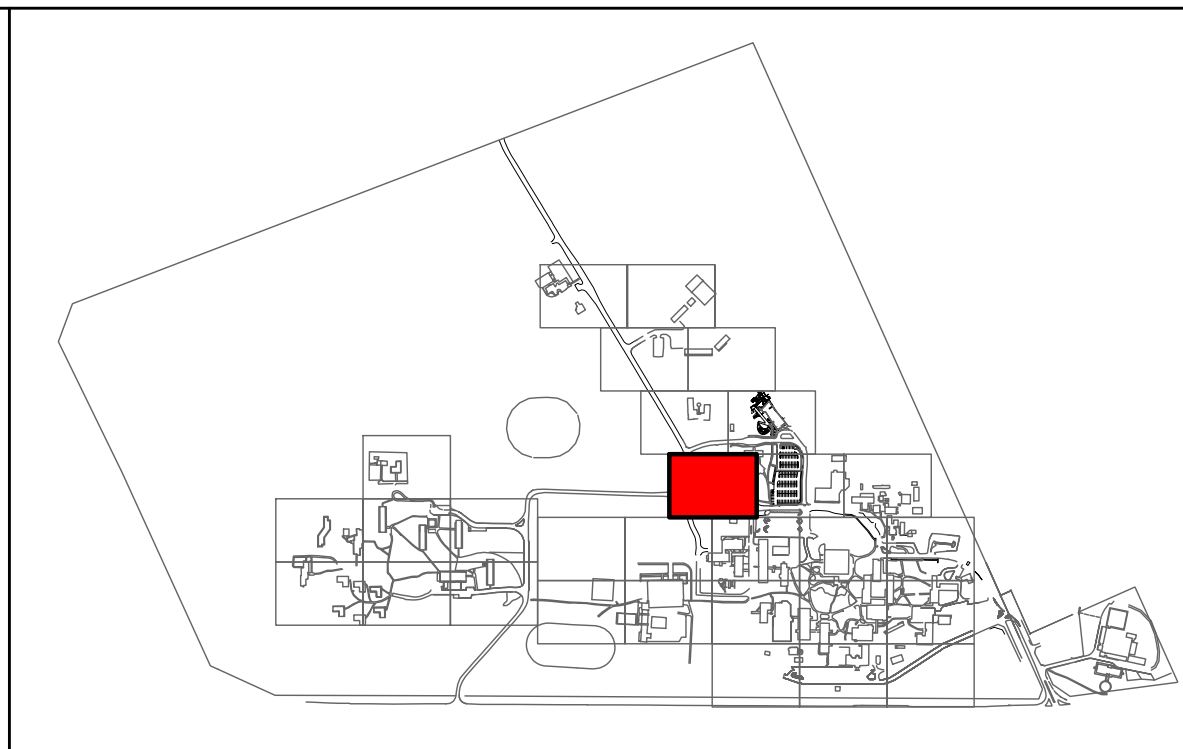
Drawing No.  
12622621-GHD-00-00-DRG-HY-00106

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00106



### LEGEND

- |  |                         |
|--|-------------------------|
|  | NEW FIRE SERVICE        |
|  | NEW WATER SERVICE       |
|  | EXISTING COMMUNICATIONS |
|  | EXISTING ELECTRICAL     |
|  | EXISTING GAS            |
|  | EXISTING SEWER          |
|  | EXISTING STORMWATER     |
|  | EXISTING WATER          |
|  | EXISTING FIRE SERVICE   |
|  | NEW DUAL PILLAR HYDRANT |
|  | ISOLATION VALVE         |
|  | EXISTING HYDRANT        |

NOTES:

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FOR CONTINUATION REFER HY-00108

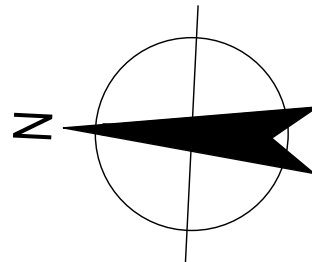
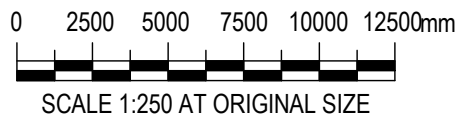
EXISTING ABOVE GROUND VALVE TO BE  
DECOMMISSIONED FOLLOWING COMPLETION  
OF CUT OVER WORKS.

FOR CONTINUATION REFER HY-00111

FOR CONTINUATION REFER HY-00112

## SCALE 1 : 250

|                 |                       |                |                 |             |
|-----------------|-----------------------|----------------|-----------------|-------------|
| 0               | AS-CONSTRUCTED        |                |                 | 11.06.25    |
| P01             | REVISED TENDER ISSUE  | BP             | DL              | 19.08.24    |
| <b>Rev</b>      | <b>Description</b>    | <b>Checked</b> | <b>Approved</b> | <b>Date</b> |
| <b>Author</b>   | <b>Drafting Check</b> |                |                 |             |
| <b>Designer</b> | <b>Design Check</b>   |                |                 |             |



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Project No.  
**12622621**

Client **CENTRAL QUEENSLAND UNIVERSITY**

Project **FIRE MAIN UPGRADE**

Status **AS-CONSTRUCTED**

Drawing Title **HYDRAULIC SERVICES**  
**SITE SERVICES LAYOUT - SHEET 07**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00107

Size  
**A1**

Rev  
0

Plot Date: 13/06/2025 1:30:27 PM

File Name: Autodesk Docs://12622621 - CQU - Fire & Water Main Design/12622621-GHD-00-00-MDL-HY-00001.rvt



FOR CONTINUATION REFER HY-00106

NOTE, EXACT VALVE LOCATION AND MARGIN BLOCK ARRANGEMENT TO BE CONFIRMED ON SITE DURING CONSTRUCTION. LIAISE WITH FACILITIES MANAGEMENT FOR FINAL LOCATION PRIOR TO INSTALLATION.

PROVIDE NEW DOUBLE PILLAR HYDRANT TO REPLACE EXISTING IN SAME LOCATION.

PIPEWORK INSTALLATION TO BE LOCATED WITHIN EXISTING ROADWAY ALONG KERB EDGE. ALLOW TO REINSTATE IN ACCORDANCE WITH SPECIFICATION INCLUDING THE USE OF STABILISED SAND FOR TRENCH BACKFILL WITHIN ROAD SECTIONS.

PIPEWORK TO OFF SET TO WITHIN VERGE. NOTE, EXACT ALIGNMENT AND EXTENT OF PIPEWORK IN ROAD OR VERGE TO BE DETERMINED ON SITE FOLLOWING SURVEY OF EXISTING SERVICES WITHIN AREA OF NEW WORKS.

PIPEWORK TO CROSS EXISTING ROADWAY. ALLOW TO REINSTATE IN ACCORDANCE WITH SPECIFICATION INCLUDING THE USE OF STABILISED SAND FOR TRENCH BACKFILL WITHIN ROAD SECTIONS.

## KEY PLAN

### LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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FOR CONTINUATION REFER HY-00112

FOR CONTINUATION REFER HY-00113

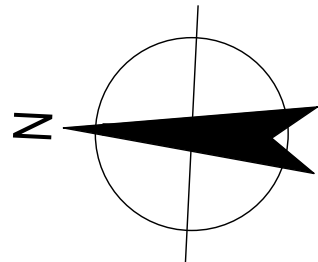
## SITE SERVICES LAYOUT - SHEET 08

SCALE 1:250

| Rev                                                | Description    | Checked | Approved | Date     |
|----------------------------------------------------|----------------|---------|----------|----------|
| 0                                                  | AS-CONSTRUCTED |         |          | 11.06.25 |
| Author: Designer      Drafting Check: Design Check |                |         |          |          |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
12622621

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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

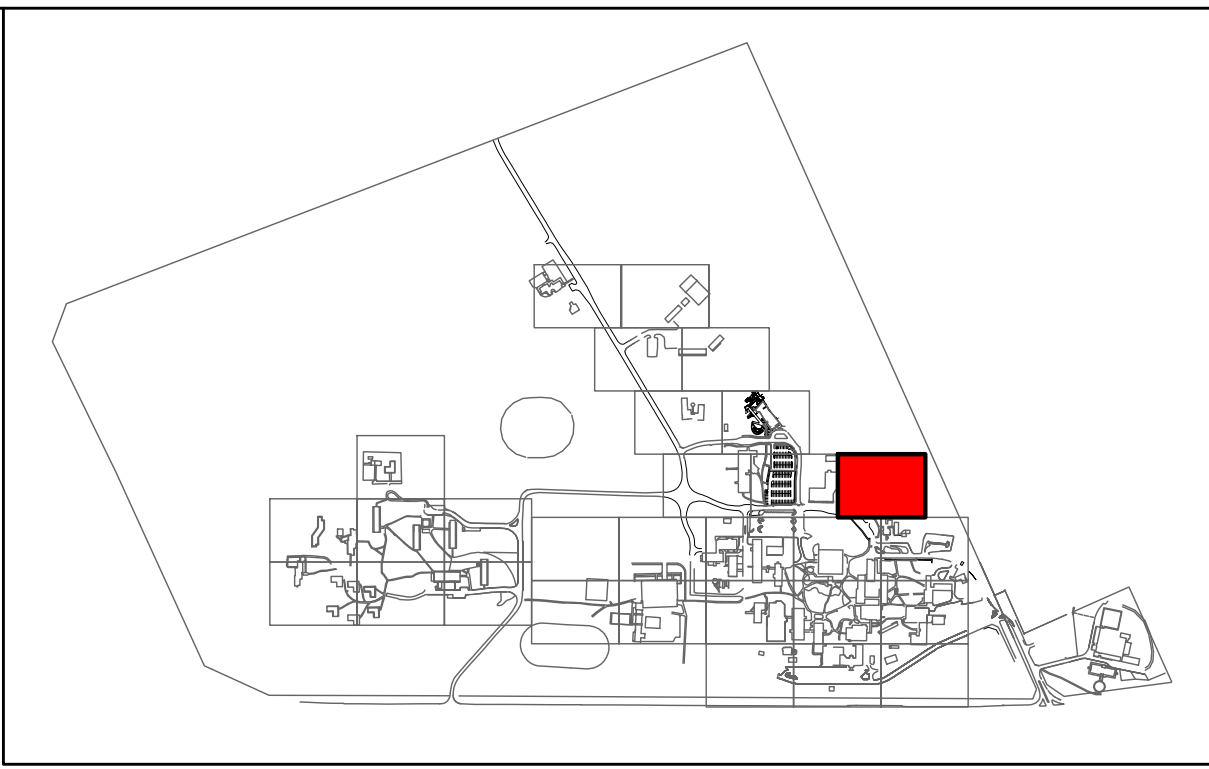
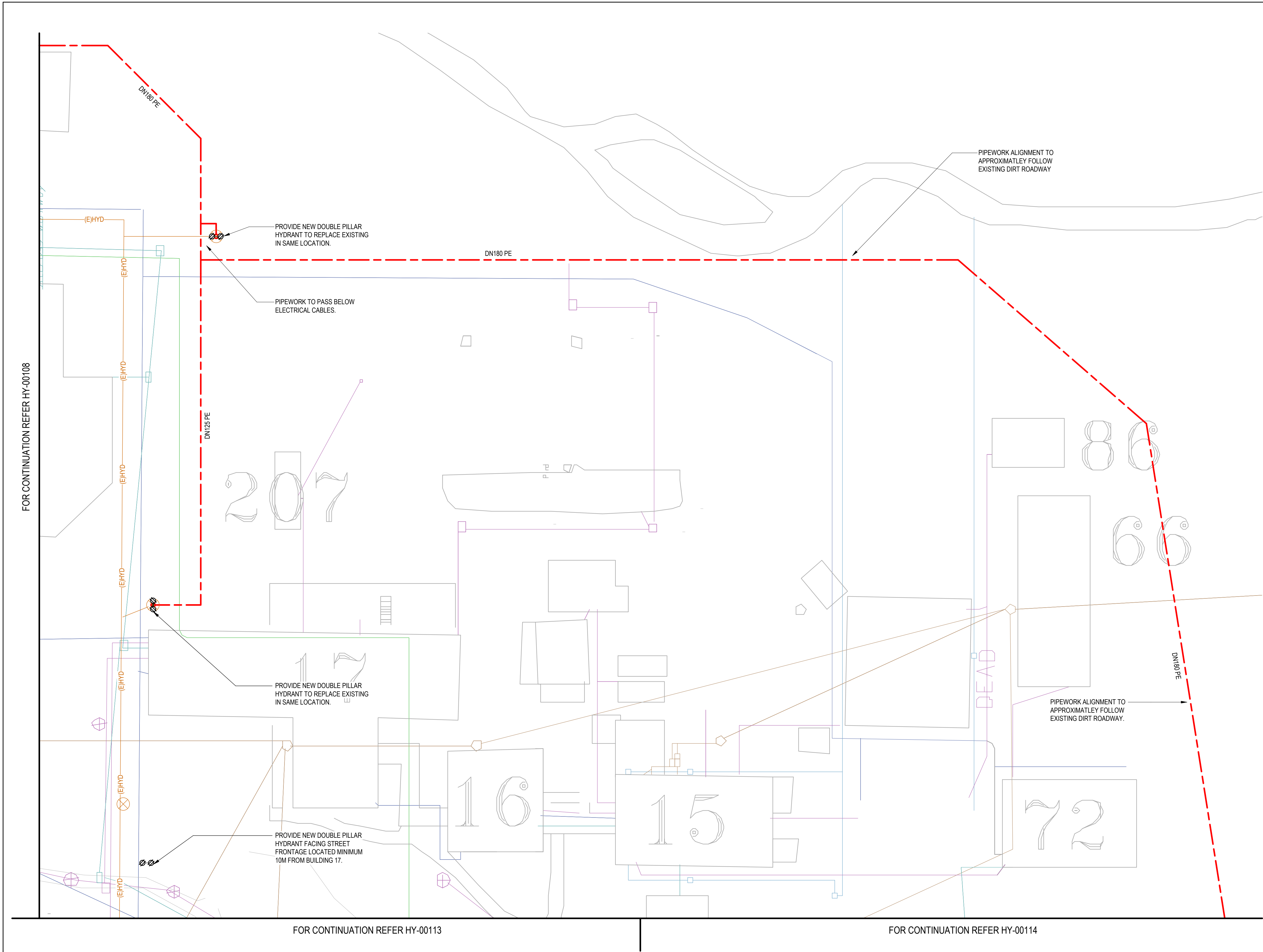
Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 08

Drawing No.  
12622621-GHD-00-00-DRG-HY-00108

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

NOTES:

- EXISTING SERVICES INCLUDING THE EXISTING FIRE HYDRANT SERVICE IS SHOWN FROM EXISTING RECORDS ONLY AND MAY NOT BE ACCURATE.
- THE ACCURACY AND EXACT LOCATION OF ALL SERVICES INCLUDING THE EXISTING FIRE HYDRANT SERVICE ARE TO BE CONFIRMED ON SITE PRIOR TO UNDERTAKING ANY EXCAVATION WORKS.
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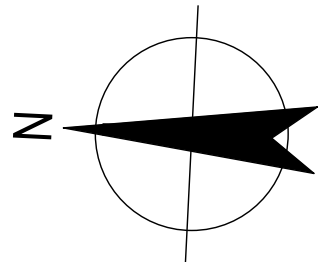
SITE SERVICES LAYOUT - SHEET 09

SCALE 1 : 250

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|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
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0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 09

Drawing No.  
12622621-GHD-00-00-DRG-HY-00109

Size  
A1

Rev  
0



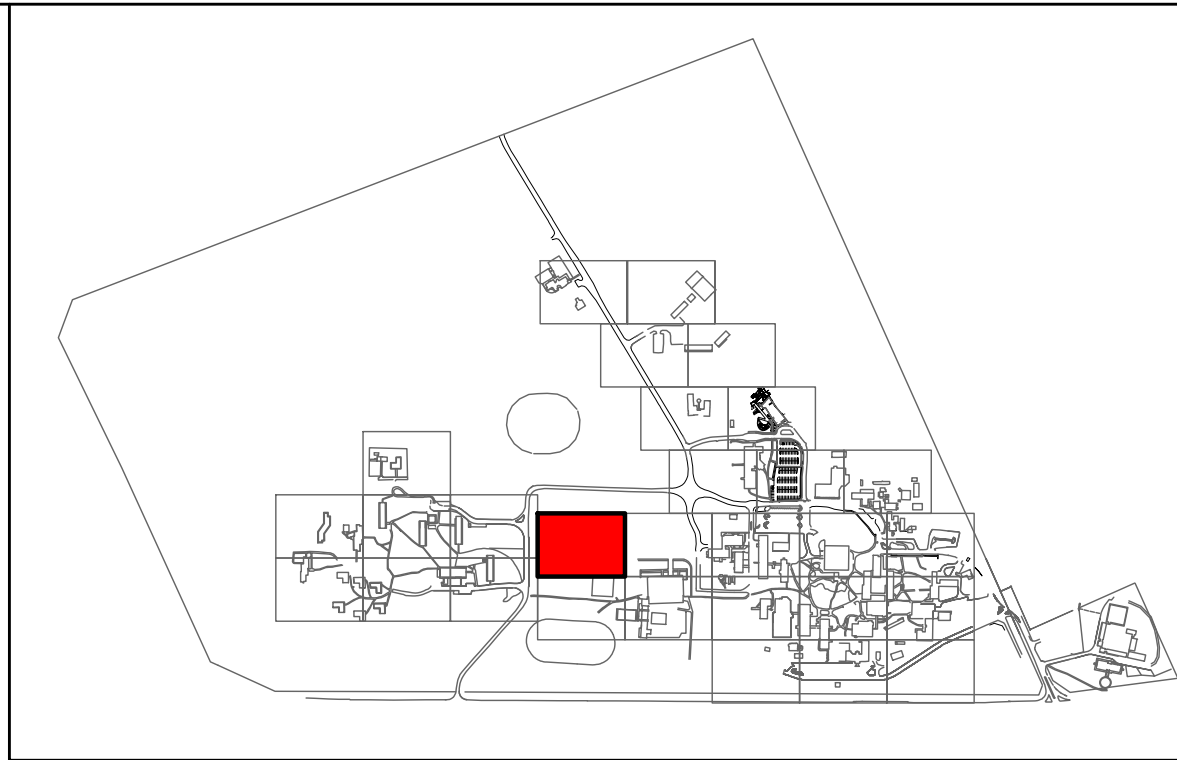
FOR CONTINUATION REFER HY-00126

FOR CONTINUATION REFER HY-00127

FOR CONTINUATION REFER HY-00115

SITE SERVICES LAYOUT - SHEET 10

SCALE 1 : 250



KEY PLAN

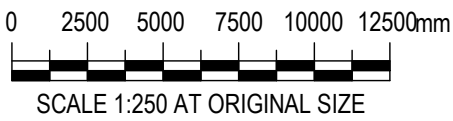
LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

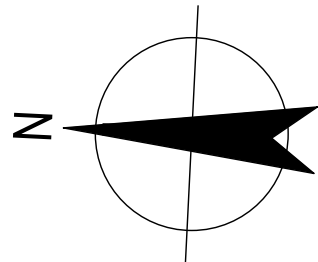
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FOR CONTINUATION REFER HY-00111



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Project No.  
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Client **CENTRAL QUEENSLAND UNIVERSITY**  
Project **FIRE MAIN UPGRADE**  
Status **AS-CONSTRUCTED**

Drawing Title **HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 10**

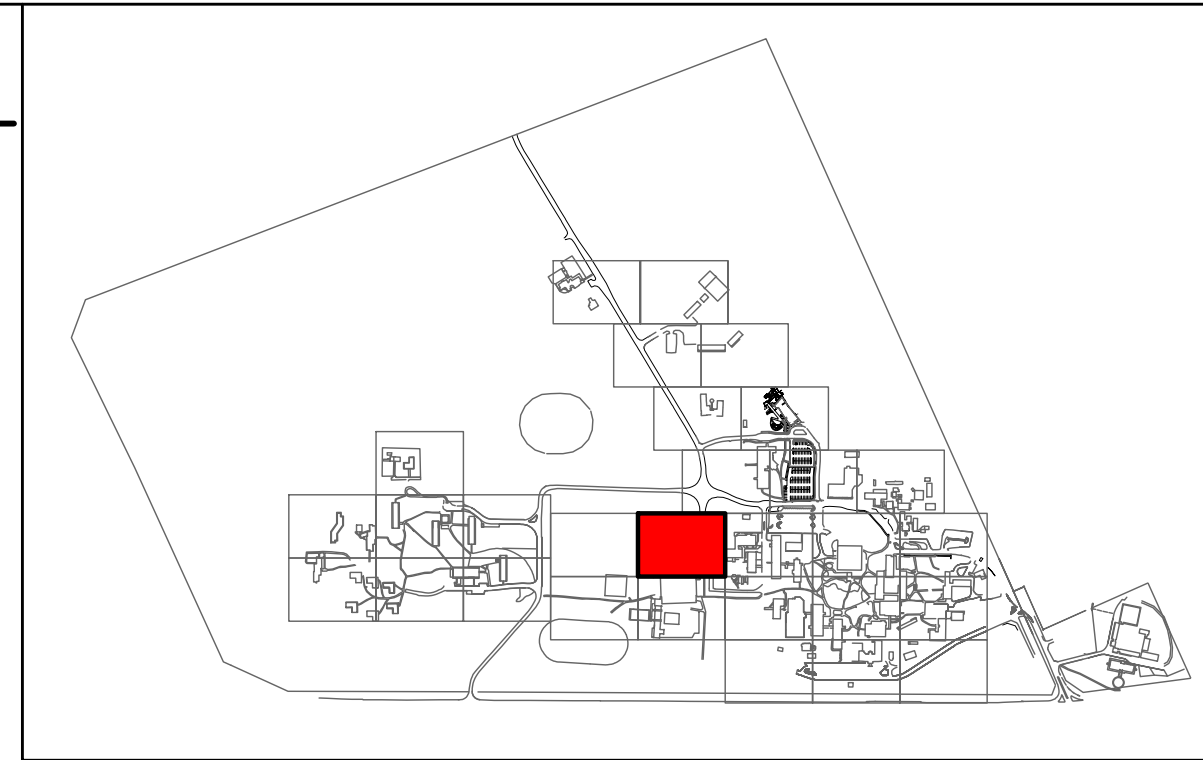
Drawing No.  
12622621-GHD-00-00-DRG-HY-00110

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00107



## KEY PLAN

### LEGEND

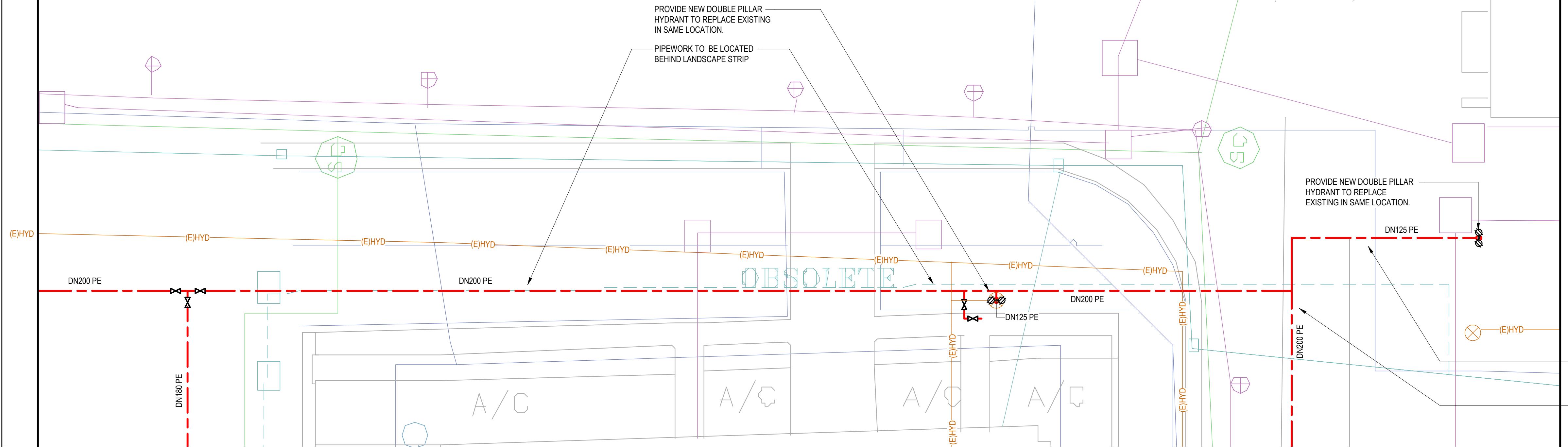
- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

### NOTES:

- EXISTING SERVICES INCLUDING THE EXISTING FIRE HYDRANT SERVICE IS SHOWN FROM EXISTING RECORDS ONLY AND MAY NOT BE ACCURATE.
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FOR CONTINUATION REFER HY-00110

FOR CONTINUATION REFER HY-00112



FOR CONTINUATION REFER HY-00116

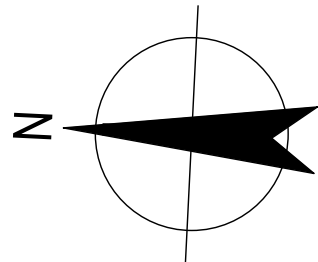
## SITE SERVICES LAYOUT - SHEET 11

SCALE 1:250

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|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 11

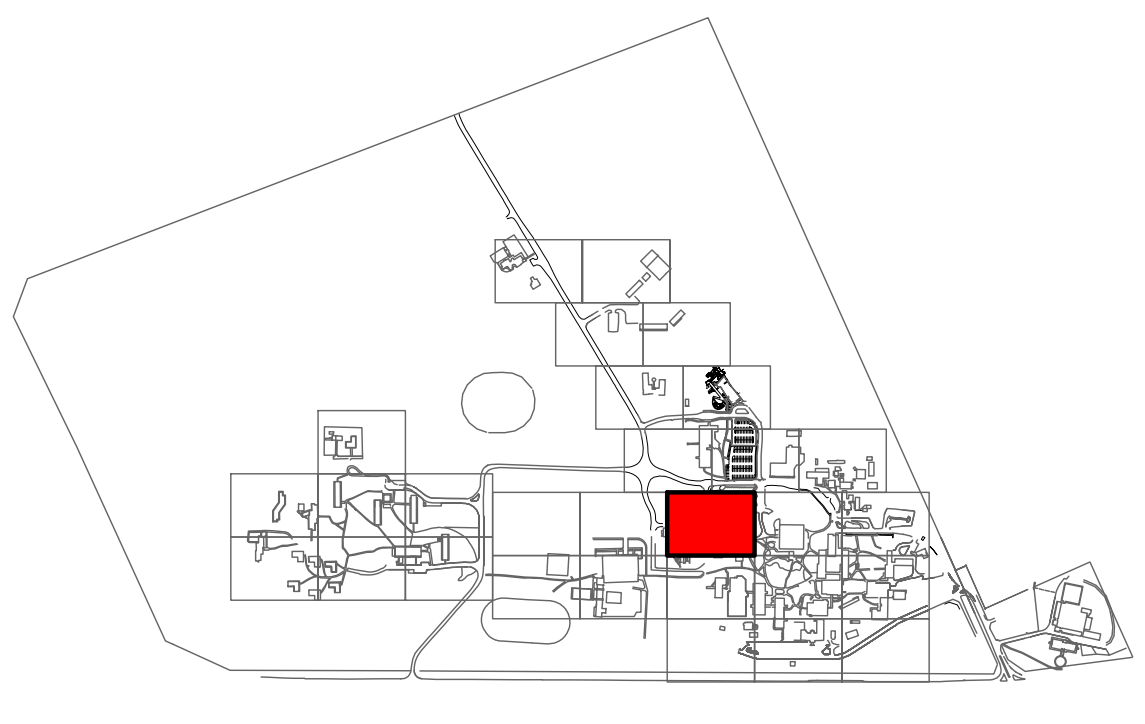
Drawing No.  
12622621-GHD-00-00-DRG-HY-00111

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00108



## KEY PLAN

### LEGEND

- |  |                         |
|--|-------------------------|
|  | NEW FIRE SERVICE        |
|  | NEW WATER SERVICE       |
|  | EXISTING COMMUNICATIONS |
|  | EXISTING ELECTRICAL     |
|  | EXISTING GAS            |
|  | EXISTING SEWER          |
|  | EXISTING STORMWATER     |
|  | EXISTING WATER          |
|  | EXISTING FIRE SERVICE   |
|  | NEW DUAL PILLAR HYDRANT |
|  | ISOLATION VALVE         |
|  | EXISTING HYDRANT        |

NOTES:

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4. THE EXISTING FIRE HYDRANT PIPEWORK AND ASSOCIATED INFRASTRUCTURE IS TO BE MAINTAINED THROUGHOUT COMPLETION OF ALL NEW WORKS. REFER TO THE PROJECT SPECIFICATION FOR DECOMMISSIONING DETAILS AND REQUIREMENTS.

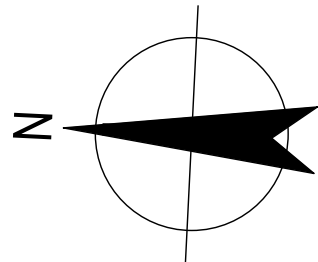
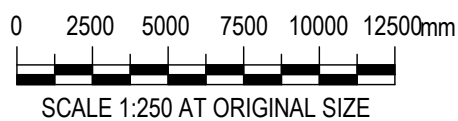
FOR CONTINUATION REFER HY-00111

FOR CONTINUATION REFER HY-00113

FOR CONTINUATION REFER HY-00117

SITE SERVICES LAYOUT - SHEET 12

SCALE 1 : 250



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Client **CENTRAL QUEENSLAND UNIVERSITY**

Project **FIRE MAIN UPGRADE**

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| Status | AS-CONSTRUCTED |
|--------|----------------|

Drawing Title **HYDRAULIC SERVICES**  
**SITE SERVICES LAYOUT - SHEET 12**

Drawing No.  
**12622621-GHD-00-00-DRG-HY-00112**

**Size**  
**A1**

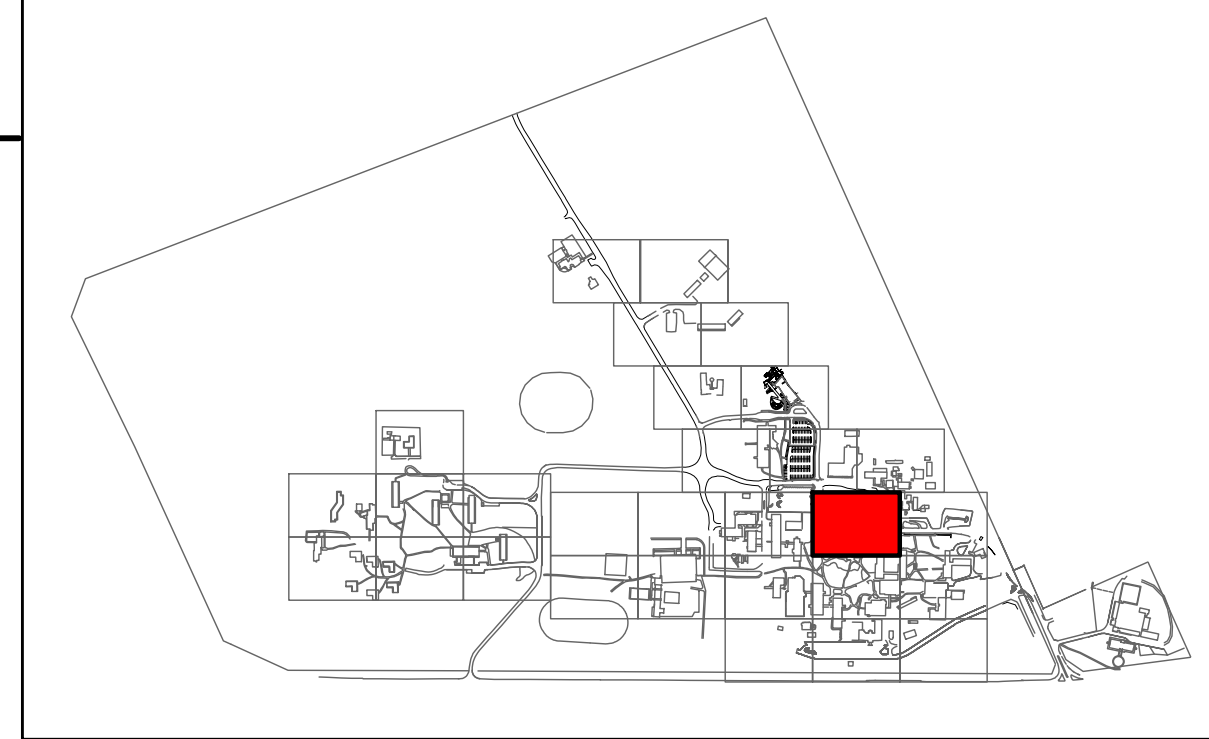
Rev  
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Plot Date: 13/06/2025 1:30:44 PM

File Name: Autodesk Docs://12622621 - CQU - Fire & Water Main Design/12622621-GHD-00-00-MDL-HY-00001.rvt



FOR CONTINUATION REFER HY-00109



LEGEND

- NOTES:

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FOR CONTINUATION REFER HY-00112

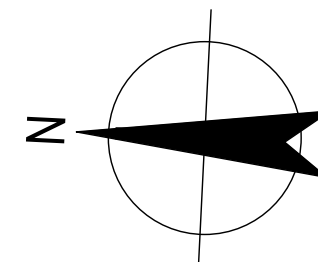
EOB CONTINUATION BEEER HV 00114

FOR CONTINUATION REFER HY-00118

SITE SERVICES LAYOUT - SHEET 13

SCALE 1 : 250

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| 0        | AS-CONSTRUCTED |         |          | 11.06.25 |
| Rev      | Description    | Checked | Approved | Date     |
| Author   | Drafting Check |         |          |          |
| Designer | Design Check   |         |          |          |



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Project No  
12622621

Project **FIRE MAIN UPGRADE**

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| No. | Status | AS-CONSTRUCTED |
|-----|--------|----------------|

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Drawing<br>Title | <b>HYDRAULIC SERVICES<br/>SITE SERVICES LAYOUT - SHEET 13</b> |
|------------------|---------------------------------------------------------------|

Drawing No.  
12622621-GHD-00-00-DRG-HY-00113

A1

Rev  
0

Plot Date: 13/06/2025 1:30:48 PM

File Name: Autodesk Docs://12622621 - CQU - Fire & Water Main Design/12622621-GHD-00-00-MDL-HY-00001.rvt



FOR CONTINUATION REFER HY-001113



SCALE 1 : 250

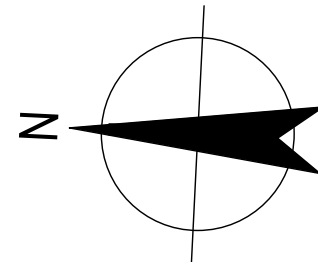


|  |                         |
|--|-------------------------|
|  | NEW FIRE SERVICE        |
|  | NEW WATER SERVICE       |
|  | EXISTING COMMUNICATIONS |
|  | EXISTING ELECTRICAL     |
|  | EXISTING GAS            |
|  | EXISTING SEWER          |
|  | EXISTING STORMWATER     |
|  | EXISTING WATER          |
|  | EXISTING FIRE SERVICE   |
|  | NEW DUAL PILLAR HYDRANT |
|  | ISOLATION VALVE         |
|  | EXISTING HYDRANT        |

1. EXISTING SERVICES INCLUDING THE EXISTING FIRE HYDRANT SERVICE IS SHOWN FROM EXISTING RECORDS ONLY AND MAY NOT BE ACCURATE.
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0 2500 5000 7500 10000 12500mm

SCALE 1:250 AT ORIGINAL SIZE



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Drawing No.  
12622621-GHD-00-00-DRG-HY-00114

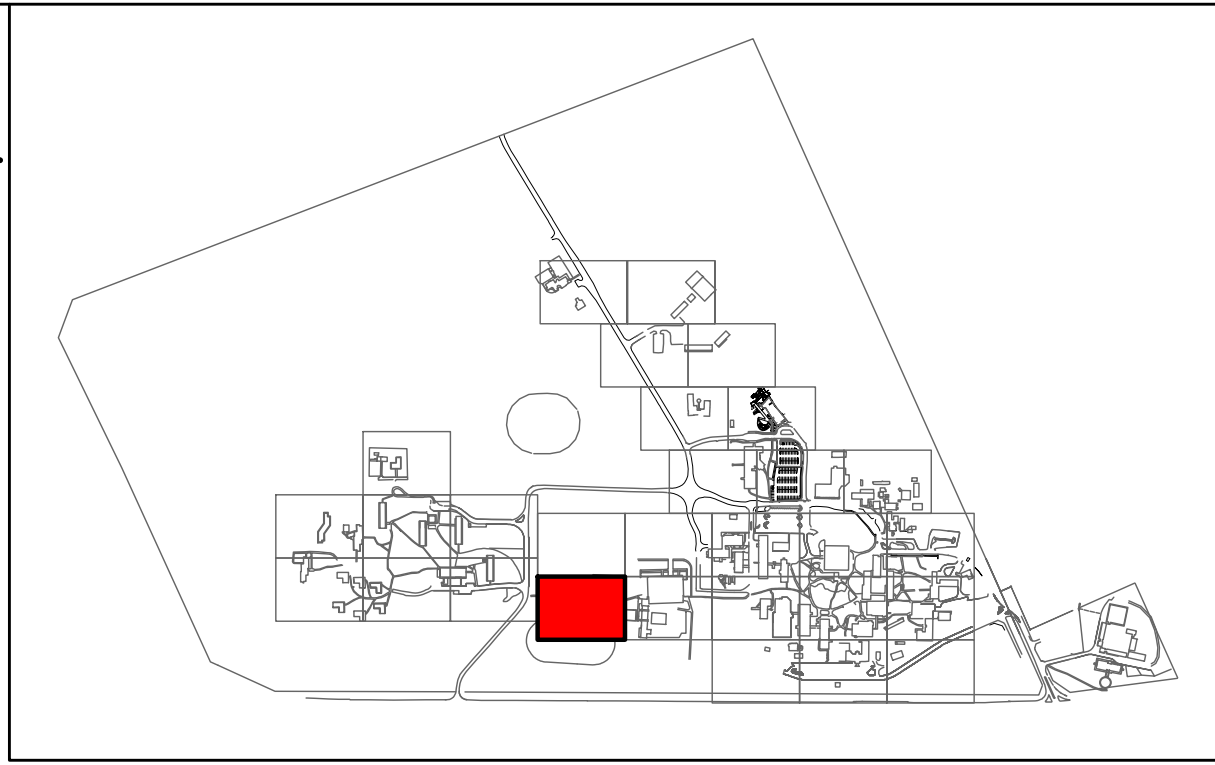
Rev  
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Plot Date: 13/06/2025 1:30:52 PM

File Name: Autodesk Docs://12622621 - CQU - Fire & Water Main Design/12622621-GHD-00-00-MDL-HY-00001.rvt



FOR CONTINUATION REFER HY-00110



KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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FOR CONTINUATION REFER HY-00129

FOR CONTINUATION REFER HY-00116

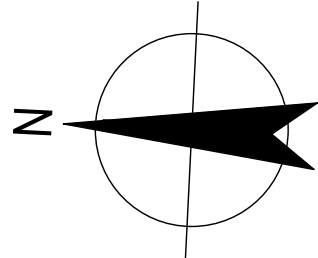
SITE SERVICES LAYOUT - SHEET 15

SCALE 1 : 250

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| Rev      | Description    | Checked        | Approved | Date |          |
| Author   |                | Drafting Check |          |      |          |
| Designer |                | Design Check   |          |      |          |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

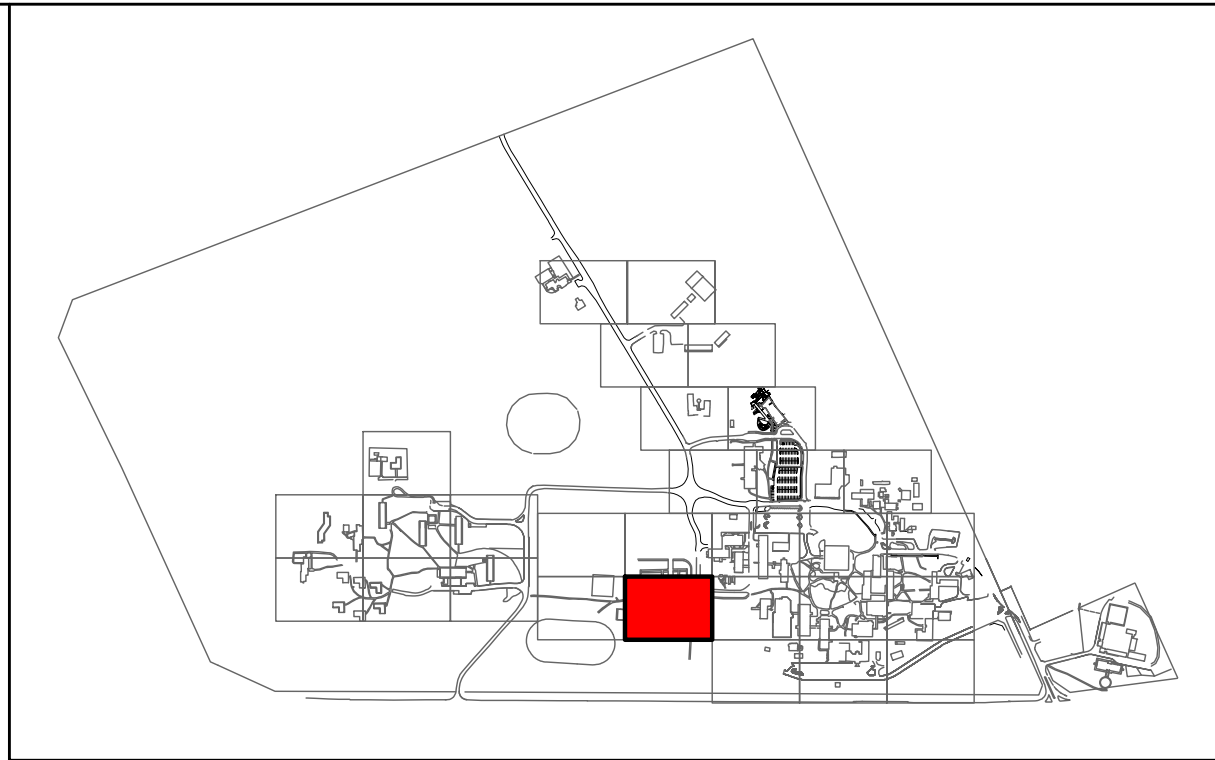
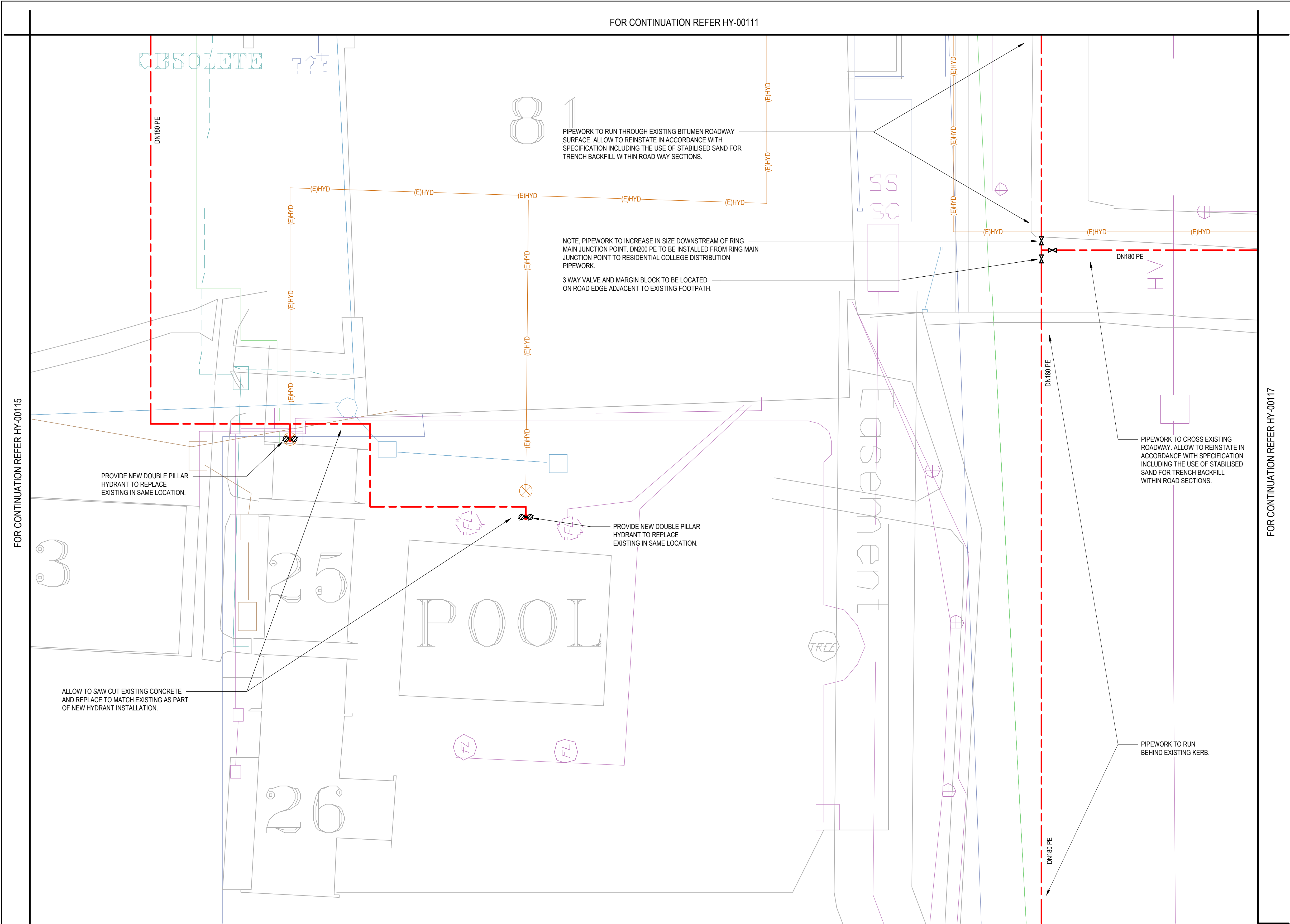
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HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 15

Drawing No.  
12622621-GHD-00-00-DRG-HY-00115

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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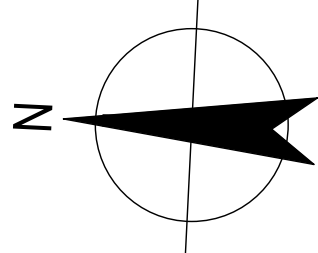
SITE SERVICES LAYOUT - SHEET 16

SCALE 1 : 250

|          |                |                       |
|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
12622621

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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

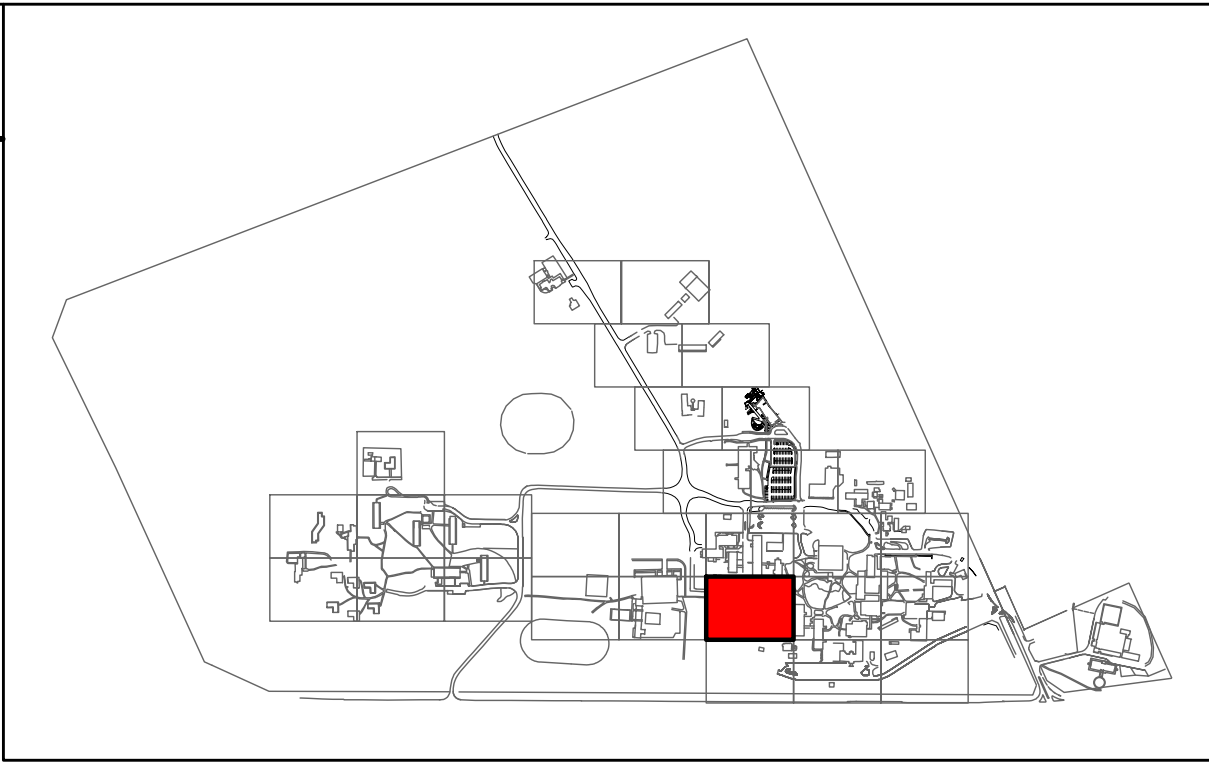
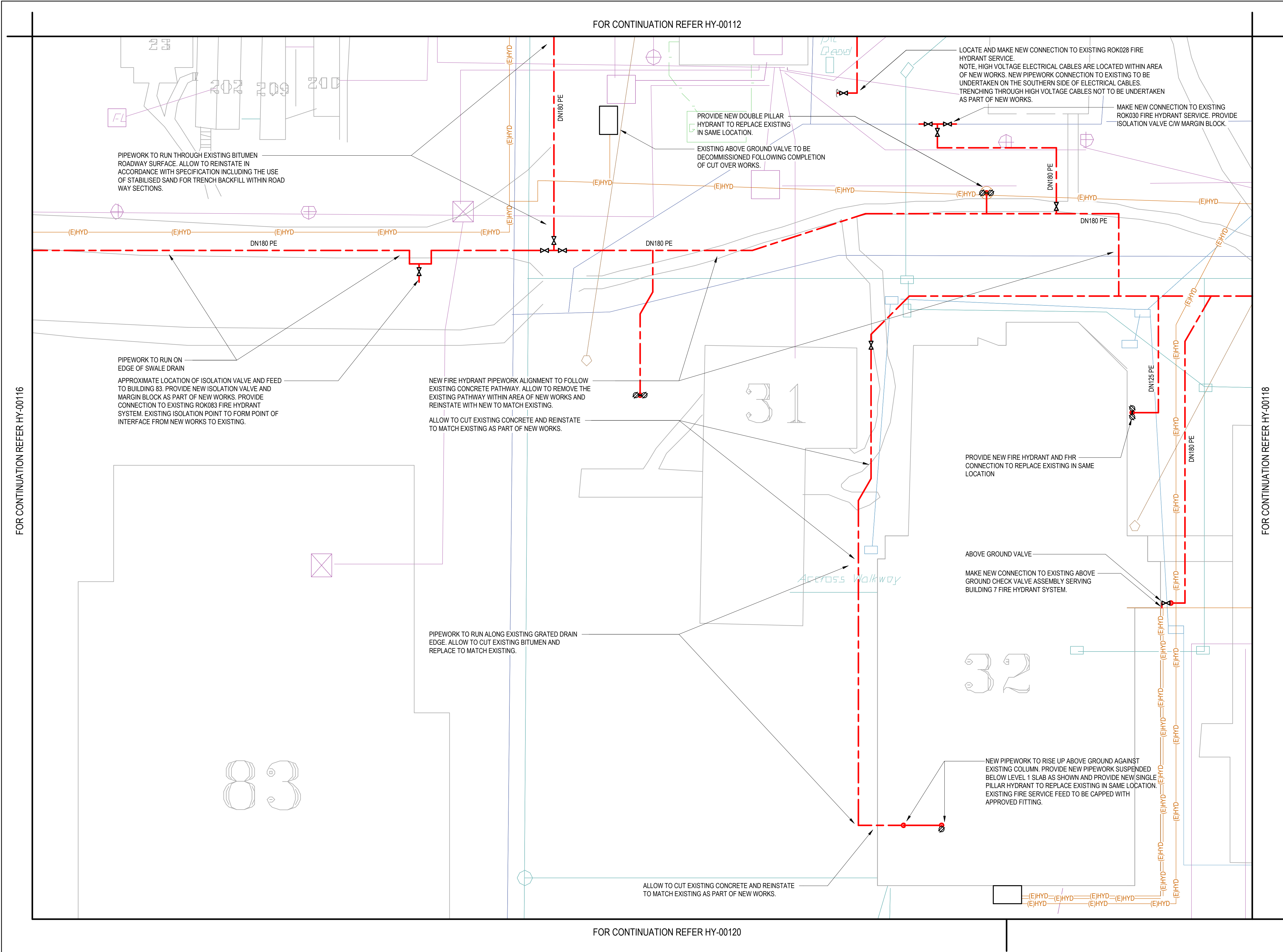
Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 16

Drawing No.  
12622621-GHD-00-00-DRG-HY-00116

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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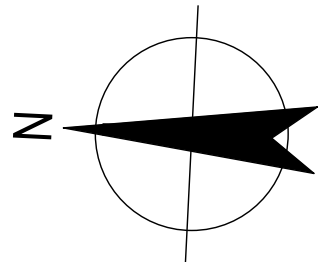
SITE SERVICES LAYOUT - SHEET 17

SCALE 1 : 250

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|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
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Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 17

12622621-GHD-00-00-DRG-HY-00117

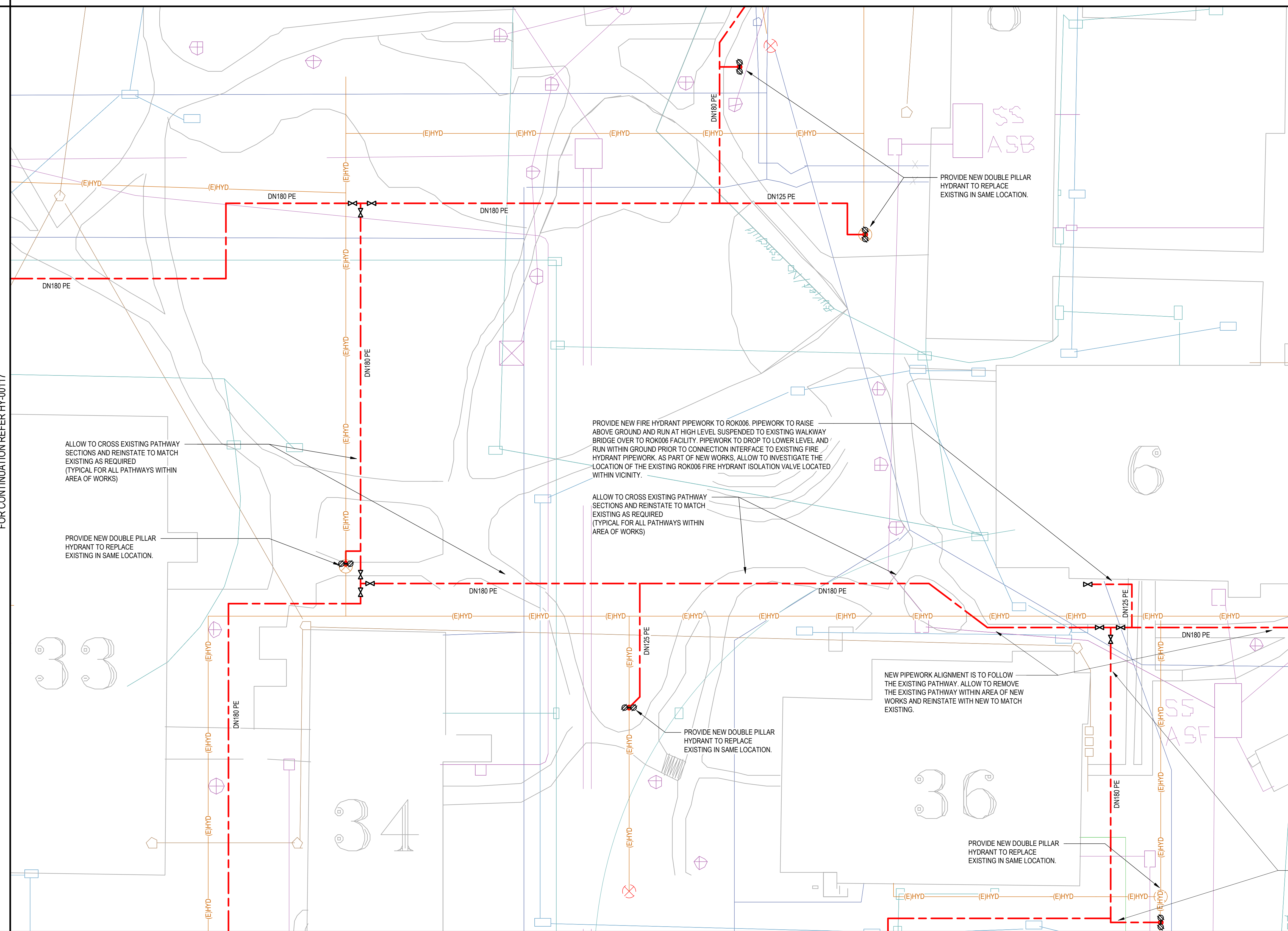
Drawing No.

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00113



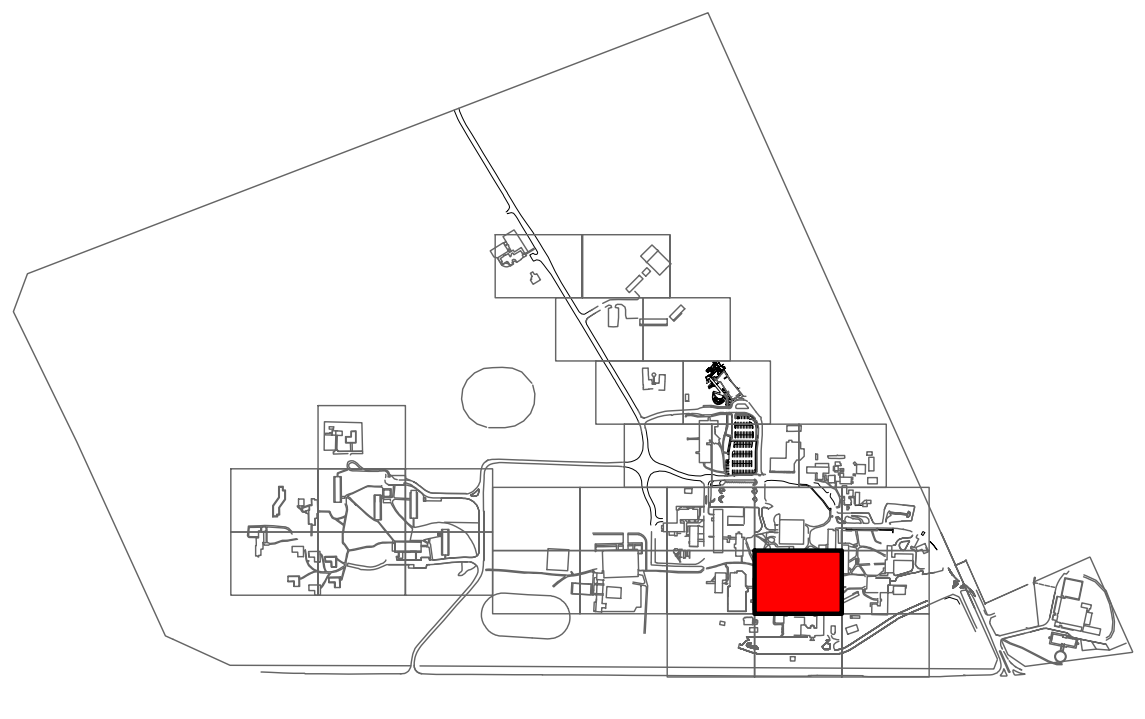
FOR CONTINUATION REFER HY-00117

FOR CONTINUATION REFER HY-00119

FOR CONTINUATION REFER HY-00121

SITE SERVICES LAYOUT - SHEET 18

SCALE 1 : 250



KEY PLAN

LEGEND

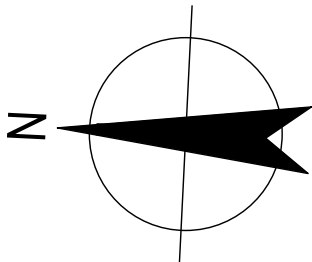
- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE



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Project No.  
12622621

Client **CENTRAL QUEENSLAND UNIVERSITY**  
Project **FIRE MAIN UPGRADE**  
Status **AS-CONSTRUCTED**

Drawing Title **HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 18**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00118

Size  
A1

Rev  
0



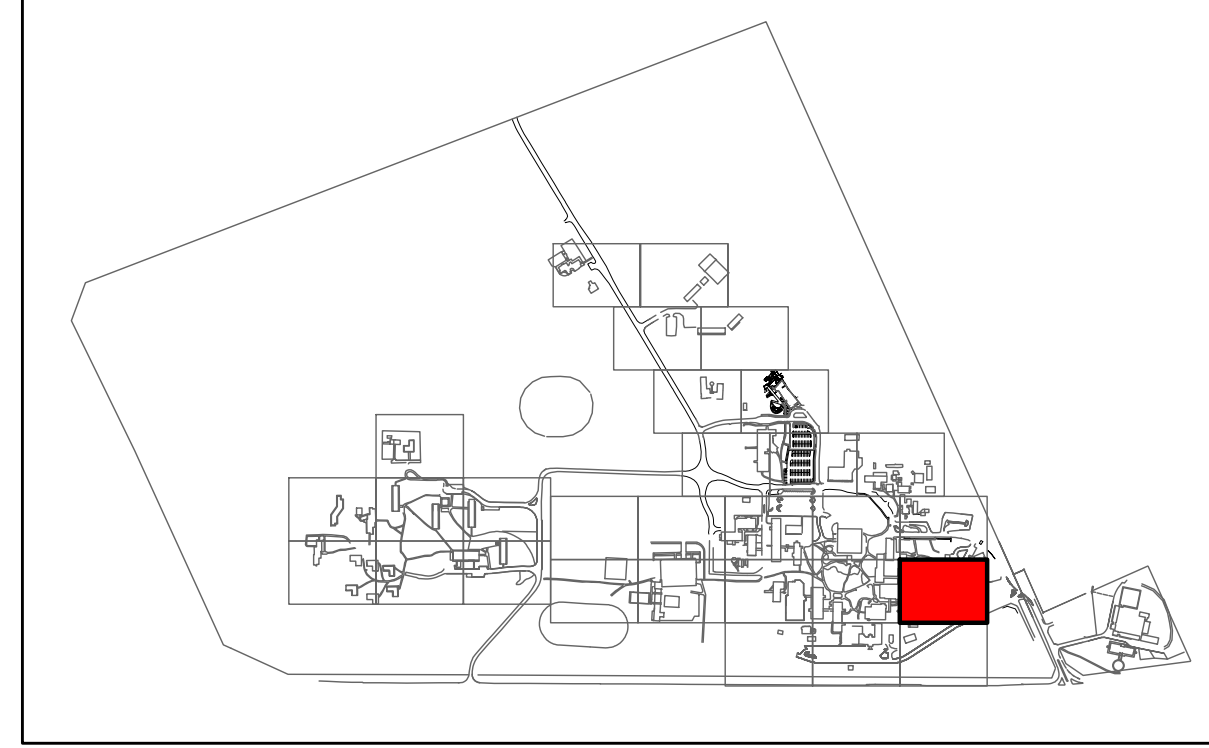
FOR CONTINUATION REFER HY-00114

FOR CONTINUATION REFER HY-00118

FOR CONTINUATION REFER HY-00122

SITE SERVICES LAYOUT - SHEET 19

SCALE 1 : 250



KEY PLAN

LEGEND

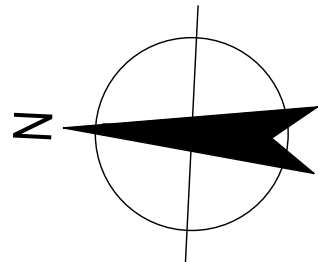
- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
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Client **CENTRAL QUEENSLAND UNIVERSITY**  
Project **FIRE MAIN UPGRADE**  
Status **AS-CONSTRUCTED**

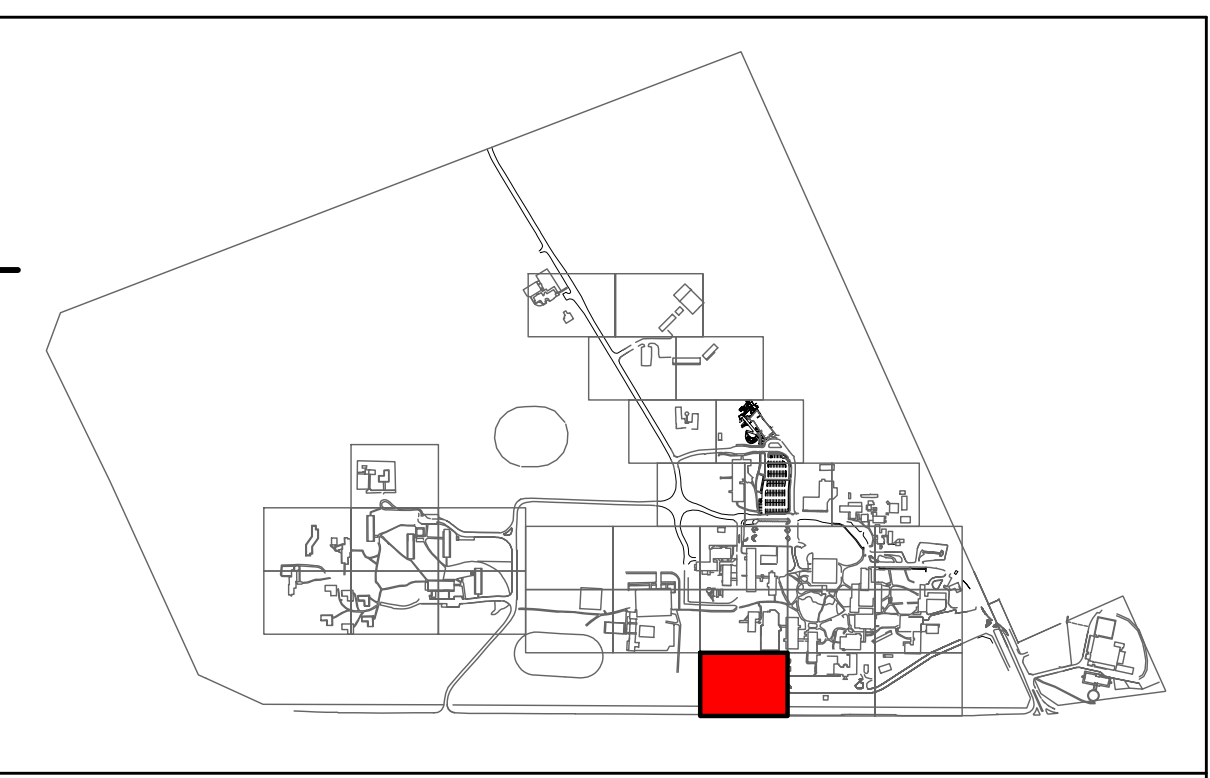
Drawing Title **HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 19**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00119

Size  
A1

Rev  
0





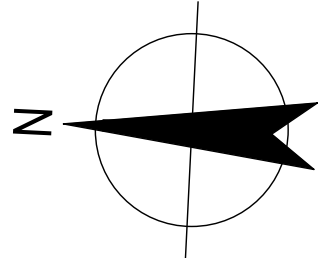
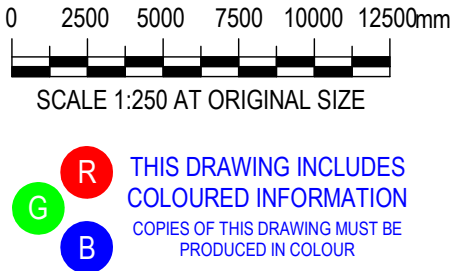
### LEGEND

-  NEW FIRE SERVICE  
 NEW WATER SERVICE  
 EXISTING COMMUNICATIONS  
 EXISTING ELECTRICAL  
 EXISTING GAS  
 EXISTING SEWER  
 EXISTING STORMWATER  
 EXISTING WATER  
 (E)HYD EXISTING FIRE SERVICE  
 NEW DUAL PIPER HYDRANT  
 ISOLATION VALVE  
 EXISTING HYDRANT

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**SITE SERVICES LAYOUT - SHEET 20**  
SCALE 1 : 250

|                  |                |         |          |      |          |  |
|------------------|----------------|---------|----------|------|----------|--|
| 0 AS-CONSTRUCTED |                |         |          |      | 11.06.25 |  |
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| Author           | Drafting Check |         |          |      |          |  |
| Designer         | Design Check   |         |          |      |          |  |



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Project No.  
**12622621**

Client **CENTRAL QUEENSLAND UNIVERSITY**

Project **FIRE MAIN UPGRADE**

Status **AS-CONSTRUCTED**

Drawing Title **HYDRAULIC SERVICES**  
**SITE SERVICES LAYOUT - SHEET 20**

Drawing No.  
**12622621-GHD-00-00-DRG-HY-00120**

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00118

PROVIDE NEW DOUBLE PILLAR  
HYDRANT TO REPLACE  
EXISTING IN SAME LOCATION.

Cooling Pipes

PROVIDE NEW DOUBLE PILLAR  
HYDRANT TO REPLACE  
EXISTING IN SAME LOCATION.

PROVIDE NEW DOUBLE PILLAR  
HYDRANT TO REPLACE  
EXISTING IN SAME LOCATION.

PIPEWORK TO RUN THROUGH  
EXISTING GARDEN BED TO  
COMPLETE RING MAIN CONNECTION.

PIPEWORK TO RUN THROUGH EXISTING BITUMEN  
CAR PARK SURFACE. ALLOW TO REINSTATE IN  
ACCORDANCE WITH SPECIFICATION INCLUDING  
THE USE OF STABILISED SAND FOR TRENCH  
BACKFILL WITHIN CAR PARK SECTIONS.

PIPEWORK INSTALLATION TO BE LOCATED WITHIN EXISTING ROADWAY  
ALONG KERB EDGE. ALLOW TO REINSTATE IN ACCORDANCE WITH  
SPECIFICATION INCLUDING THE USE OF STABILISED SAND FOR  
TRENCH BACKFILL WITHIN ROAD SECTIONS.

PIPEWORK TO TRANSITION FROM CAR PARK EDGE TO  
BACK OF KERB. ALLOW TO COORDINATE NEW PIPEWORK  
INSTALLATION WITH EXISTING SERVICES WITHIN AREA.

PIPEWORK INSTALLATION TO BE LOCATED WITHIN EXISTING ROADWAY  
ALONG KERB EDGE. ALLOW TO REINSTATE IN ACCORDANCE WITH  
SPECIFICATION INCLUDING THE USE OF STABILISED SAND FOR  
TRENCH BACKFILL WITHIN ROAD SECTIONS.

Switch Room

Lead in  
to  
P4136018

P4136016

P4136018

SITE SERVICES LAYOUT - SHEET 21  
SCALE 1:250

KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

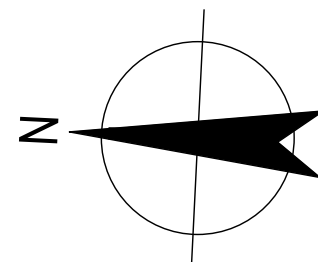
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| Rev      | Description    | Checked | Approved | Date     |
|----------|----------------|---------|----------|----------|
| 0        | AS-CONSTRUCTED |         |          | 11.06.25 |
| Author   | Drafting Check |         |          |          |
| Designer | Design Check   |         |          |          |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
12622621

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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

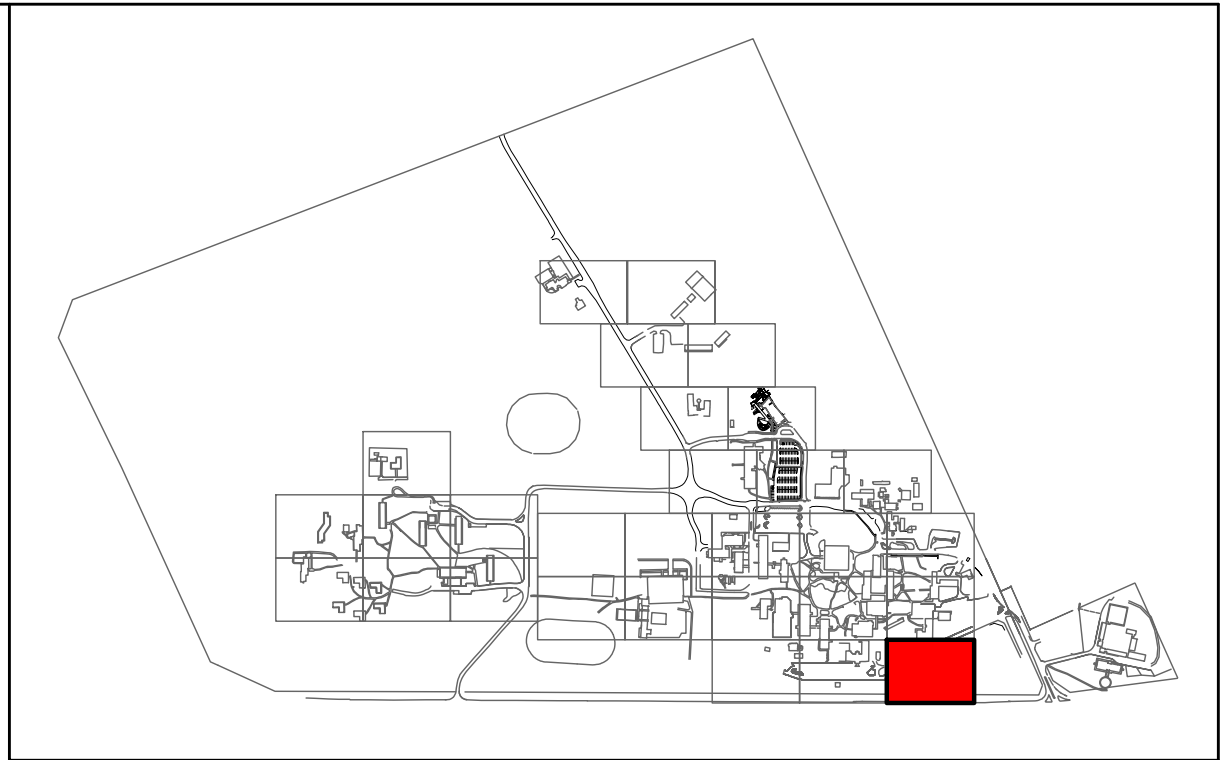
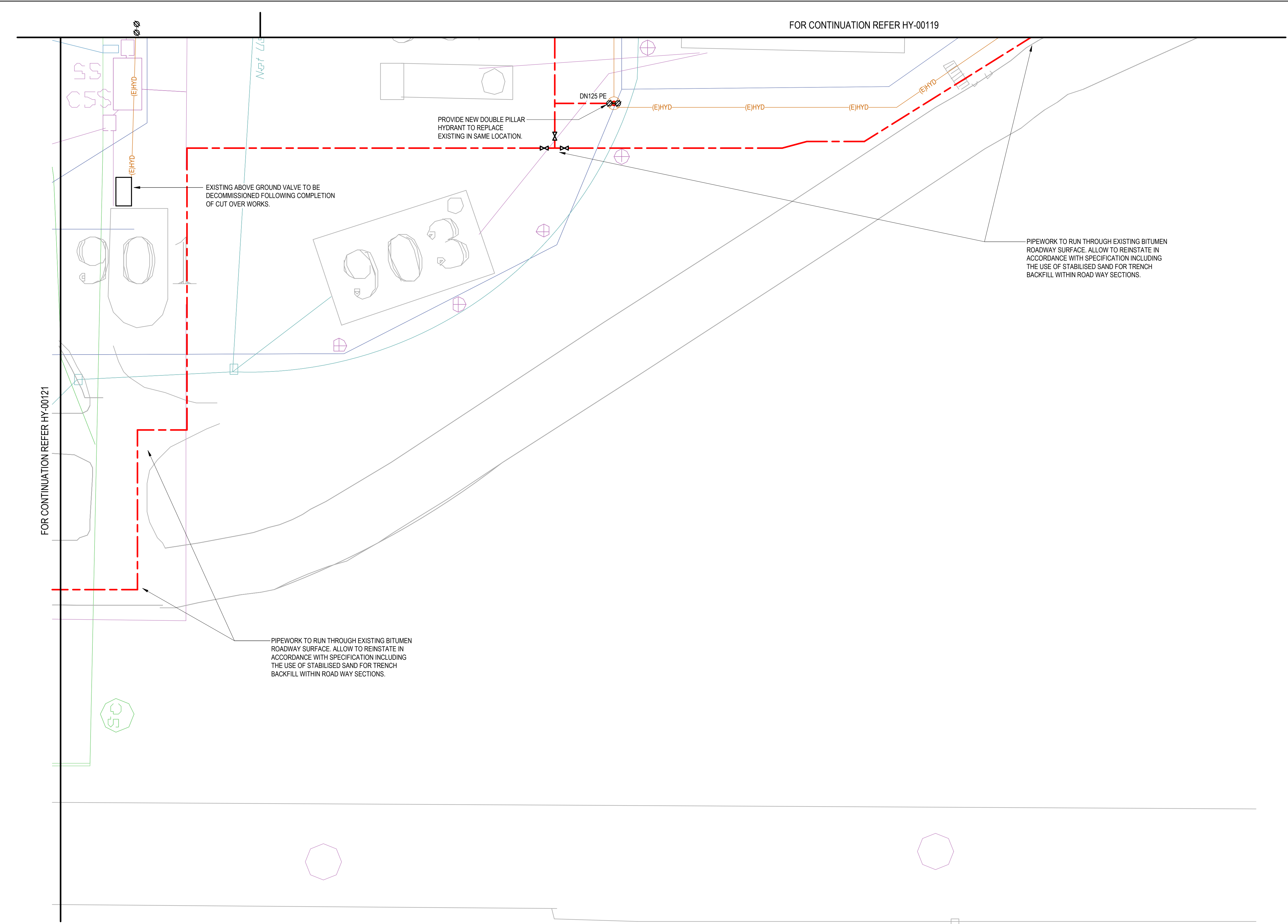
Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 21

Drawing No.  
12622621-GHD-00-00-DRG-HY-00121

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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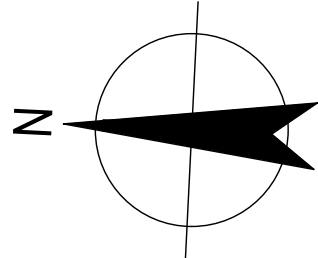
SITE SERVICES LAYOUT - SHEET 22

SCALE 1 : 250

|          |                |                       |
|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Designing      | Designing             |
| Designer | Designing      | Designing             |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

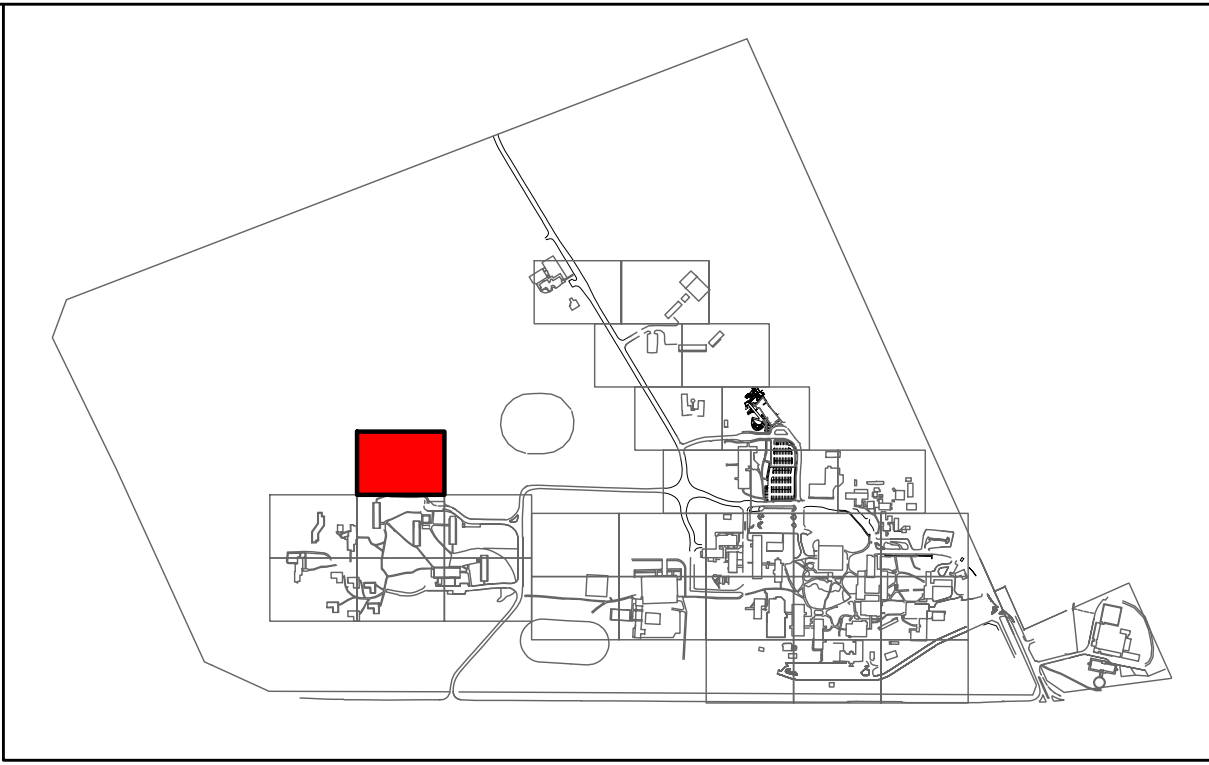
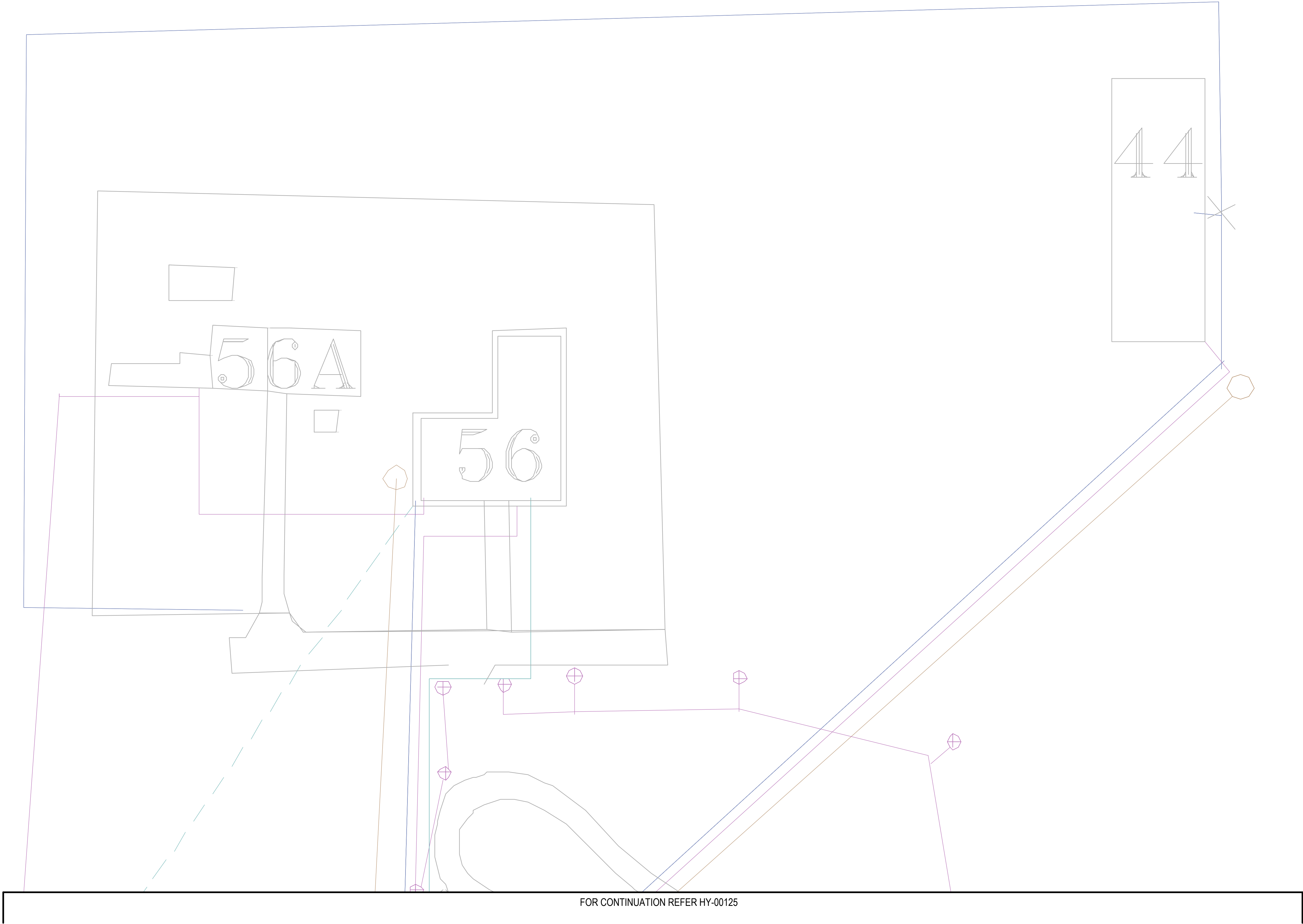
Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 22

Drawing No.  
12622621-GHD-00-00-DRG-HY-00122

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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FOR CONTINUATION REFER HY-00125

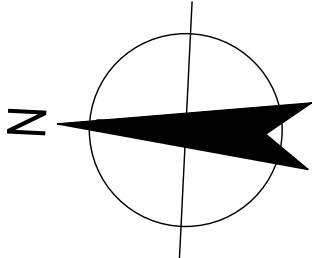
SITE SERVICES LAYOUT - SHEET 23

SCALE 1 : 250

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| 0        | AS-CONSTRUCTED |         |          | 11.06.25 |
| Rev      | Description    | Checked | Approved | Date     |
| Author   | Drafting Check |         |          |          |
| Designer | Design Check   |         |          |          |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Project No.  
12622621

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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

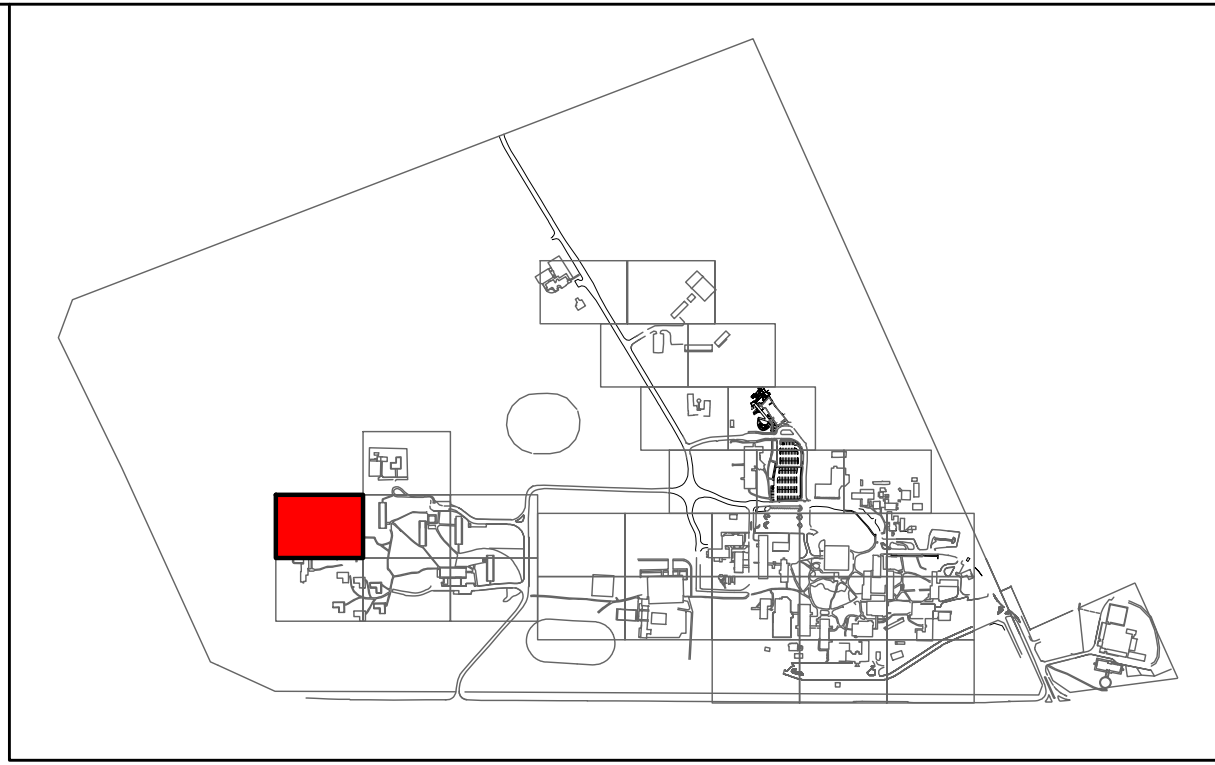
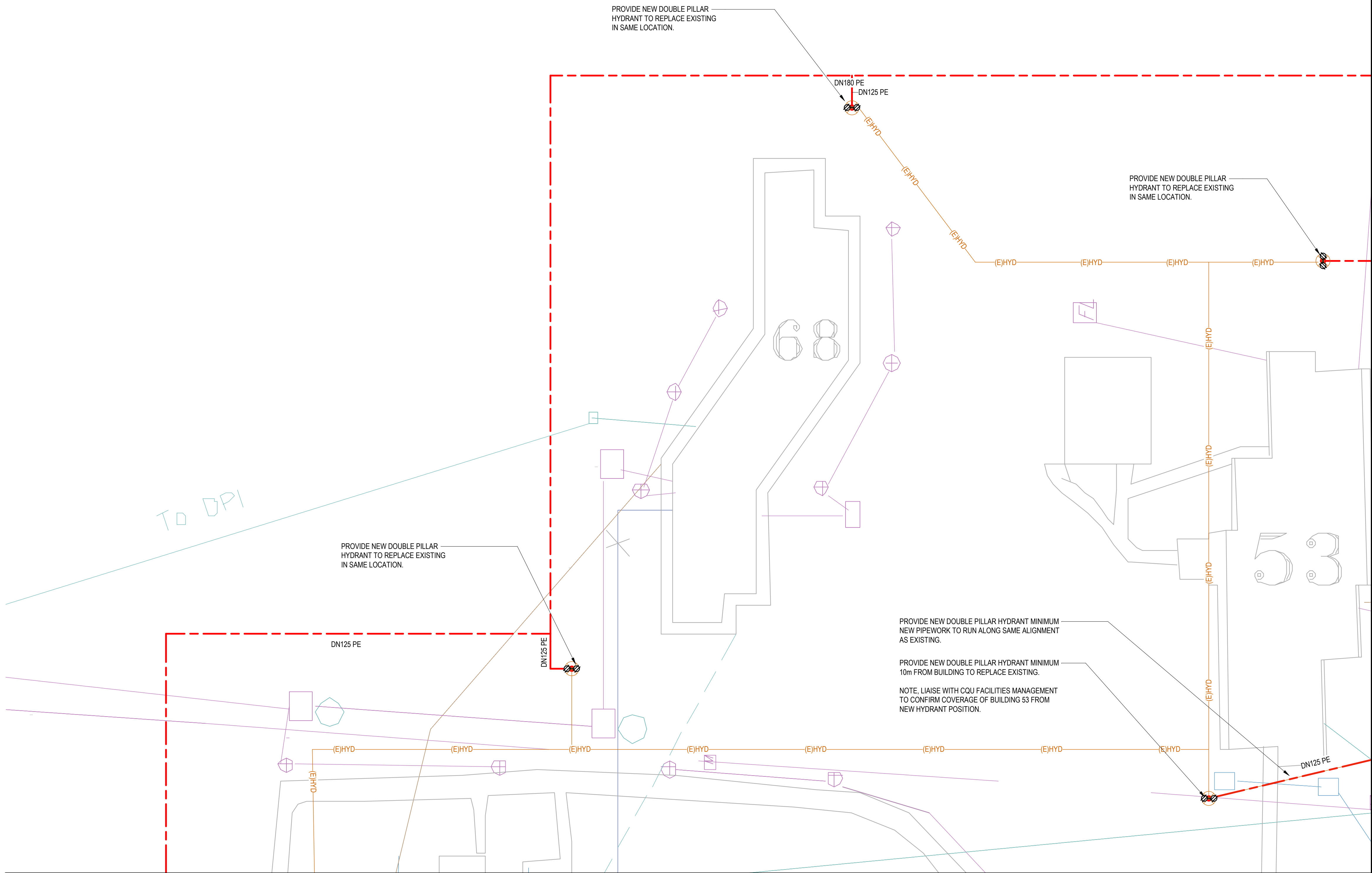
Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 23

Drawing No.  
12622621-GHD-00-00-DRG-HY-00123

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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FOR CONTINUATION REFER HY-00127

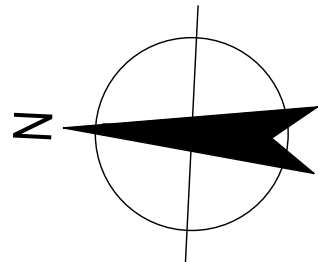
SITE SERVICES LAYOUT - SHEET 24

SCALE 1 : 250

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| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 24

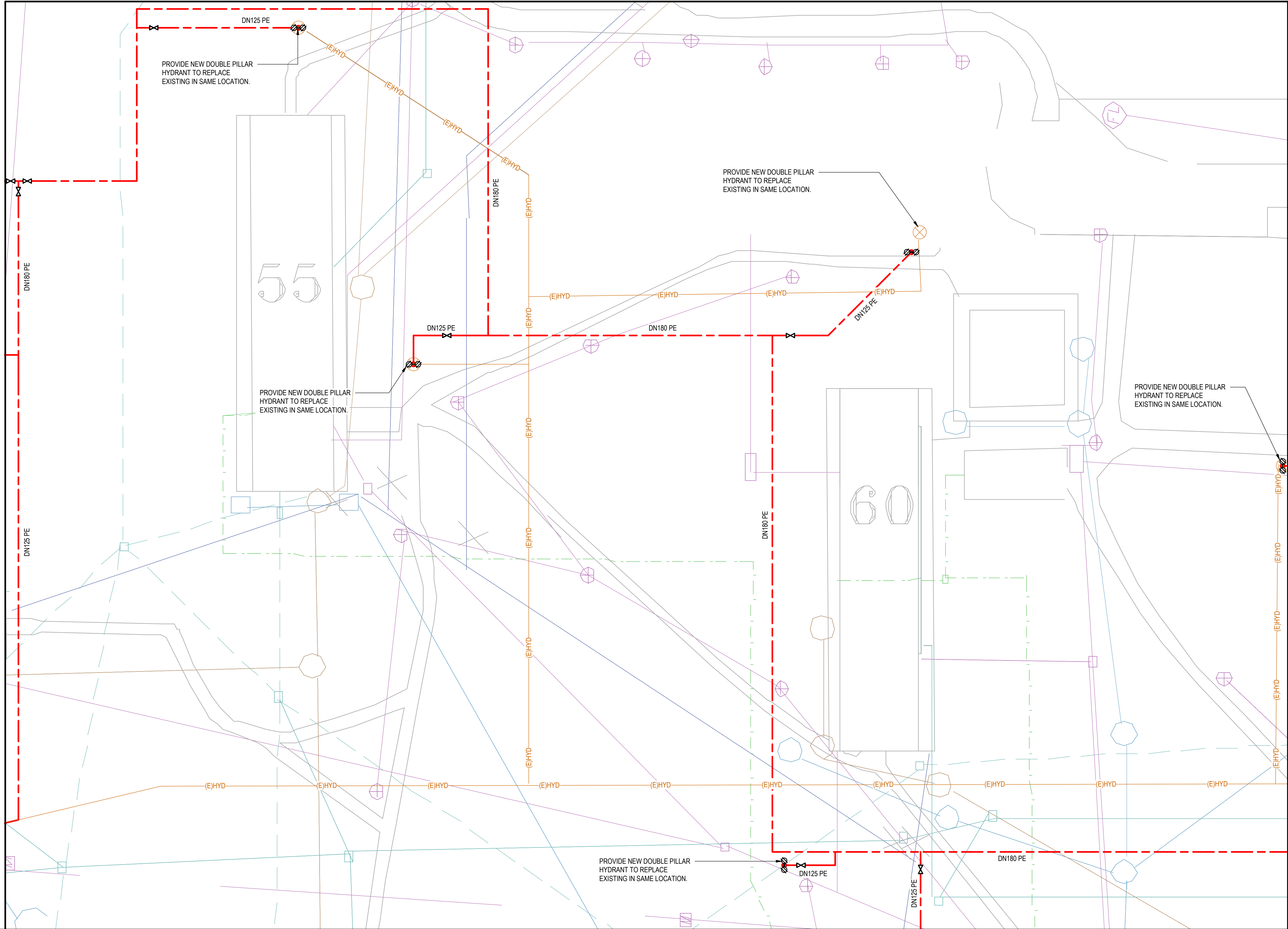
Drawing No.  
12622621-GHD-00-00-DRG-HY-00124

Size  
A1

Rev  
0



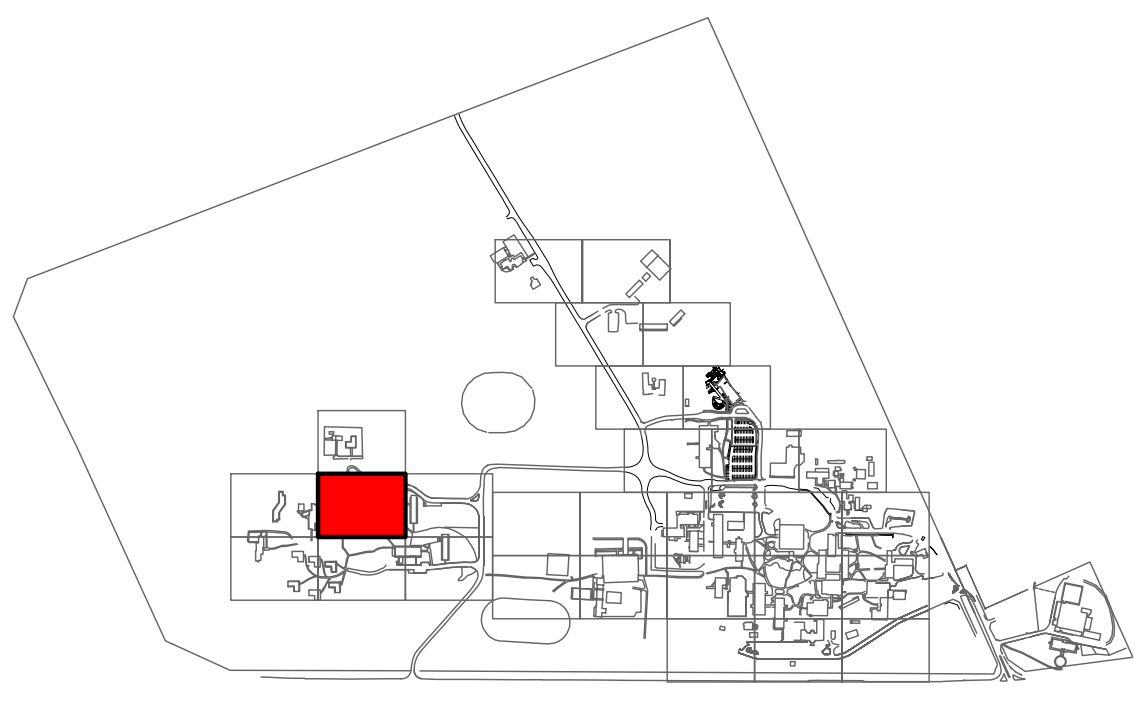
FOR CONTINUATION REFER HY-00123



FOR CONTINUATION REFER HY-00128

**SITE SERVICES LAYOUT - SHEET 25**

SCALE 1 : 250



**KEY PLAN**

**LEGEND**

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

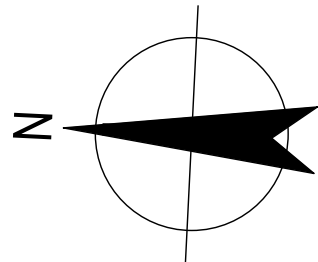
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| Rev      | Description    | Checked Approved Date |
| Author   | Drafting Check |                       |
| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Client **CENTRAL QUEENSLAND UNIVERSITY**  
Project **FIRE MAIN UPGRADE**  
Status **AS-CONSTRUCTED**

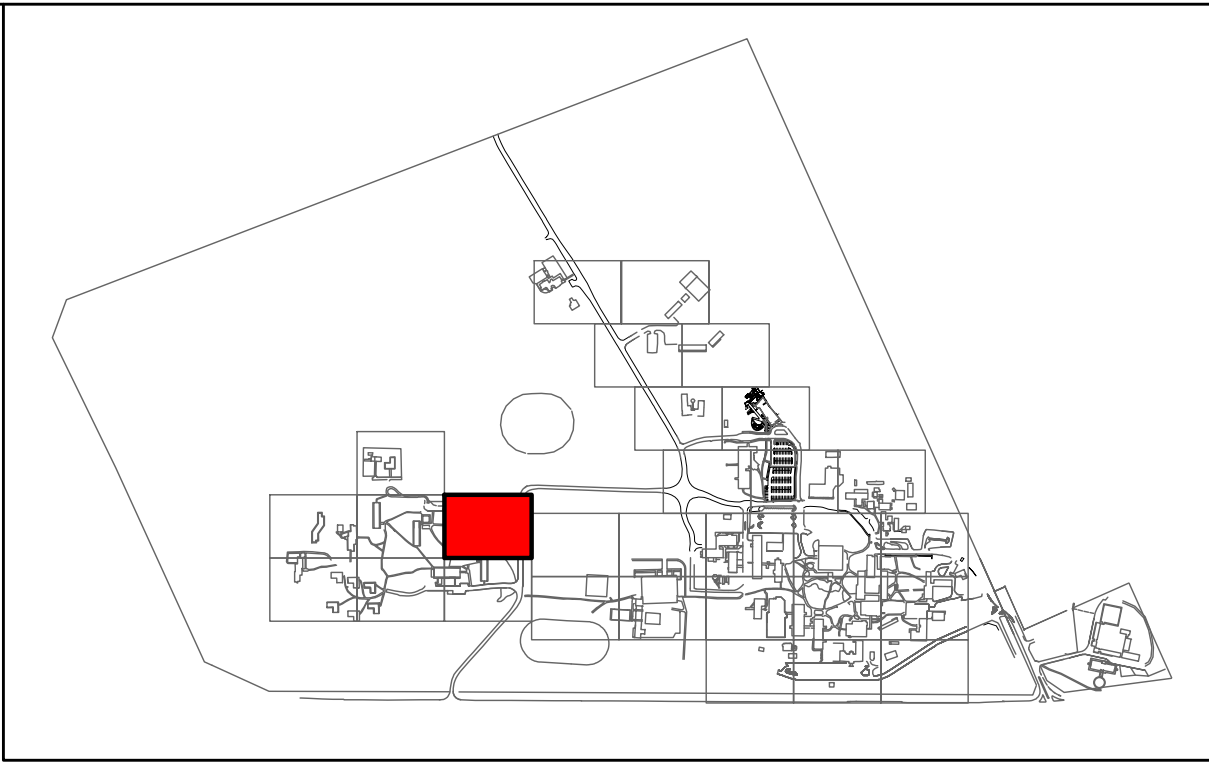
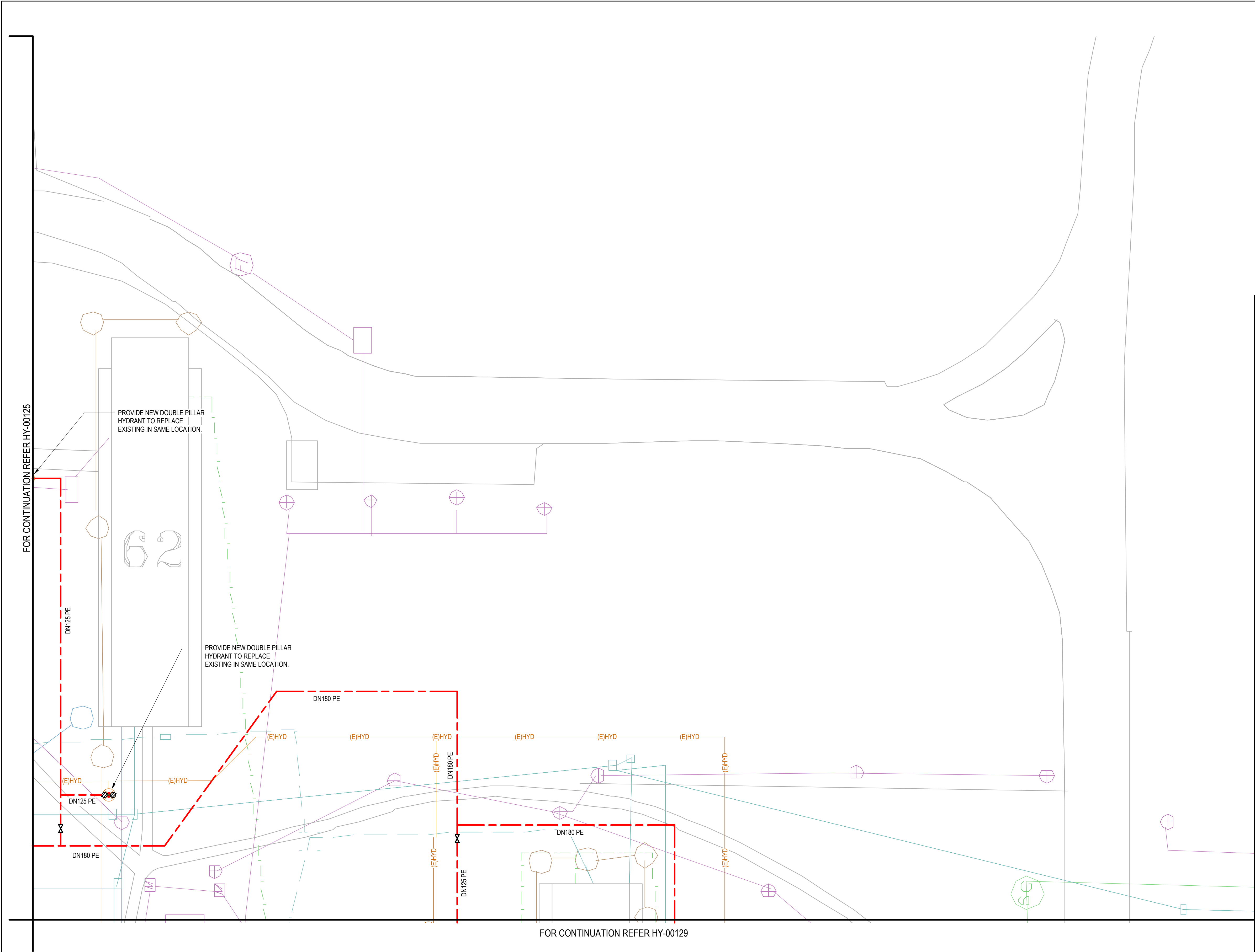
Drawing Title **HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 25**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00125

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

NOTES:

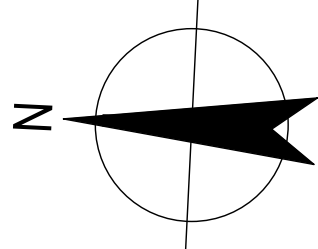
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SITE SERVICES LAYOUT - SHEET 26  
SCALE 1 : 250

|          |                |                       |
|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
| Rev      | Description    | Checked Approved Date |
| Author   | Designing      | Designing             |
| Designer | Designing      | Designing             |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 26

Drawing No.  
12622621-GHD-00-00-DRG-HY-00126

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00124

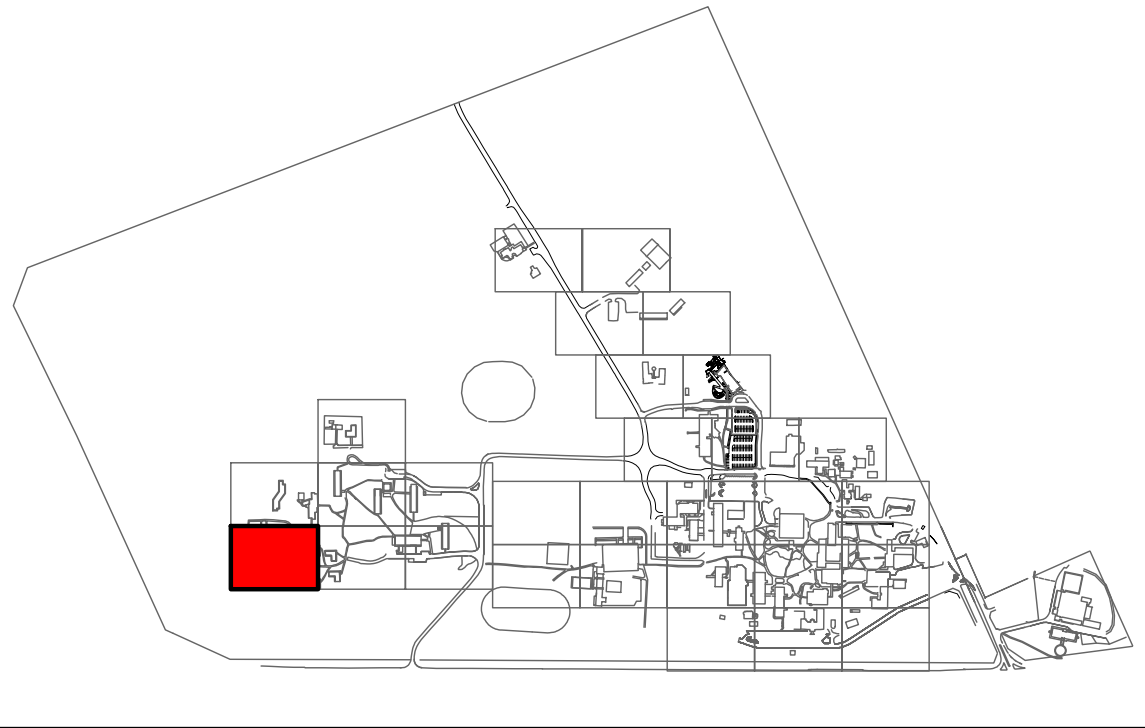
PROVIDE NEW DOUBLE PILLAR HYDRANT MINIMUM 10m FROM BUILDING TO REPLACE EXISTING.

NOTE, LIAISE WITH CQU FACILITIES MANAGEMENT TO CONFIRM COVERAGE OF BUILDING 48 FROM NEW HYDRANT POSITION.

SITE SERVICES LAYOUT - SHEET 27

SCALE 1 : 250

FOR CONTINUATION REFER HY-00128



KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

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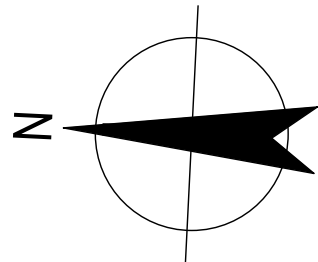
| Rev                         | Description    | Checked | Approved | Date     |
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| 0                           | AS-CONSTRUCTED |         |          | 11.06.25 |
| Author Designer             |                |         |          |          |
| Drafting Check Design Check |                |         |          |          |

0 2500 5000 7500 10000 12500mm

SCALE 1:250 AT ORIGINAL SIZE

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| Client  | CENTRAL QUEENSLAND UNIVERSITY |
| Project | FIRE MAIN UPGRADE             |
| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 27

Drawing No.  
12622621-GHD-00-00-DRG-HY-00127

Size  
A1

Rev  
0



FOR CONTINUATION REFER HY-00125

Outlet

DN125 PE

DN125 PE

PROVIDE NEW DOUBLE PILLAR  
HYDRANT TO REPLACE EXISTING  
IN SAME LOCATION.

FOR CONTINUATION REFER HY-00129

## KEY PLAN

### LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
- (E)HYD
- NEW DUAL PILLAR HYDRANT
- ISOLATION VALVE
- EXISTING HYDRANT

### NOTES:

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## SITE SERVICES LAYOUT - SHEET 28

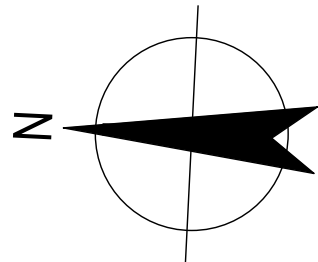
SCALE 1 : 250

FOR CONTINUATION REFER HY-00127

| Rev                         | Description    | Checked | Approved | Date     |
|-----------------------------|----------------|---------|----------|----------|
| 0                           | AS-CONSTRUCTED |         |          | 11.06.25 |
| Author Designer             |                |         |          |          |
| Drafting Check Design Check |                |         |          |          |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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Client **CENTRAL QUEENSLAND UNIVERSITY**  
Project **FIRE MAIN UPGRADE**  
Status **AS-CONSTRUCTED**

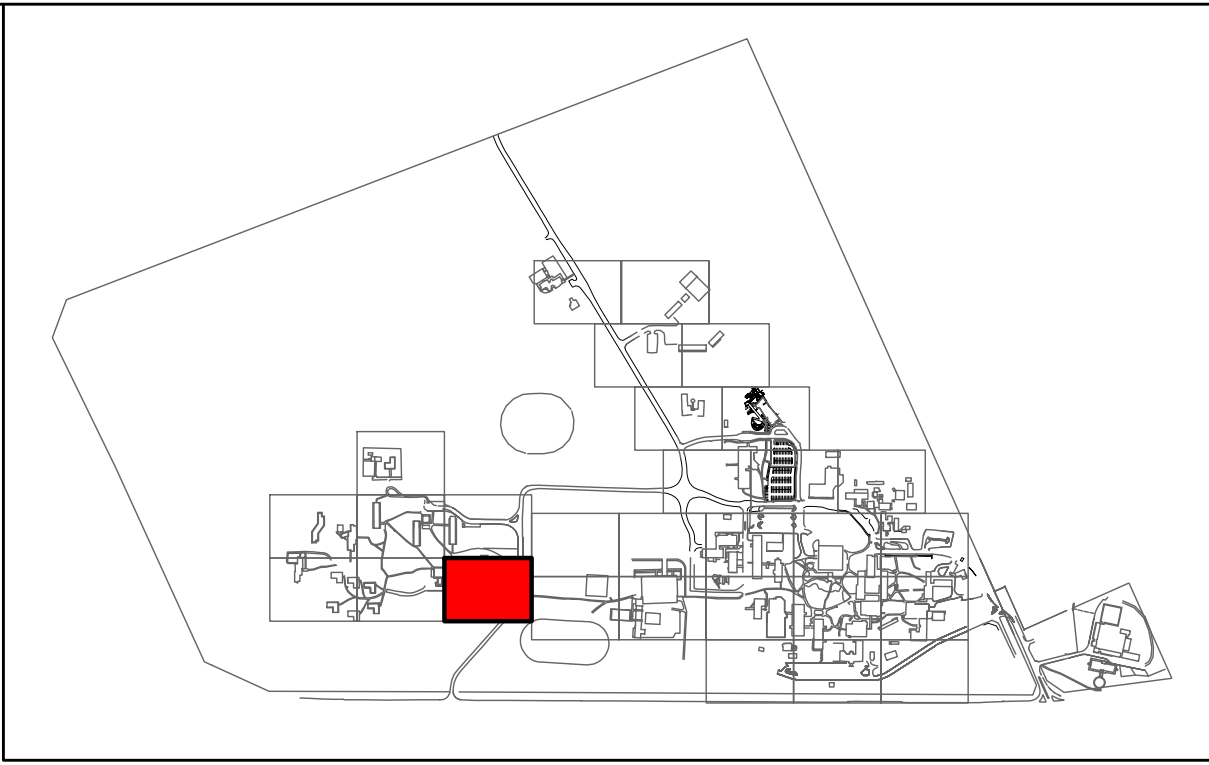
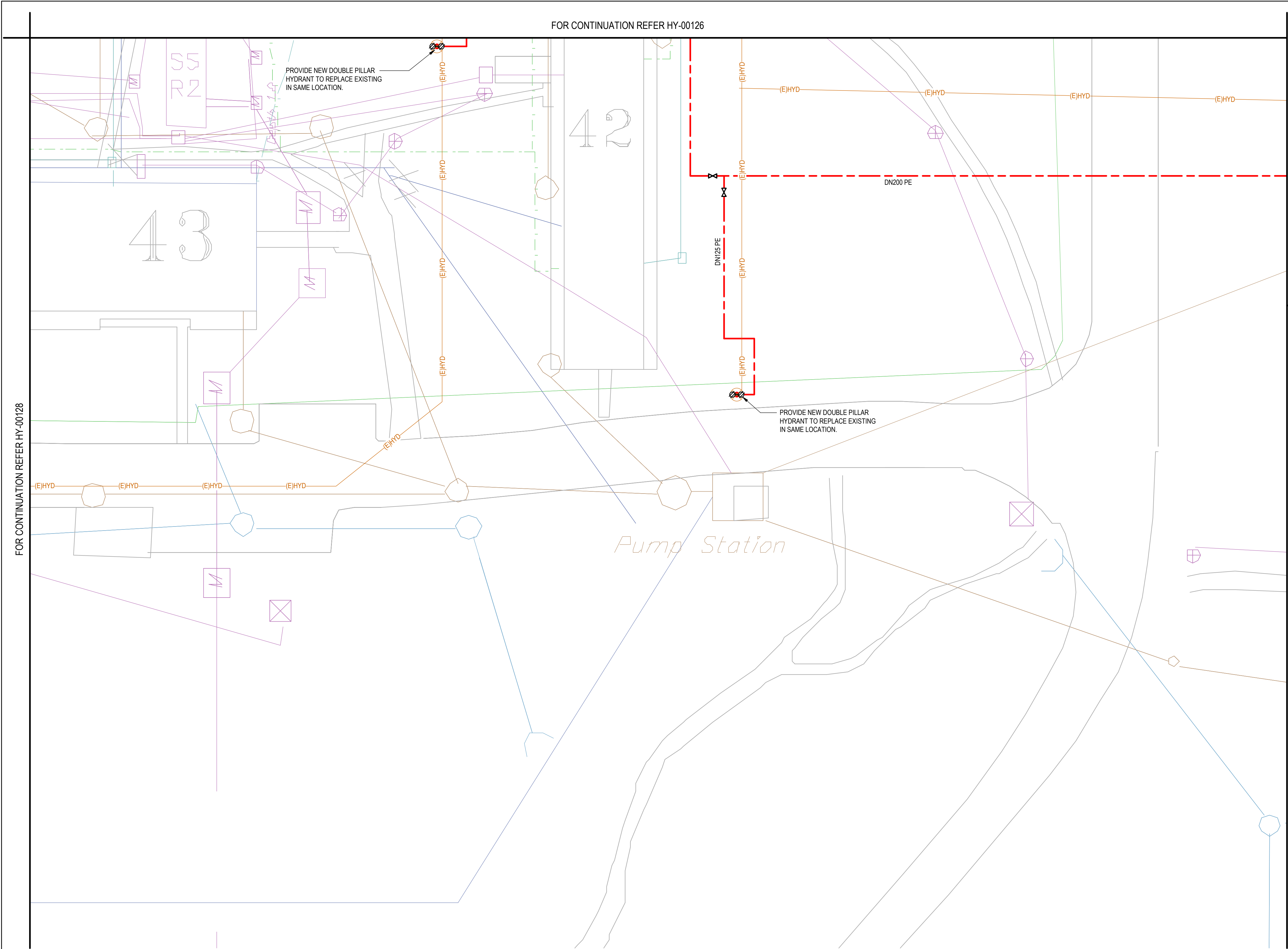
Drawing Title **HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 28**

Drawing No.  
12622621-GHD-00-00-DRG-HY-00128

Size  
A1

Rev  
0





KEY PLAN

LEGEND

- NEW FIRE SERVICE
- NEW WATER SERVICE
- EXISTING COMMUNICATIONS
- EXISTING ELECTRICAL
- EXISTING GAS
- EXISTING SEWER
- EXISTING STORMWATER
- EXISTING WATER
- EXISTING FIRE SERVICE
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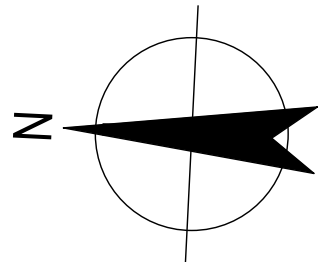
SITE SERVICES LAYOUT - SHEET 29

SCALE 1 : 250

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|----------|----------------|-----------------------|
| 0        | AS-CONSTRUCTED | 11.06.25              |
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| Designer | Design Check   |                       |

0 2500 5000 7500 10000 12500mm  
SCALE 1:250 AT ORIGINAL SIZE

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| Status  | AS-CONSTRUCTED                |

Drawing Title  
HYDRAULIC SERVICES  
SITE SERVICES LAYOUT - SHEET 29

Drawing No.  
12622621-GHD-00-00-DRG-HY-00129

Size  
A1  
Rev  
0